



IT3021 – Data Warehousing & Business Intelligence

Assignment 01

**Submitted By:
Dulshini W.G.S.
IT23720824**

Step 1: Data set selection

Dataset Identification

- **Dataset Name:** Zomato Cart Add-On Sessions Dataset
- **Source URL:** <https://www.kaggle.com/datasets/abdullahsafwan333/zomato-cart-addon-sessions-dataset>

Description of the Dataset The selected dataset consists of **100,000 simulated food delivery sessions** modeled after real-world behavior on platforms like Zomato in major Indian cities including Mumbai, Hyderabad, Chennai, and Bengaluru. Each record represents a single cart-screen session at checkout and captures a comprehensive set of features across **50+ attributes**. These include:

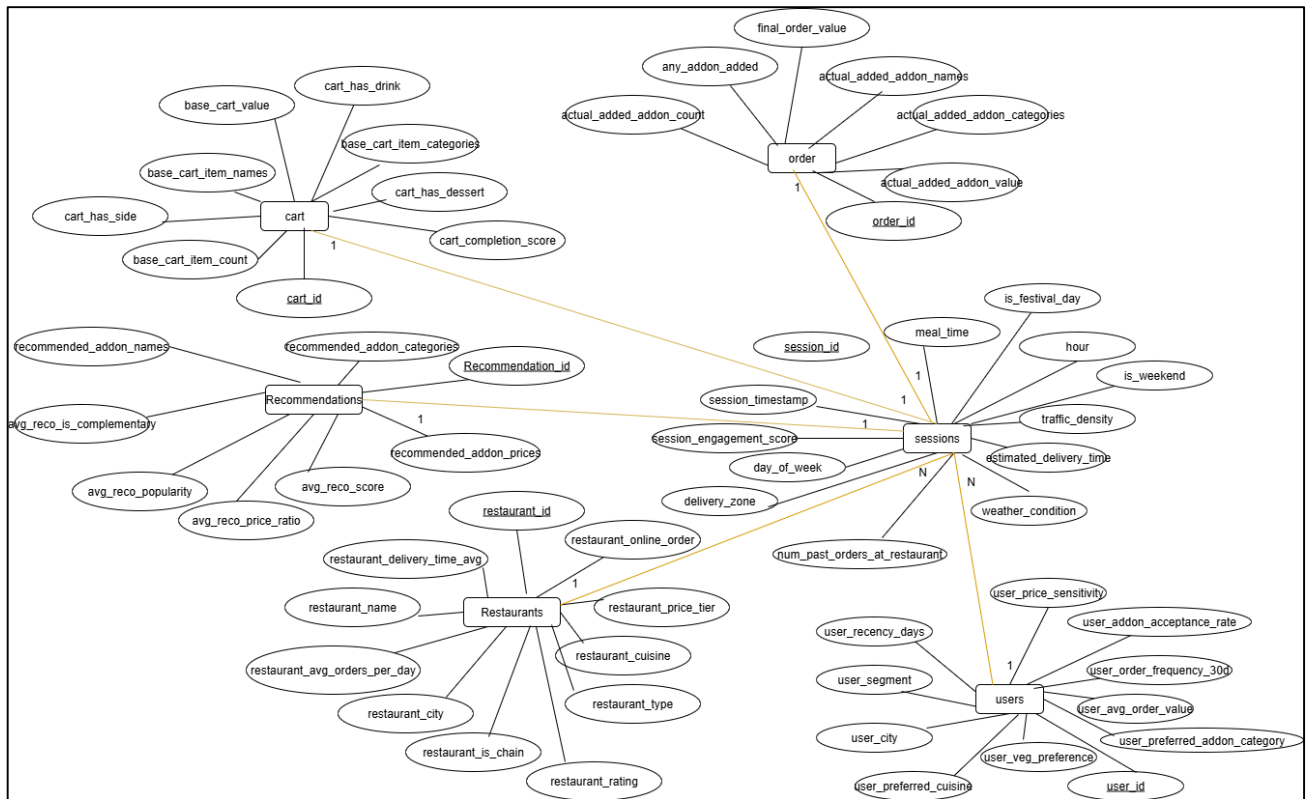
- **Session Metadata:** Unique identifiers and timestamps.
- **User Profiles:** Spending tiers (budget, mid, premium), city, and historically preferred cuisines.
- **Restaurant Details:** Cuisine types (North Indian, South Indian, Chinese, etc.), ratings, and whether the restaurant is a chain.
- **Contextual Signals:** Time of day, day of week, weather conditions, and traffic density.
- **Transaction Context:** Items currently in the cart, base cart value, and suggested add-ons.
- **Outcomes:** Actual add-ons accepted by the user and the final order value.

Justification for Selection This dataset was chosen because it strictly adheres to the "Excellent" marking criteria for Task 1:

- **Novel OLTP Scenario:** This is a novel food delivery scenario that has not been used in laboratory sessions or the AdventureWorks dataset.
- **Data Volume and Duration:** The dataset contains **100,000 records** and covers a **full-year simulation for 2025**, satisfying the requirement for at least one year of data.
- **Complexity for DW/BI:** With over 50 columns, the data provides sufficient complexity to demonstrate multiple sources, dimensional modeling, and a rich set of ETL tasks.
- **Analytical Potential:** The variety of attributes allows for the creation of meaningful hierarchies (e.g., City > Restaurant), facts (order values, scores), and dimensions (Date, User, Restaurant) required for SSAS Cubes and reporting in later stages of the project

The screenshot shows the Kaggle dataset page for 'Zomato Cart Add-On Sessions Dataset' by abdullahsafwan333. The page includes a search bar, a sidebar with navigation icons, and a main content area. The dataset is described as 'Food delivery data with user profiles, cart context & add-on recommendations'. It has 26 versions, a 'Code' button, and a 'Download' button. The 'About Dataset' section states: '100,000 simulated food delivery sessions modelled after real Indian food delivery platform behaviour (Zomato-style). Each row is one cart-screen session capturing the full context at checkout — who the user is, what's in their cart, what add-ons were recommended, and which ones were actually accepted.' The 'Usability' is 10.00, the 'License' is CC BY-SA 4.0, and the 'Expected update frequency' is 'Never'. The page also features a 'zomato' logo and a small image of a food delivery app interface.

ER Diagram



Step 2: Preparation of data sources

Initial Data State The original dataset was a single `csao_session_dataset.csv` containing 100,000 records. To fulfil the requirement for **multiple source types** and demonstrate ETL capabilities, the data has been separated into two distinct formats: **Flat Files (CSV)** and a **SQL Server Source Database**

Description of Sources

1. **Source Type 1: Flat Files (CSV format)** Reference data that is typically managed by external departments or third-party providers was kept in CSV format

Users.csv – Contain details of users including user_segment, user_city, and user_avg_order_value

Restaurants.csv - Contains details for the various restaurants, such as restaurant_cuisine, restaurant_rating, and restaurant_is_chain

2. **Source Type 2: SQL Server Database (DWBI_Source)** The high-volume transactional data and environmental variables were loaded into a relational database to simulate a live OLTP environment
 - **dbo. Cart** - This source contains cart-level details
 - **dbo.Order** - This source contains order-related information including add-ons and final order values
 - **dbo. Recommendations** - This source contains recommendation-related data used for analysis of recommendation effectiveness.
 - **dbo.sessions** - This is the main transactional dataset containing session-level information

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	user_id	user_segment	user_city	user_preferred_cuisine	user_veg	user_price	user_order	user_avg	user_rece	user_pref	user_addon	acceptance_rate						
2	2342	mid	mumbai	chinese	0	0.797196	3	439.071	4	drink	0.1251							
3	1564	mid	hyderabad	chinese	1	0.496768	8	589.9231	29	dip	0.1032							
4	3159	mid	chennai	south_indian	0	0.392278	7	586.3488	2	dip	0.118							
5	1022	budget	bengaluru	north_indian	0	0.623821	7	260.4252	27	side	0.05							
6	3883	premium	hyderabad	south_indian	1	0.522527	10	550.0849	4	starter	0.2986							
7	496	mid	hyderabad	continental	1	0.366646	7	601.4902	9	bread	0.1658							
8	863	mid	chennai	south_indian	0	0.467163	4	1411.36	5	dip	0.1411							
9	1439	budget	pune	biryani	0	0.460017	4	675.1448	34	side	0.1437							
10	2611	budget	bengaluru	chinese	1	0.613512	2	700.6033	22	dip	0.05							
11	2281	budget	hyderabad	desserts	1	0.737509	4	718.8233	7	bread	0.091							
12	468	mid	hyderabad	italian	0	0.436595	9	246.1263	3	dip	0.246							
13	790	mid	bengaluru	chinese	0	0.543539	11	1054.958	2	bread	0.1897							
14	1559	mid	mumbai	italian	1	0.05	8	624.7084	33	side	0.1887							
15	2207	budget	bengaluru	italian	0	0.65964	4	1148.152	6	dip	0.05							
16	1265	premium	bengaluru	fast_food	0	0.435019	10	344.5236	29	starter	0.2292							
17	3738	mid	mumbai	arabian	0	0.52387	5	803.8931	31	side	0.1615							
18	230	budget	bengaluru	continental	0	0.05	4	611.1093	2	bread	0.2227							
19	3257	budget	bengaluru	biryani	0	0.668244	8	742.0484	11	bread	0.0894							
20	1739	budget	delhi	italian	1	0.553246	4	713.1837	24	bread	0.05							
21	3440	budget	bengaluru	arabian	1	0.585225	5	248.2939	6	side	0.0928							
22	2013	mid	bengaluru	continental	0	0.581237	3	159.9361	2	drink	0.2093							
23	3674	premium	delhi	italian	0	0.05	9	1003.653	31	dessert	0.2914							

Users.csv

Restaurants - Microsoft Office (Product Activation Failed)

restaurant_id	restaurant_name	restaurant_city	restaurant_cuisine	restaurant_type	restaurant_avg_orders_per_day
1	428 Great Wall Restaurant - Branch 3	mumbai	chinese	Quick Bites	Yes
2	52 SodaBottleOpenerWala	hyderabad	continental	Cafe	No
3	386 SLV Corner - Branch 4	chennai	south_indian	Casual Dining	Yes
4	95 The Bigg Small Cafe - Branch 2	bengaluru	north_indian	Fine Dining	Yes
5	393 Nandhini Restaurant - Branch 4	hyderabad	south_indian	Food Court	No
6	623 Biryani By Kilo - Branch 4	hyderabad	biryani	Casual Dining	No
7	104 BOX8	chennai	fast_food	Quick Bites	Yes
8	536 Nandhini Restaurant - Branch 5	pune	south_indian	Delivery	Yes
9	282 Eat South - Branch 3	bengaluru	south_indian	Quick Bites	Yes
10	25 Sugar & Spice	hyderabad	desserts	Cafe	No
11	70 Saffron House	hyderabad	north_indian	Casual Dining	Yes
12	568 Gino's Pizza - Branch 4	bengaluru	italian	Fine Dining	Yes
13	331 The Great Kabab Factory - Branch 4	mumbai	north_indian	Casual Dining	Yes
14	538 Moti Mahal - Branch 6	bengaluru	north_indian	Casual Dining	Yes
15	635 Rebel Foods - Branch 4	bengaluru	fast_food	Delivery	Yes
16	64 Napoli Ristorante	mumbai	italian	Casual Dining	Yes
17	627 Hotel Janatha Deluxe - Branch 6	bengaluru	south_indian	Casual Dining	Yes
18	50 Authentic Kebab	bengaluru	arabian	Quick Bites	Yes
19	200 La Piazza	delhi	italian	Delivery	Yes
20	427 Mama Mia - Branch 3	bengaluru	italian	Casual Dining	Yes
21	363 Bistro 57 - Branch 2	bengaluru	continental	Casual Dining	No
22	467 Hotel Dorchini - Branch 4	delhi	south_indian	Casual Dining	Yes

Restaurants.csv

DWBI_Source
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.Cart
dbo.Order
dbo.Recommendations
dbo.Sessions
Dropped Ledger Tables
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
Storage
Security

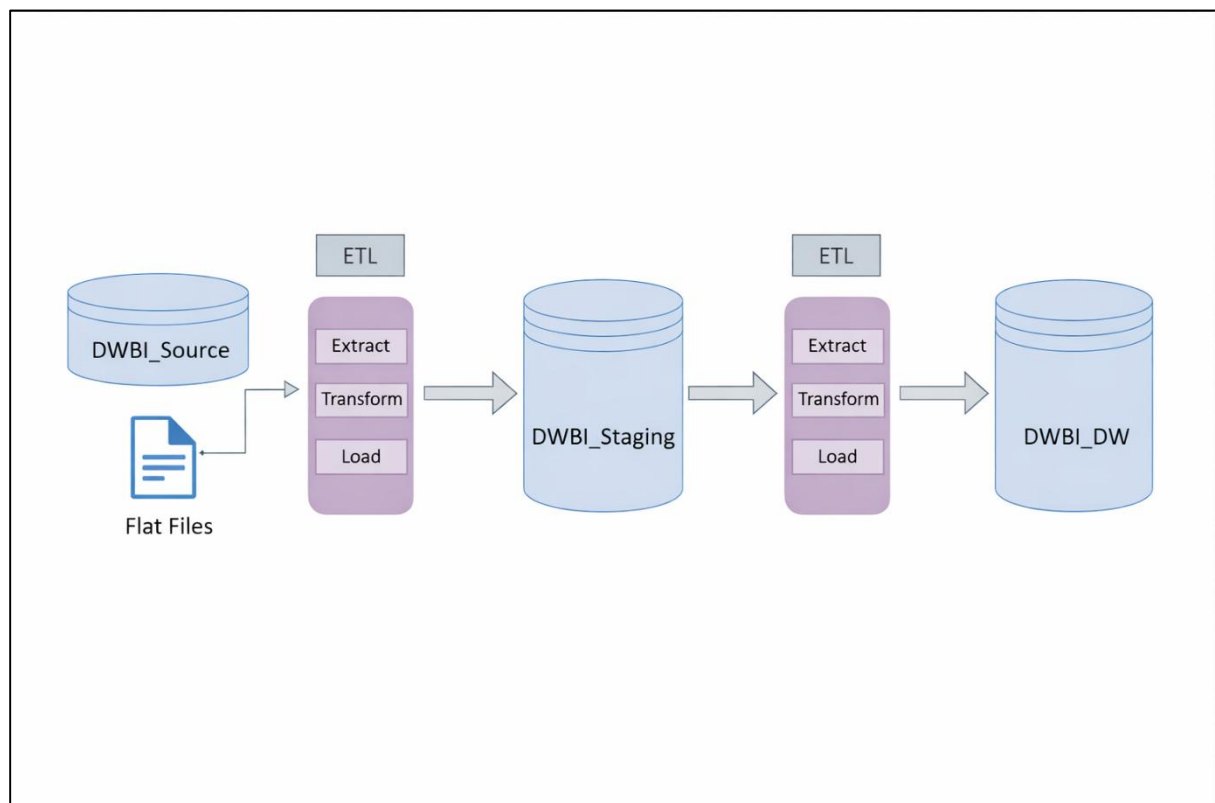
DWBI_Source

Step 3: Solution architecture

The solution architecture for the Zomato Cart Add-On analysis project is designed to manage data from heterogeneous sources, transforming raw session logs and master data into actionable business insights. The architecture follows a multi-tier approach, ensuring data integrity and scalability for multi-dimensional analysis.

Architectural Components

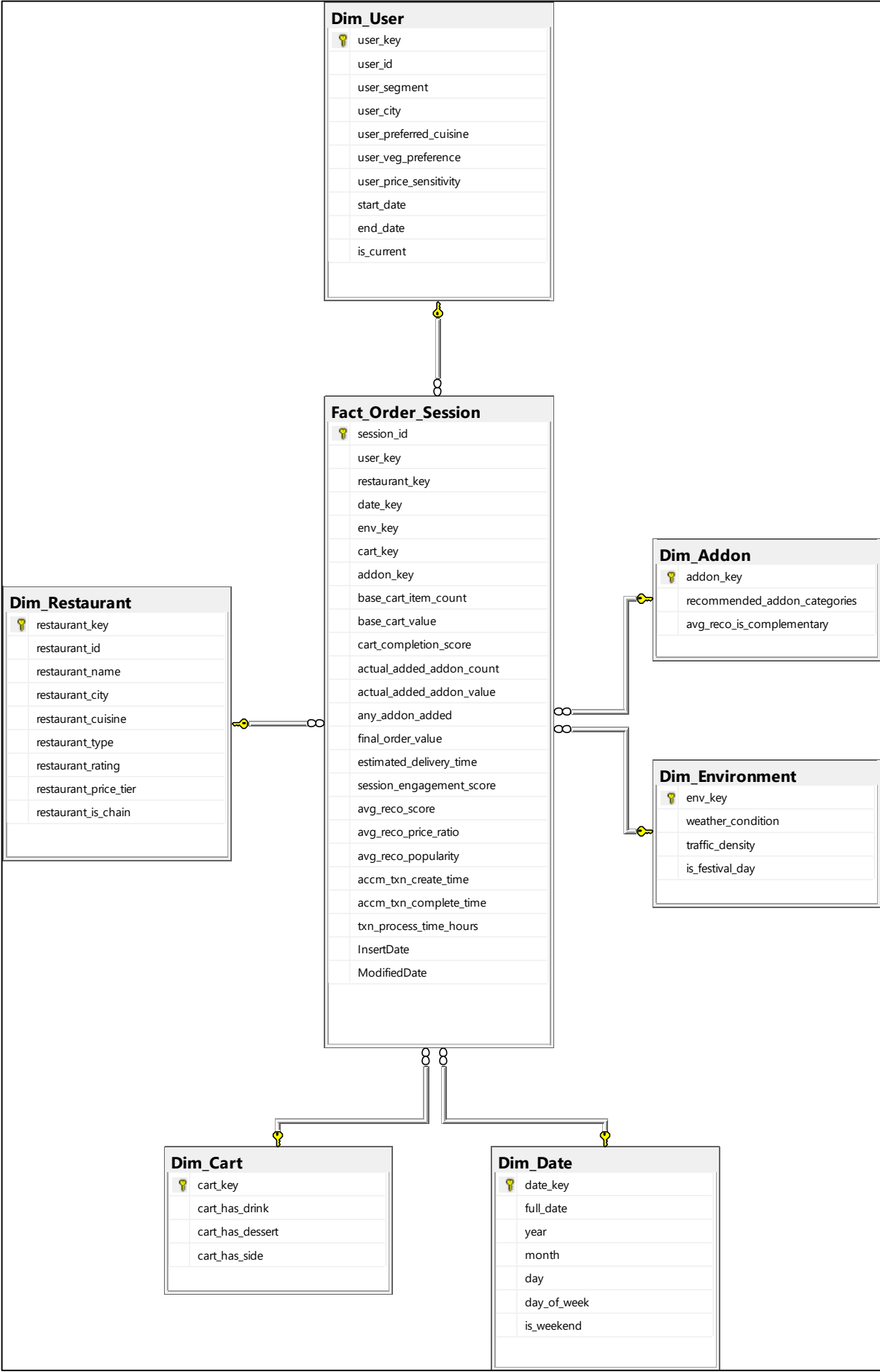
1. **Data Sources (Multiple Formats)** The foundation consists of the two source types prepared in Step 2:
 - **Flat Files:** Users.csv and Restaurants.csv.
 - **Relational Database (DWBI_Surce):** dbo.Cart , dbo.Order, dbo.Recommendations, dbo.Sessions
2. **Staging Area (DWBI_Staging)** A dedicated staging database is used as a landing zone. During the first ETL phase, data is extracted from the sources and loaded here. The primary purpose of this layer is to **standardize data types** and perform initial cleansing without affecting the production sources.
3. **Core ETL Process (SSIS)** The ETL (Extract, Transform, Load) logic is developed using **SQL Server Integration Services (SSIS)**. This component is split into two main pipelines:
 - **Extraction Pipeline:** Moves data from the CSV/SQL sources to the Staging DB.
 - **Transformation Pipeline:** Performs complex lookups to replace natural keys with surrogate keys, implements data cleansing (e.g., using REPLACENULL to handle missing values), and manages **Slowly Changing Dimensions (SCD)** for Users
4. **Data Warehouse (DWBI_DW)** The central repository where cleaned data is stored in a **dimensional model** (Star Schema). It consists of dimension tables (e.g., Dim_User, Dim_Restaurant, Dim_Date) and the Fact_Order_Session table, which stores transaction measures like final_order_value



Step 4: Data warehouse design & development

Assumptions

- Star schema is used to design the Dataware house design. There is one fact table and 5 dimensions apart from the data dimension.
- DimUser was considered as a slowly changing dimension.
- **Dim_User** – Extracting data from User staging table and loading into dimension. Also this is a slowly changing dimension.
- **Dim_Restaurant** – Extracting data from Restaurant staging table and loading into dimension. Also this is a slowly changing dimension.
- **Dim_Date** – Generated date dimension and loaded into the data warehouse for time-based analysis.
- **Dim_Order** – Extracting data from Order staging table and loading into dimension. Also this is a slowly changing dimension.
- **Dim_Cart** – Extracting data from Cart staging table and loading into dimension.
- **Dim_Recommendation** – Extracting data from Recommendation staging table and loading into dimension.
- **Fact_Session** – Extract data from Cart, Order, and Recommendation staging tables and load into fact table. Also implemented as an accumulating fact table



```
33
34 CREATE TABLE Dim_Addon (
35     addon_key INT IDENTITY(1,1) PRIMARY KEY,
36     recommended_addon_categories VARCHAR(100),
37     avg_reco_is_complementary BIT
38 );
```

100 % No issues found

Connected. (1/1)

SQLQuery20.s..._DW (sa (87))		Subhani.DW...o.Dim_Addon	
Column Name	Data Type	Allow Nulls	
addon_key	int	☐	
recommended_addon_c...	varchar(100)	☒	
avg_reco_is_complemen...	bit	☒	
		☐	

Column Properties

A Z

▼ (General)

(Name)

Allow Nulls

Data Type

Default Value or Binding

(General)

Dim_Addon

```
CREATE TABLE Dim_Cart (
    cart_key INT IDENTITY(1,1) PRIMARY KEY,
    cart_has_drink BIT,
    cart_has_dessert BIT,
    cart_has_side BIT
);
```

✓ No issues found

Dim_Cart

Subhani.DWB...bo.Dim_Cart SQLQuery22.s..._DW (sa (77)) SQLQuery20.s...D

Column Name	Data Type	Allow Nulls
cart_key	int	☐
cart_has_drink	bit	☒
cart_has_dessert	bit	☒
cart_has_side	bit	☒
		☐

Column Properties

```
CREATE TABLE Dim_Date (
    date_key INT PRIMARY KEY,
    full_date DATE,
    year INT,
    month INT,
    day INT,
    day_of_week VARCHAR(20),
    is_weekend BIT
);
```

Subhani.DW...bo.Dim_Date			Subhani.DWB...dbo.Dim_Cart	SQLQuery22.s..._DW (
Column Name	Data Type	Allow Nulls		
 date_key	int	☐		
full_date	date	☒		
year	int	☒		
month	int	☒		
day	int	☒		
day_of_week	varchar(20)	☒		
is_weekend	bit	☒		
		☐		

Column Properties





▼ (General)

(Name)	date_key
Allow Nulls	No
Data Type	int
Default Value or Binding	

▼ Table Designer

(General)

Dim_Date

```

49
50 CREATE TABLE Dim_Environment (
51     env_key INT IDENTITY(1,1) PRIMARY KEY,
52     weather_condition VARCHAR(50),
53     traffic_density VARCHAR(50),
54     is_festival_day BIT
55 );

```

00 % No issues found

Connected. (1/1)

Subhani.DW...Environment		Subhani.DWB...bo.Dim_Date	
Column Name	Data Type	Allow Nulls	
env_key	int	☐	
weather_condition	varchar(50)	☒	
traffic_density	varchar(50)	☒	
is_festival_day	bit	☒	
		☐	

Column Properties

(General)

(Name)

Allow Nulls

Data Type

Default Value or Binding

Table Designer

(General)

Dim_Environment


```
SQLQuery1.sql...er (sa (55))* ✕
1 CREATE TABLE Dim_User (
2     user_key INT IDENTITY(1,1) PRIMARY KEY,
3     user_id INT,
4     user_segment VARCHAR(50),
5     user_city VARCHAR(100),
6     user_preferred_cuisine VARCHAR(100),
7     user_veg_preference VARCHAR(10),
8     user_price_sensitivity VARCHAR(50),
9
10    start_date DATETIME,
11    end_date DATETIME,
12    is_current BIT
13 );
```

Subhani.DW...bo.Dim_User ✕ Subhani.DWB...m_Restaurant SQLQuery2.sql..._DW (sa (72))

Column Name	Data Type	Allow Nulls
user_key	int	☐
user_id	int	☒
user_segment	nvarchar(50)	☒
user_city	nvarchar(50)	☒
user_preferred_cuisine	nvarchar(50)	☒
user_veg_preference	bit	☒
user_price_sensitivity	real	☒
start_date	datetime	☒
end_date	datetime	☒
is_current	bit	☒
		☐

Column Properties

⚙️ ⚡ ⚙️

▼ (General)

(Name)	user_key
Allow Nulls	No
Data Type	int
Default Value or Binding	

▼ Table Designer

(General)

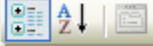
Dim_User

```

14
15 CREATE TABLE Dim_Restaurant (
16     restaurant_key INT IDENTITY(1,1) PRIMARY KEY,
17     restaurant_id INT,
18     restaurant_name VARCHAR(255),
19     restaurant_city VARCHAR(100),
20     restaurant_cuisine VARCHAR(100),
21     restaurant_type VARCHAR(100),
22     restaurant_rating DECIMAL(3,2),
23     restaurant_price_tier INT,
24     restaurant_is_chain VARCHAR(10)
25 );
26

```

Subhani.DWB..._Restaurant			SQLQuery2.sql..._DW (sa (72))	SQLQuery1.sql...ster (sa (70))
Column Name	Data Type	Allow Nulls		
 restaurant_key	int	☐		
restaurant_id	int	☒		
restaurant_name	varchar(255)	☒		
restaurant_city	varchar(100)	☒		
restaurant_cuisine	varchar(100)	☒		
restaurant_type	varchar(100)	☒		
restaurant_rating	decimal(3, 2)	☒		
restaurant_price_tier	int	☒		
restaurant_is_chain	varchar(10)	☒		
		☐		

Column Properties	
	
(General) (Name) Allow Nulls Data Type Default Value or Binding restaurant_key No int	
Table Designer (General)	

Dim_Restaurant

```
56
57 CREATE TABLE Fact_Order_Session (
58     session_id VARCHAR(50),
59
60     user_key INT,
61     restaurant_key INT,
62     date_key INT,
63
64     base_cart_item_count INT,
65     base_cart_value FLOAT,
66     cart_completion_score FLOAT,
67
68     actual_added_addon_count INT,
69     actual_added_addon_value FLOAT,
70     any_addon_added BIT,
71
72     final_order_value FLOAT,
73     estimated_delivery_time INT,
74     session_engagement_score FLOAT,
75
76     avg_reco_score FLOAT,
77     avg_reco_price_ratio FLOAT,
78     avg_reco_popularity FLOAT
79 );
```

100 % No issues found

Connected. (1/1)

Subhani.DWB...der_Session		Subhani.DWB...dbo.Dim_User	Subhani.DWB...m_Restaura
	Column Name	Data Type	Allow Nulls
🔔	session_id	varchar(50)	☐
	user_key	int	☒
	restaurant_key	int	☒
	date_key	int	☒
	env_key	int	☒
	cart_key	int	☒
	addon_key	int	☒
	base_cart_item_count	int	☒
	base_cart_value	float	☒
	cart_completion_score	float	☒
	actual_added_addon_co...	int	☒
	actual_added_addon_val...	float	☒
	any_addon_added	bit	☒
	final_order_value	float	☒

Fact_Order_Session

Column Properties

🔍

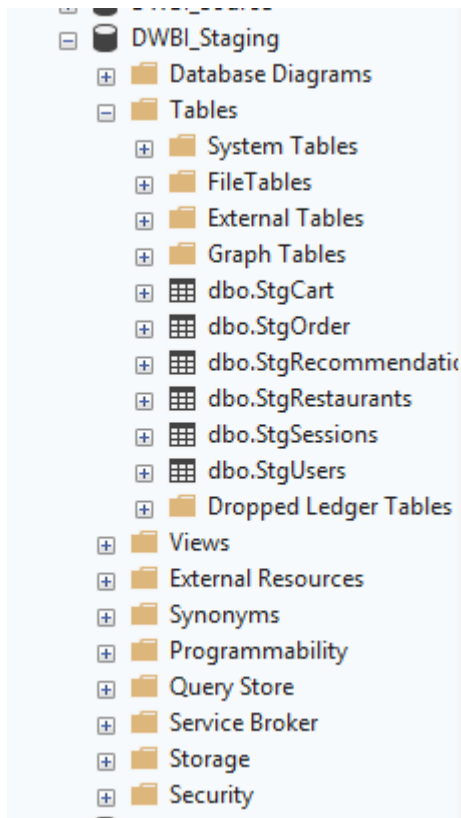
🔍

🔍

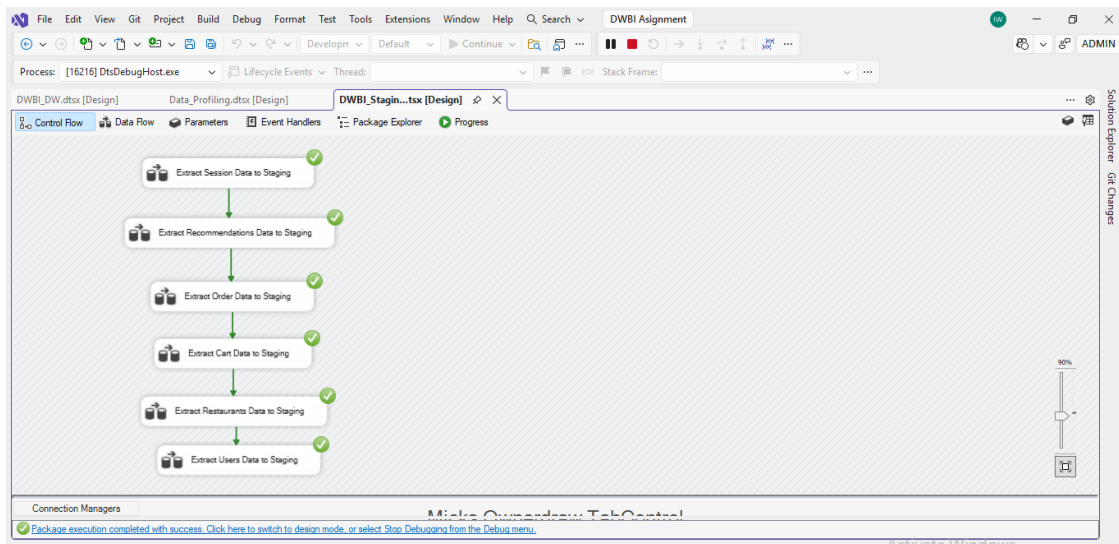
(Name)	session_id
Allow Nulls	No
Data Type	varchar
Default Value or Binding	

(General)

Step 5: ETL development

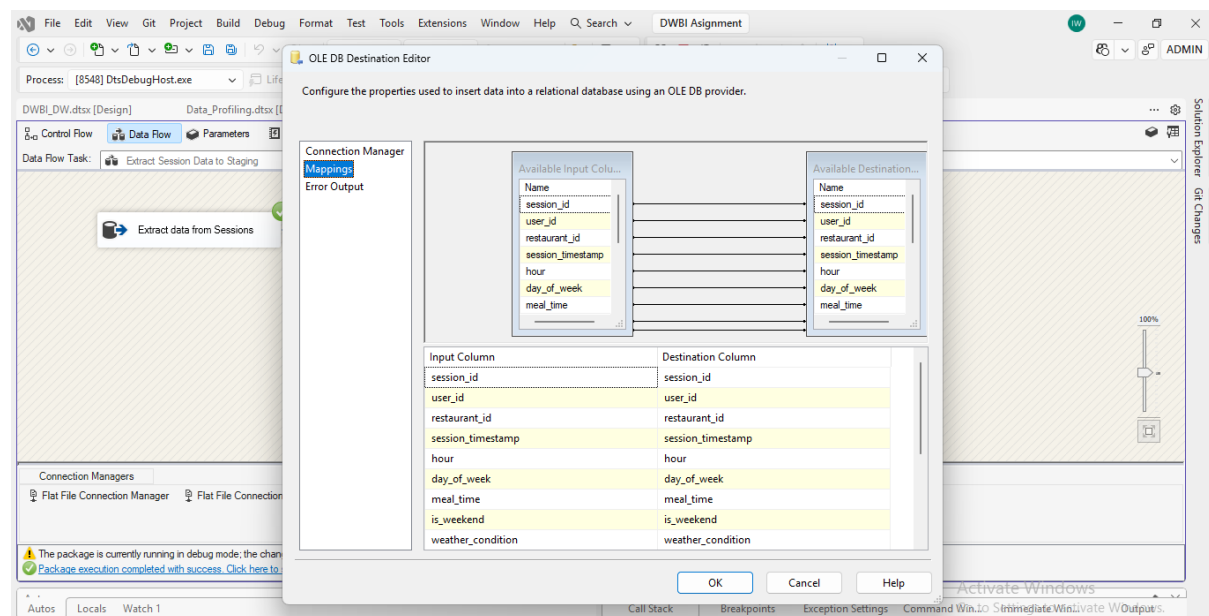
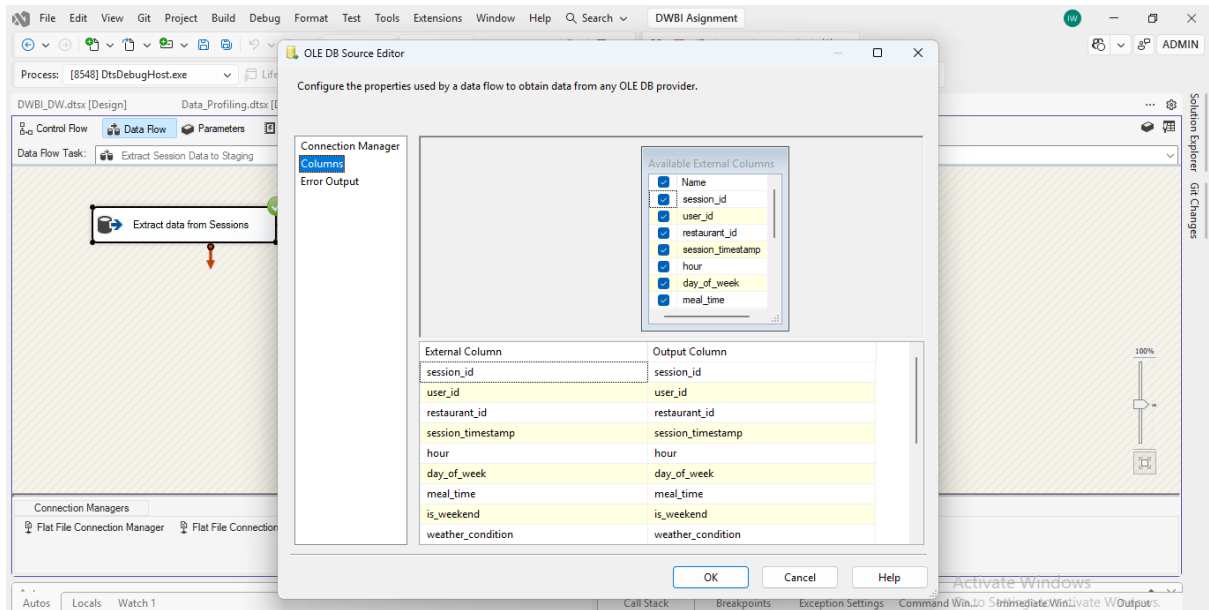
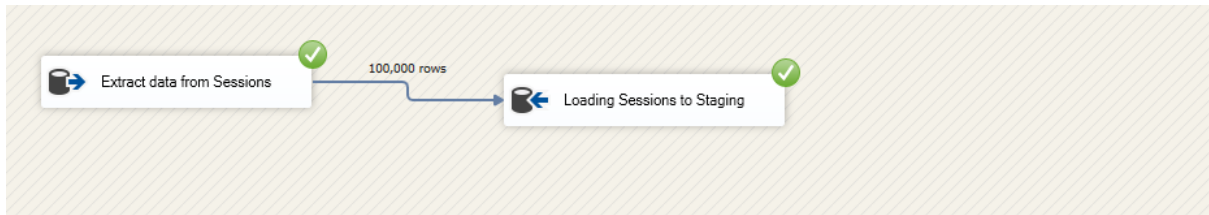


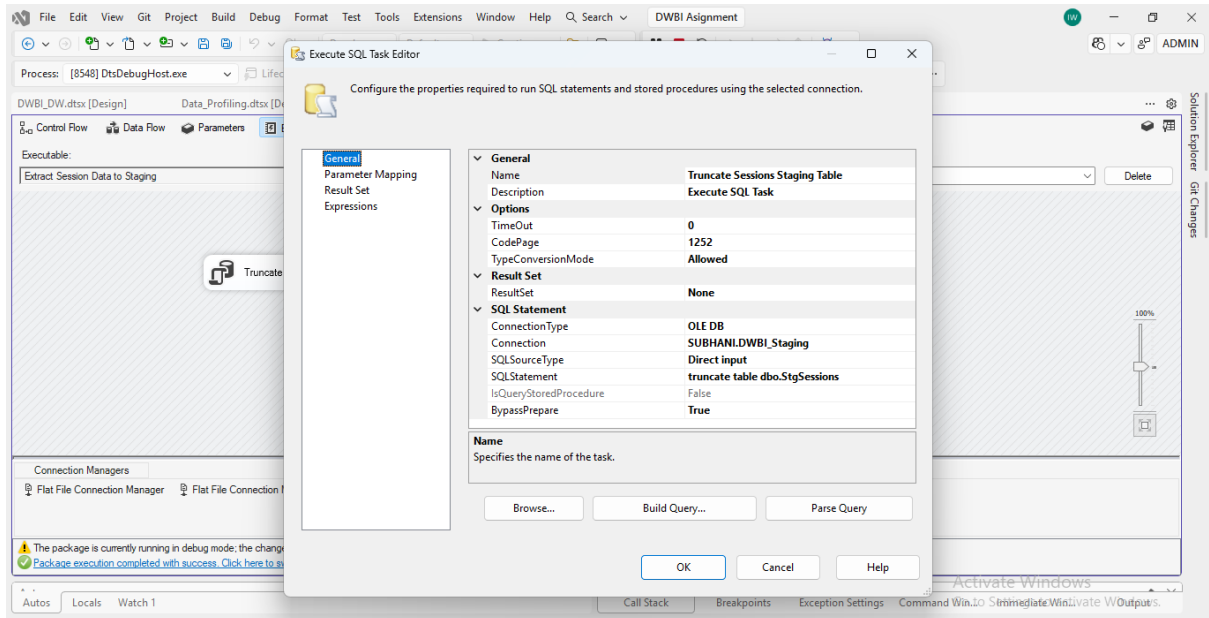
DWBI_Staging



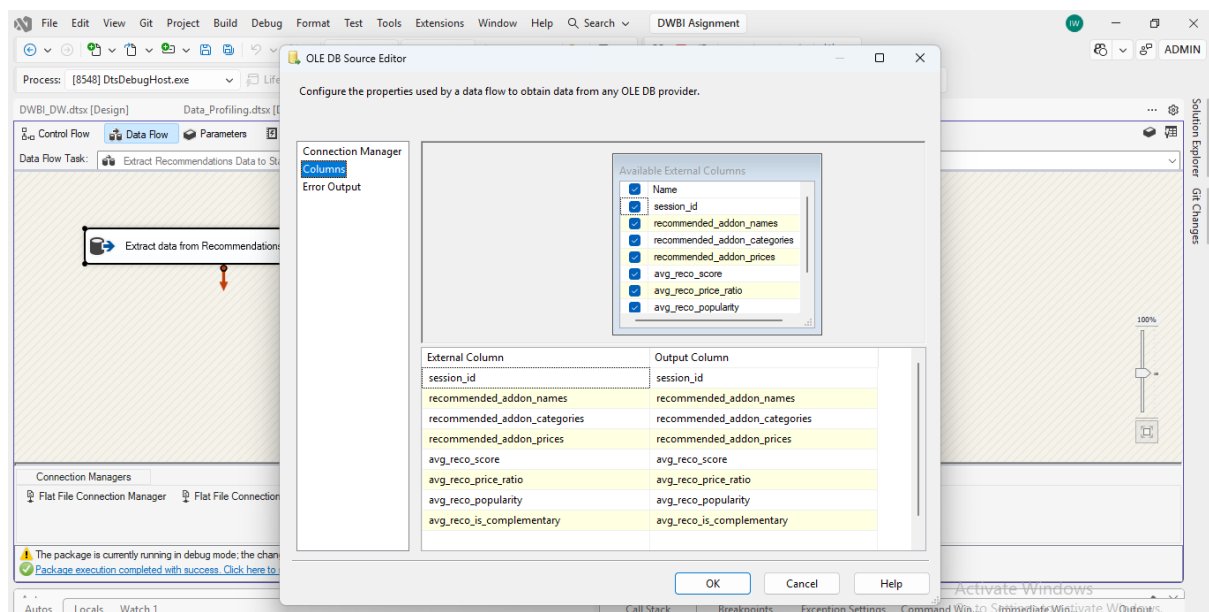
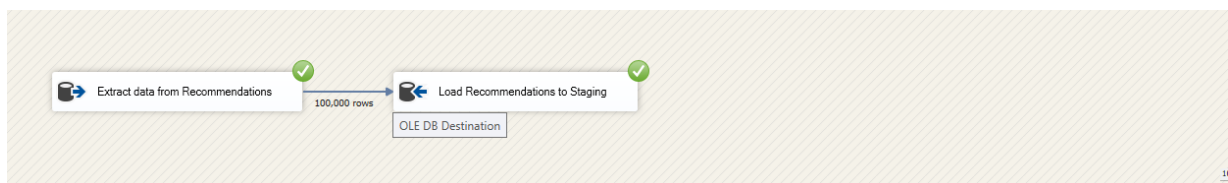
First extracted all the data from tables which are in the DWBI_Source, and flat files into separated staging DB called DWBI_Staging as shown in the below using SQL Server Integration Service Software.

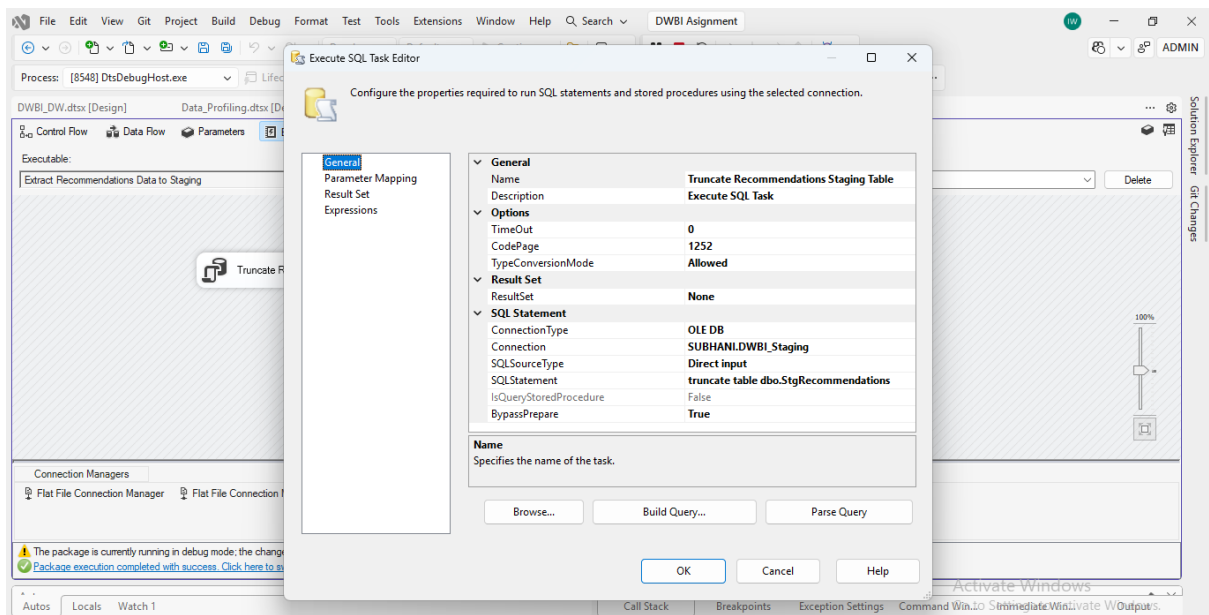
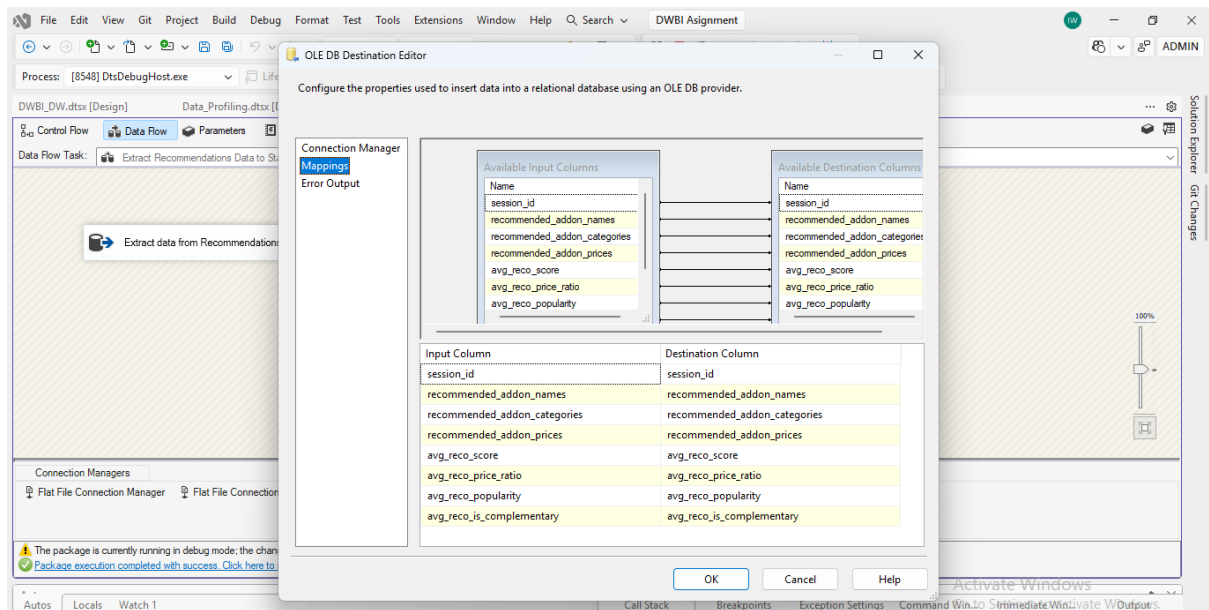
Extract Sessions Details from DWBI_Source database and load into Session Staging table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.StgSessions in DWBI_Staging database



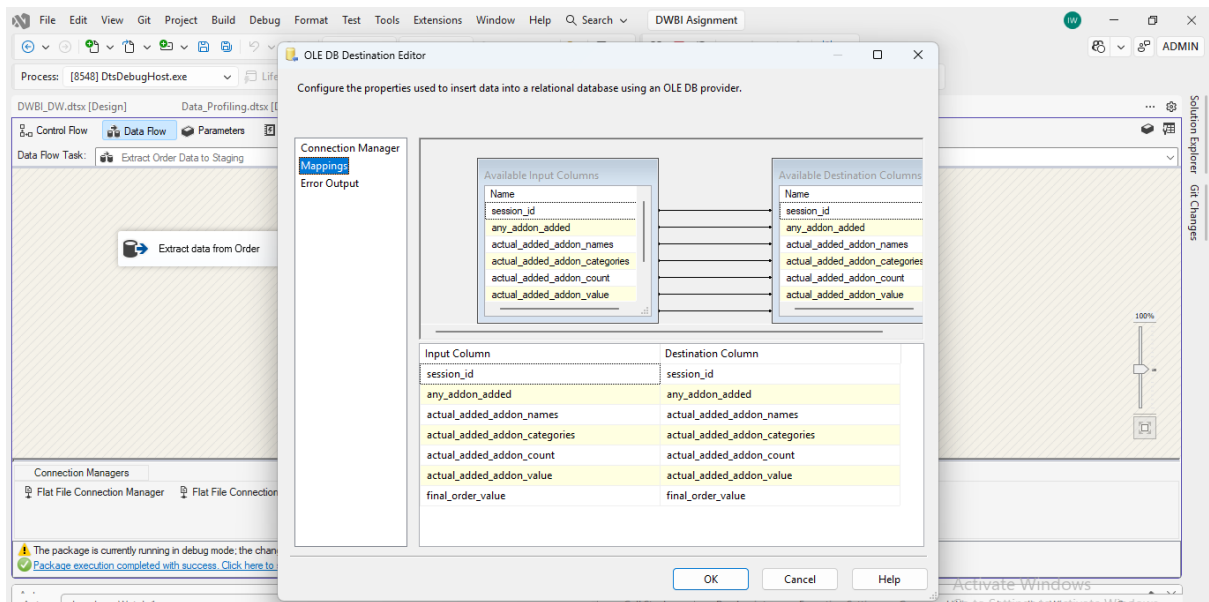
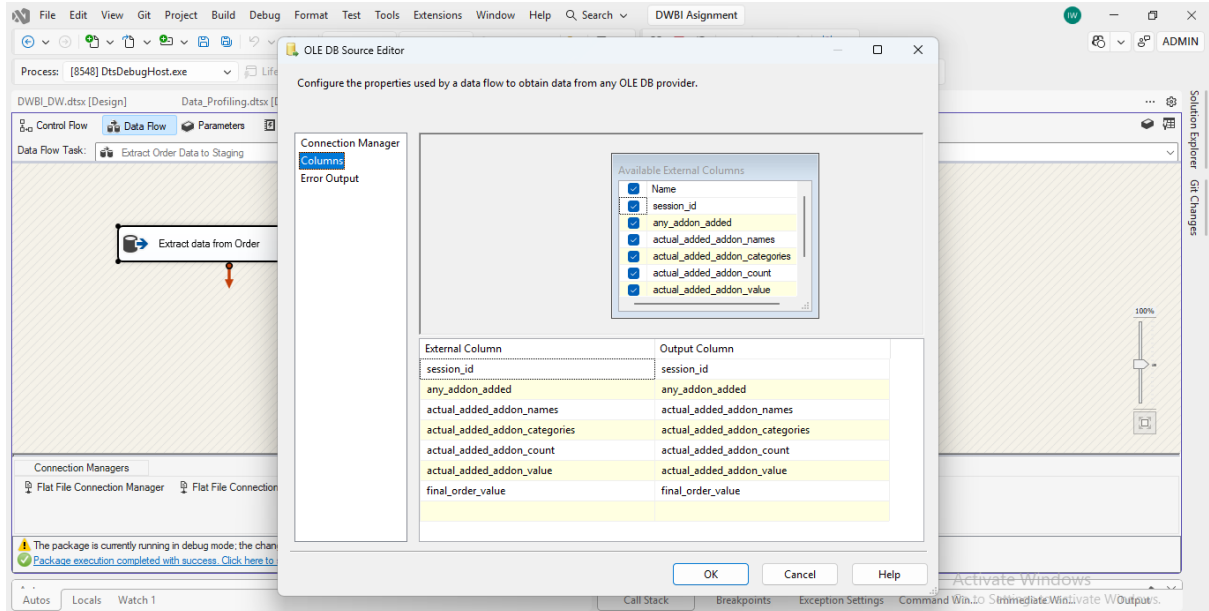
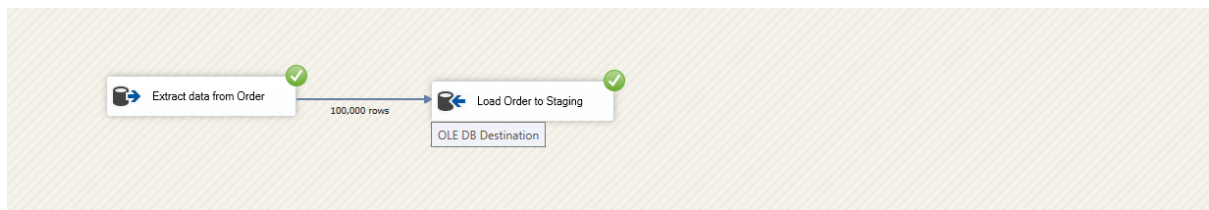


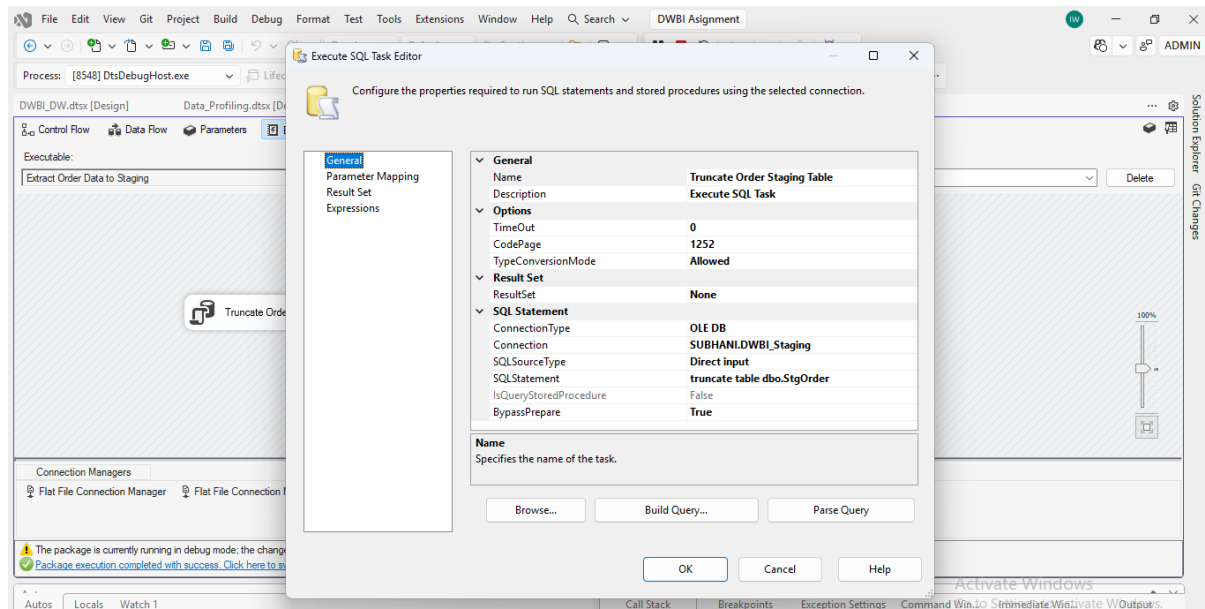
Extract Recommendations Details from DWBI_Source database and load into Customer Staging table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.StgRecommendations in DWBI_Staging database



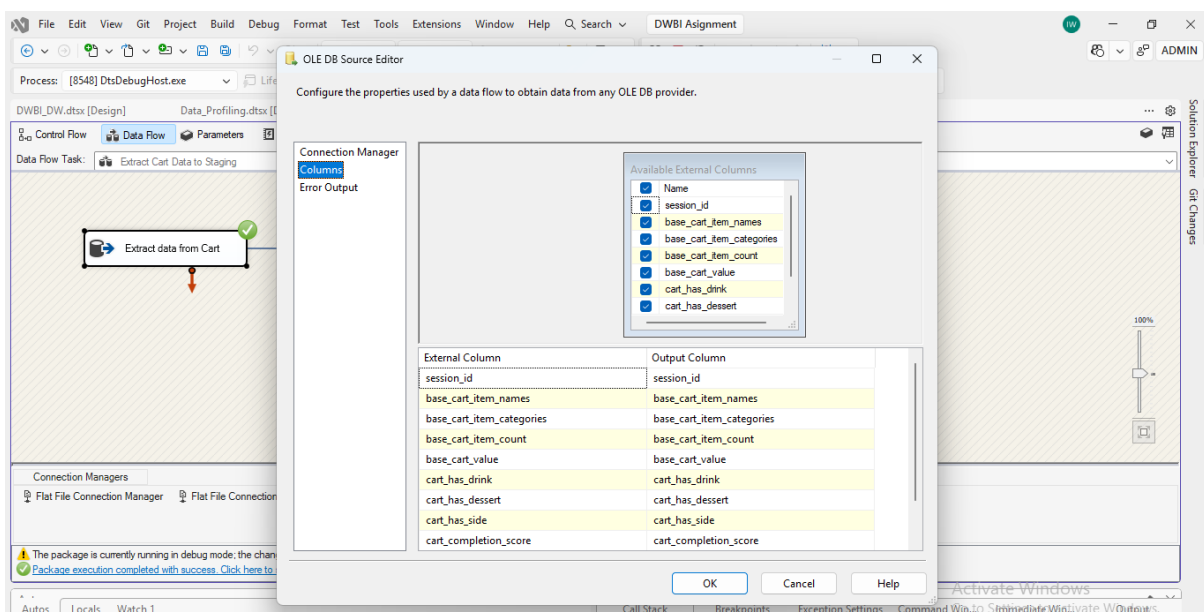
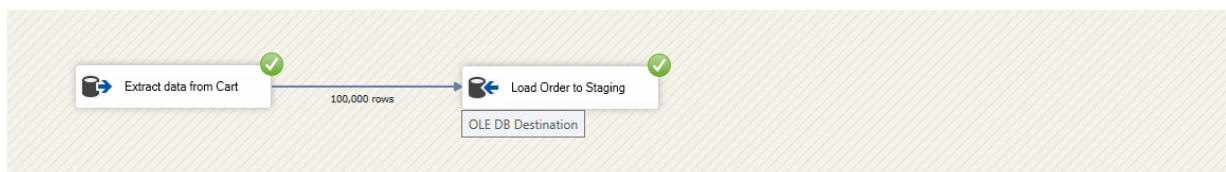


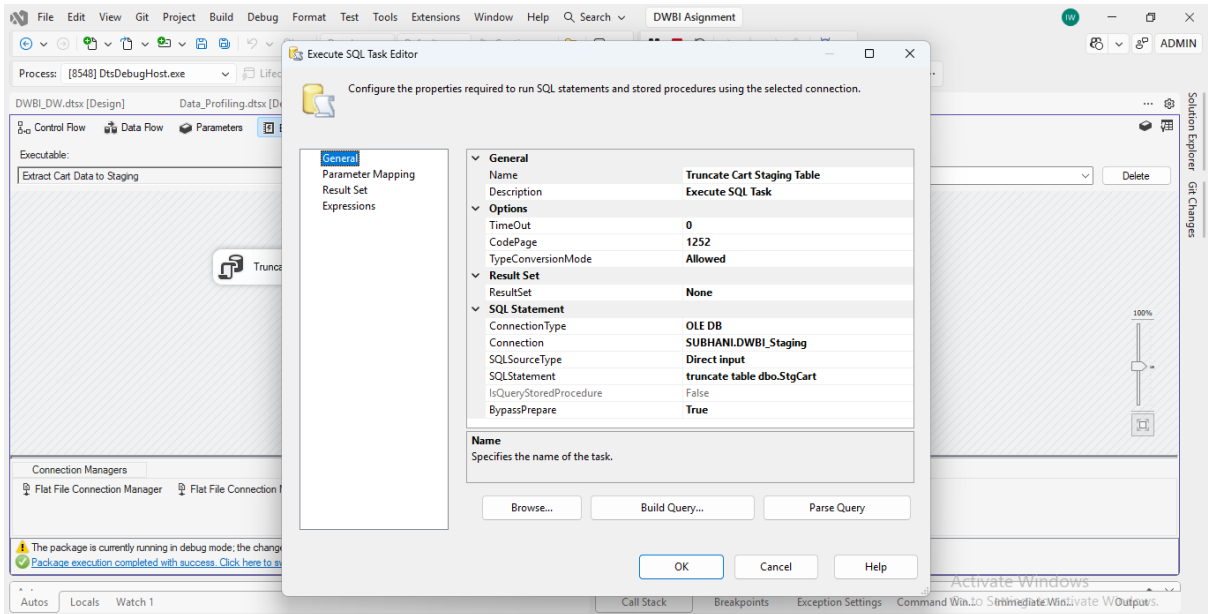
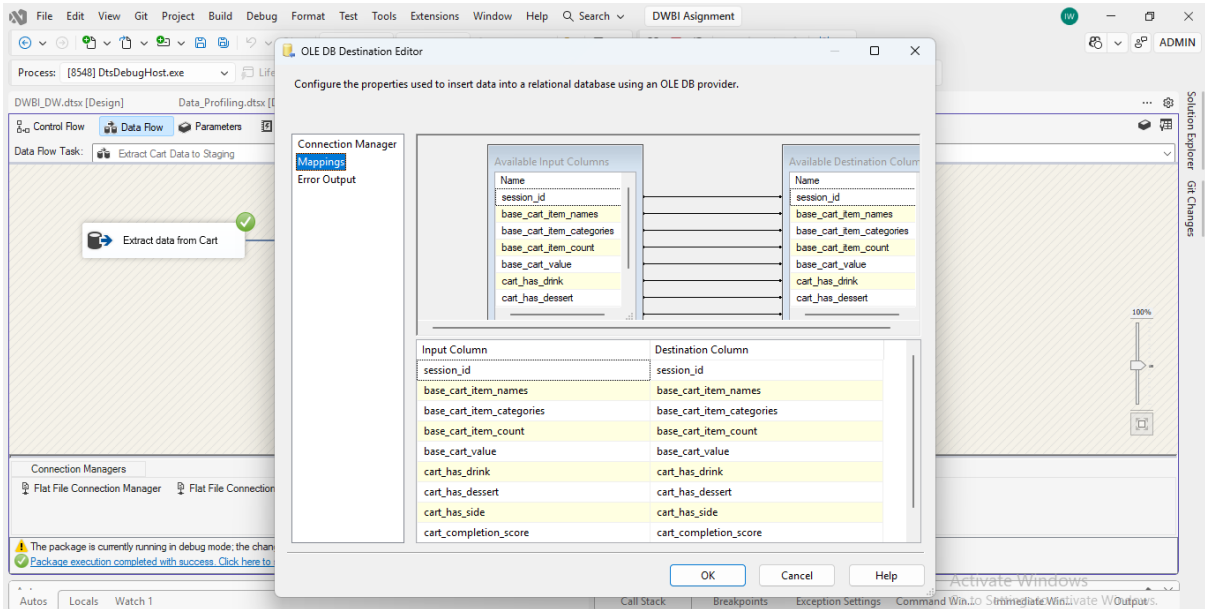
Extract Order Details from DWBI_Source and load into staging StgOrder table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.StgOrder in DWBI_Staging database



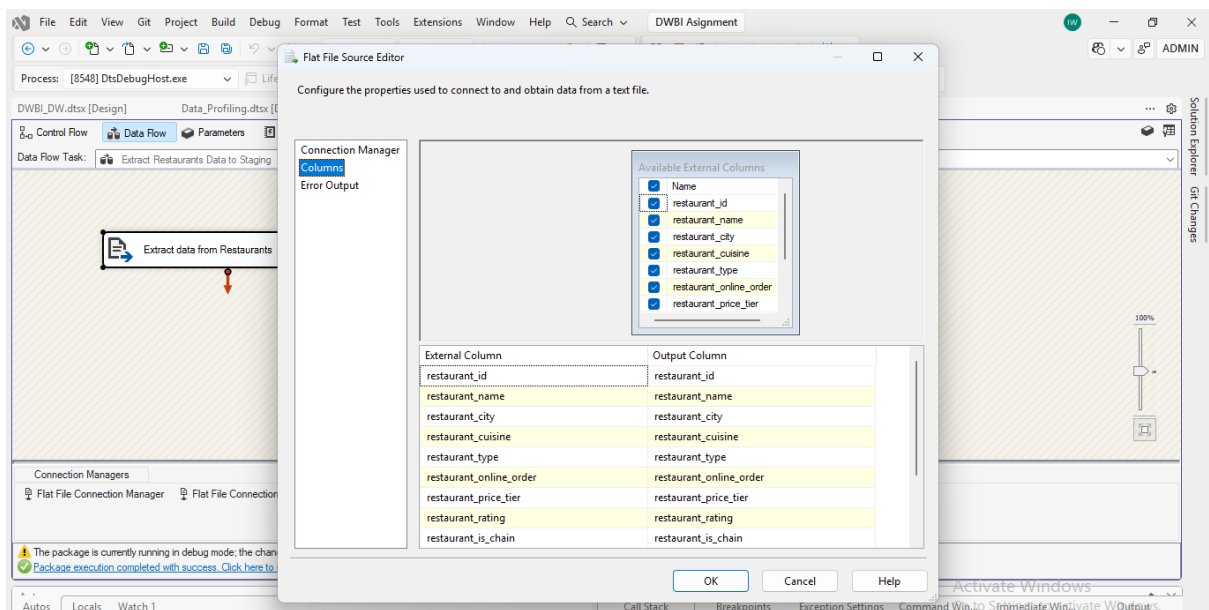
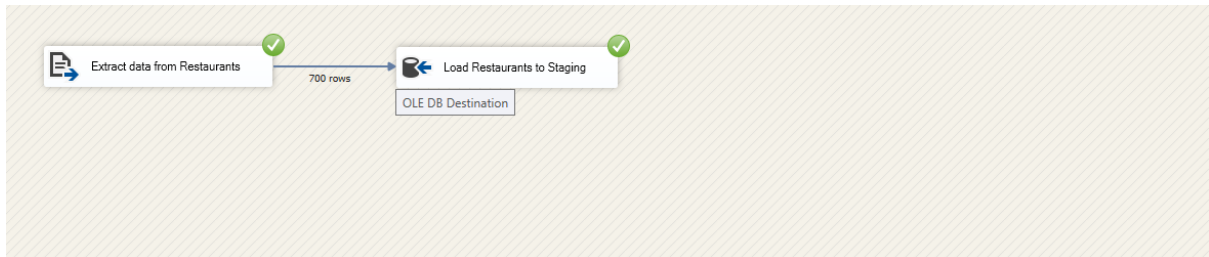


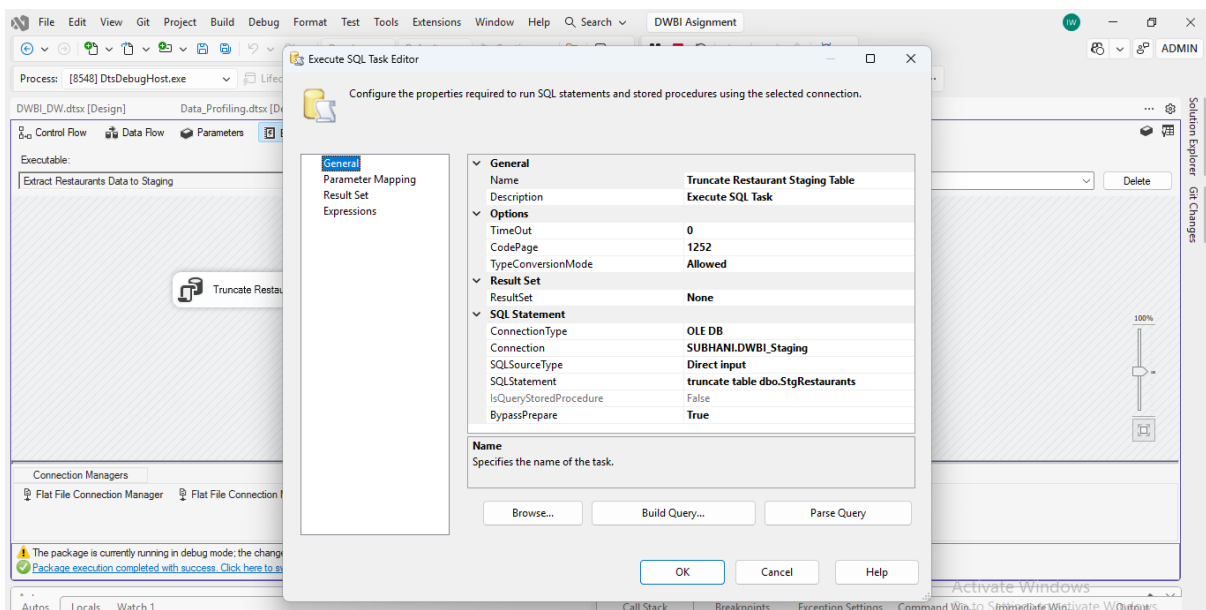
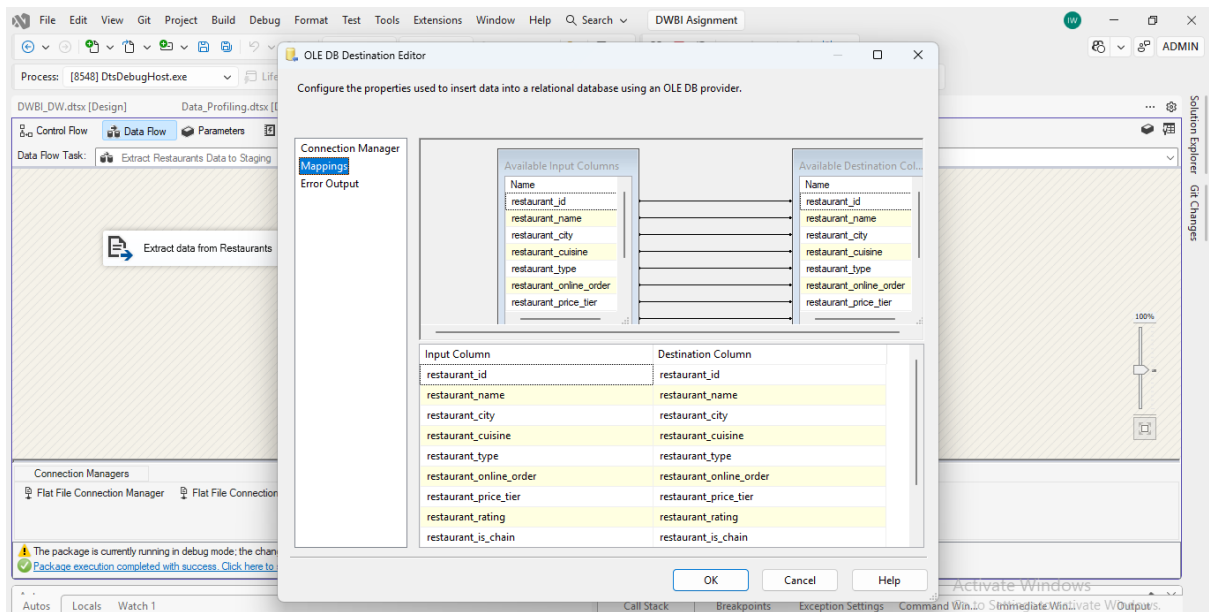
Extract cart Details from DWBI_Source database and load into staging StgCart table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.StgCart in DWBI_Staging Database.



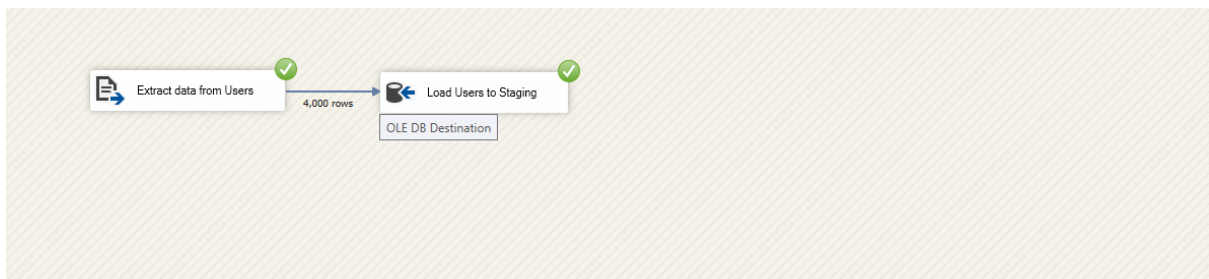


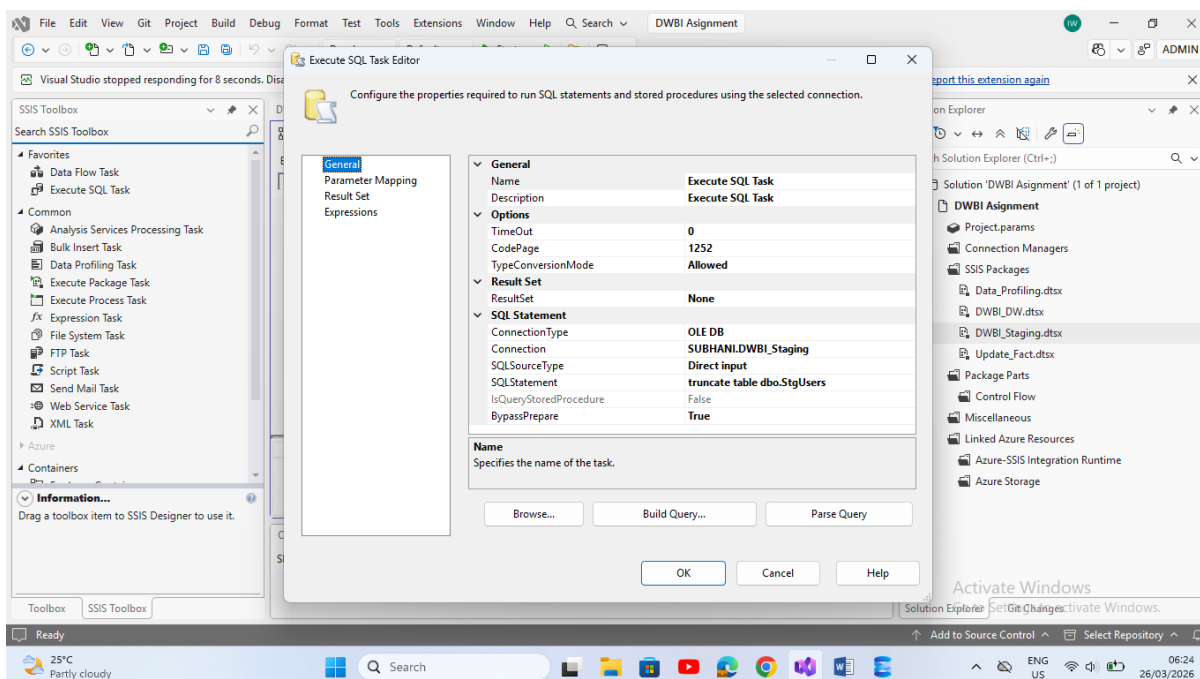
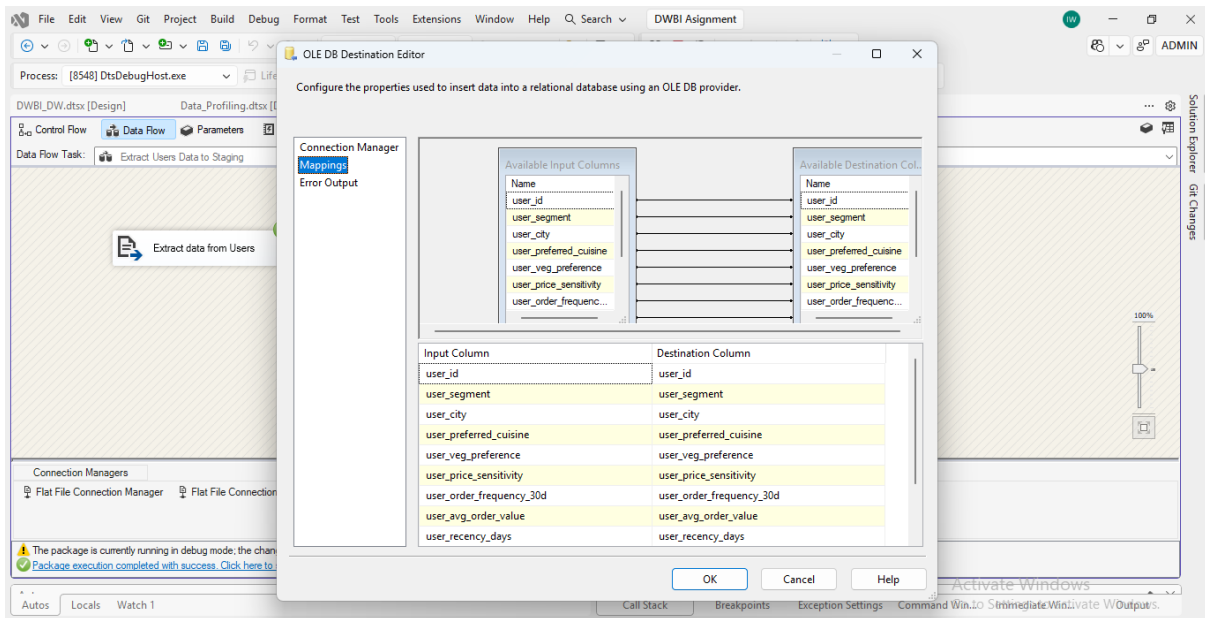
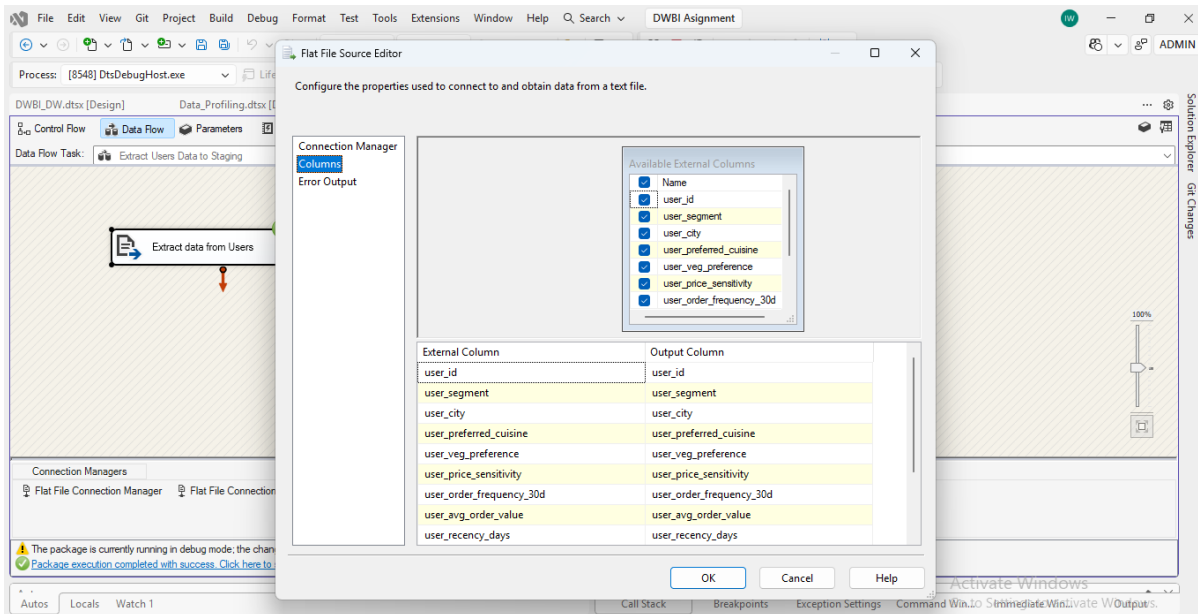
Extract cart Details from Restaurants.csv and load into staging Restaurant Staging table. Used Execute SQL Task SSIS tool Truncate table for SQL command as truncate table dbo.StgRestaurants in DWBI_Staging Database.



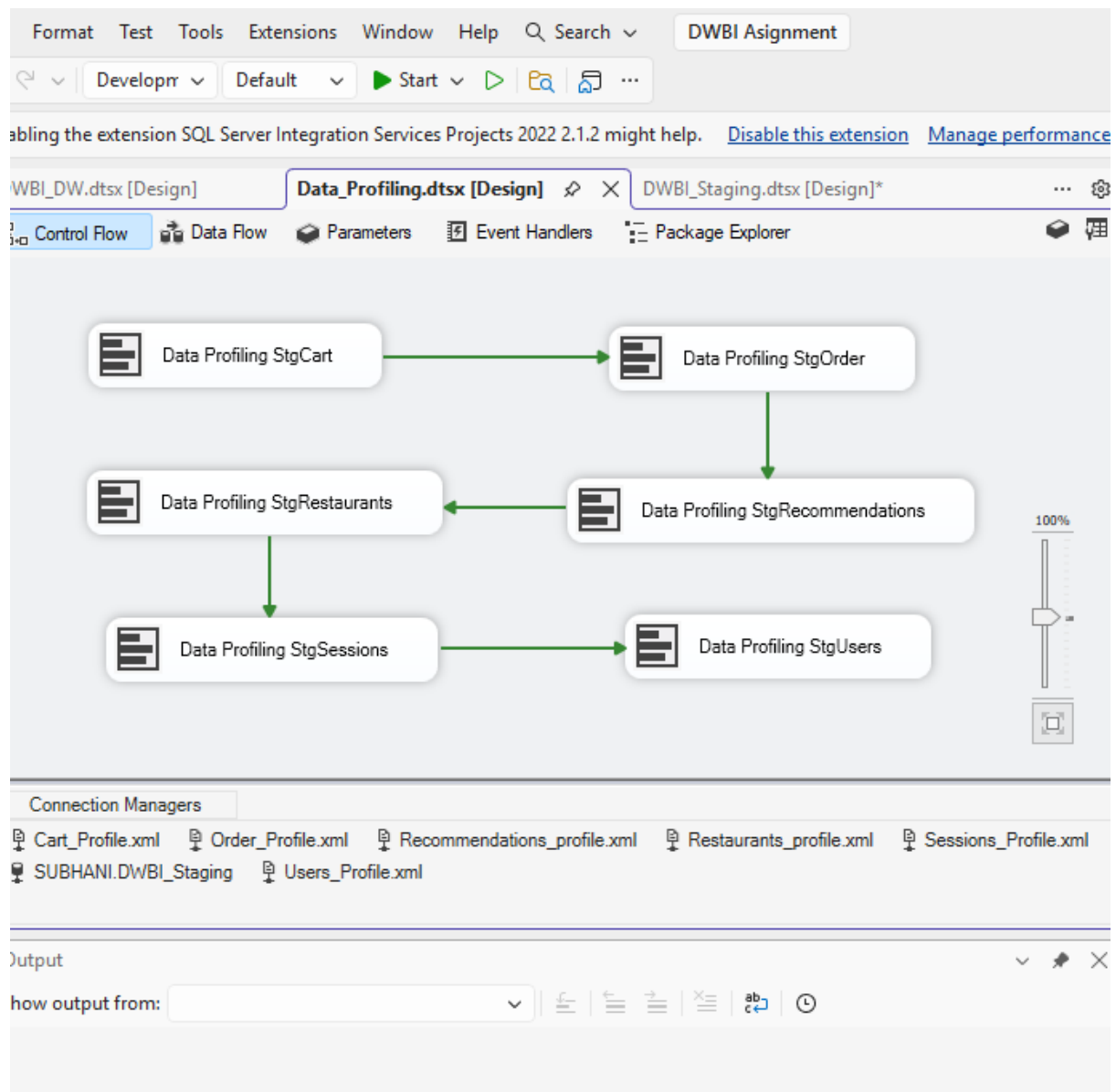


Extract Users data from Users.csv and load it into Users Stgaing Table in DWBI_Staging Database. Use execute task to truncate StgUsers





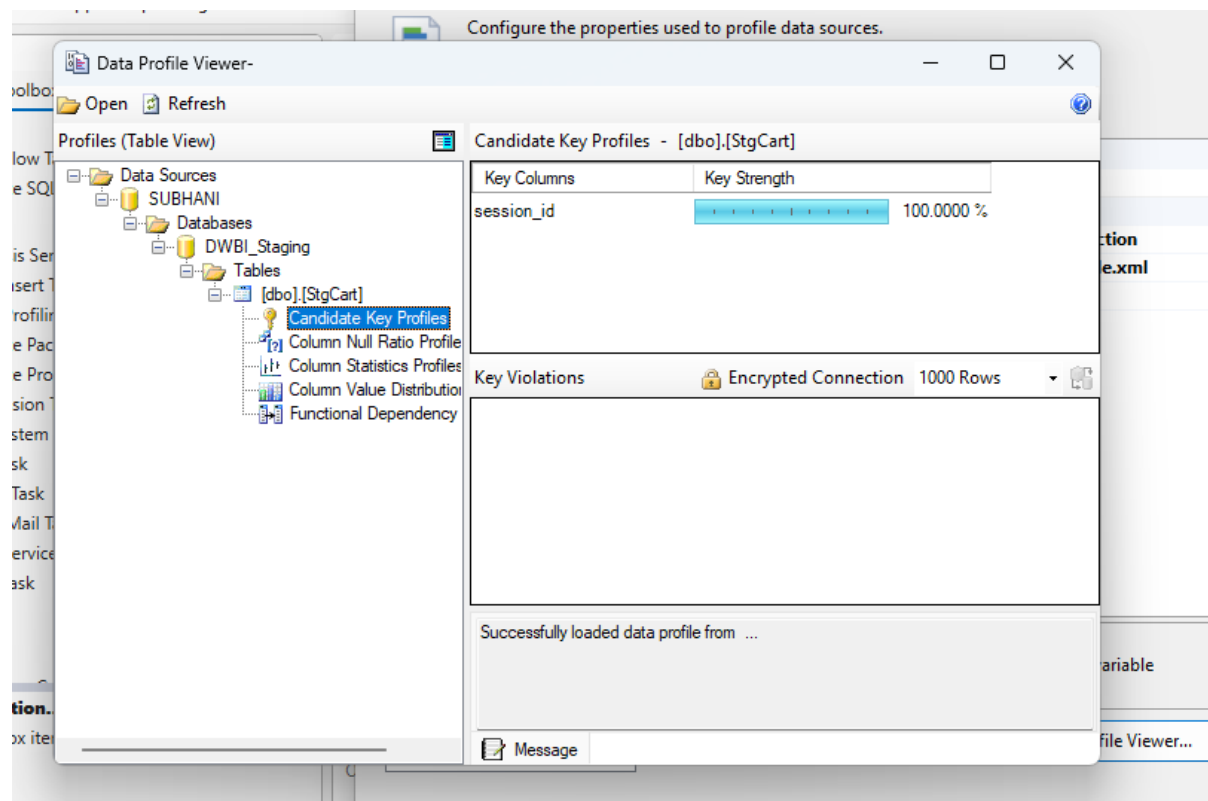
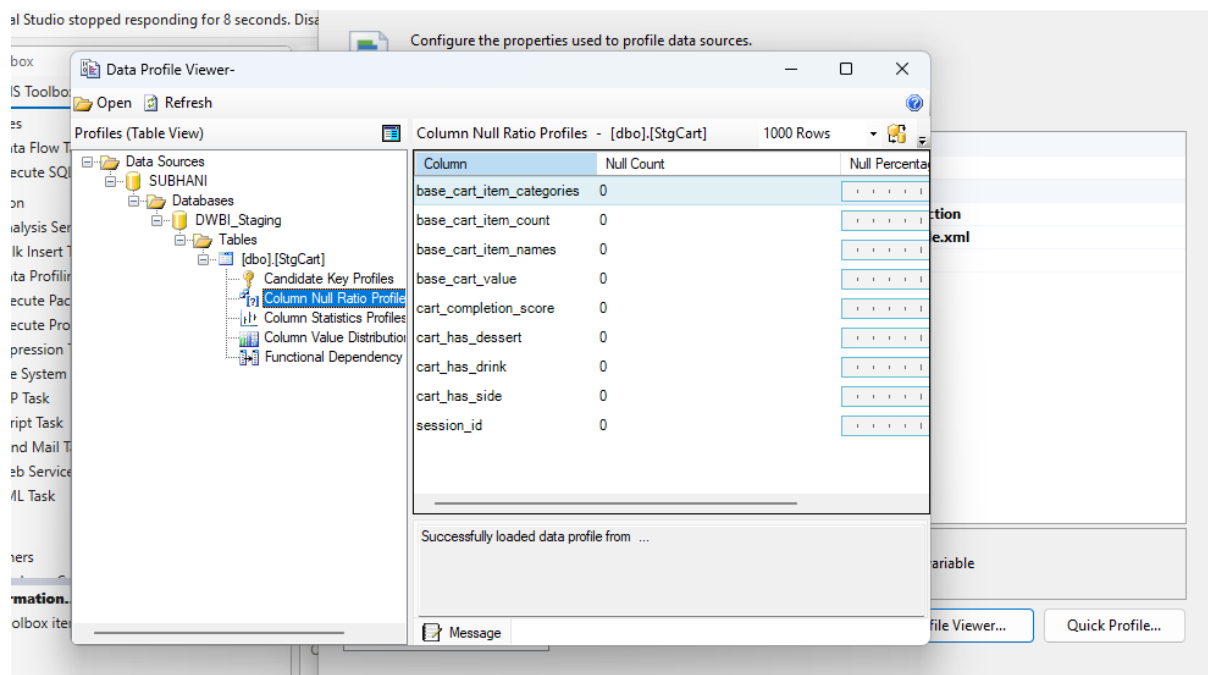
Data Profiling

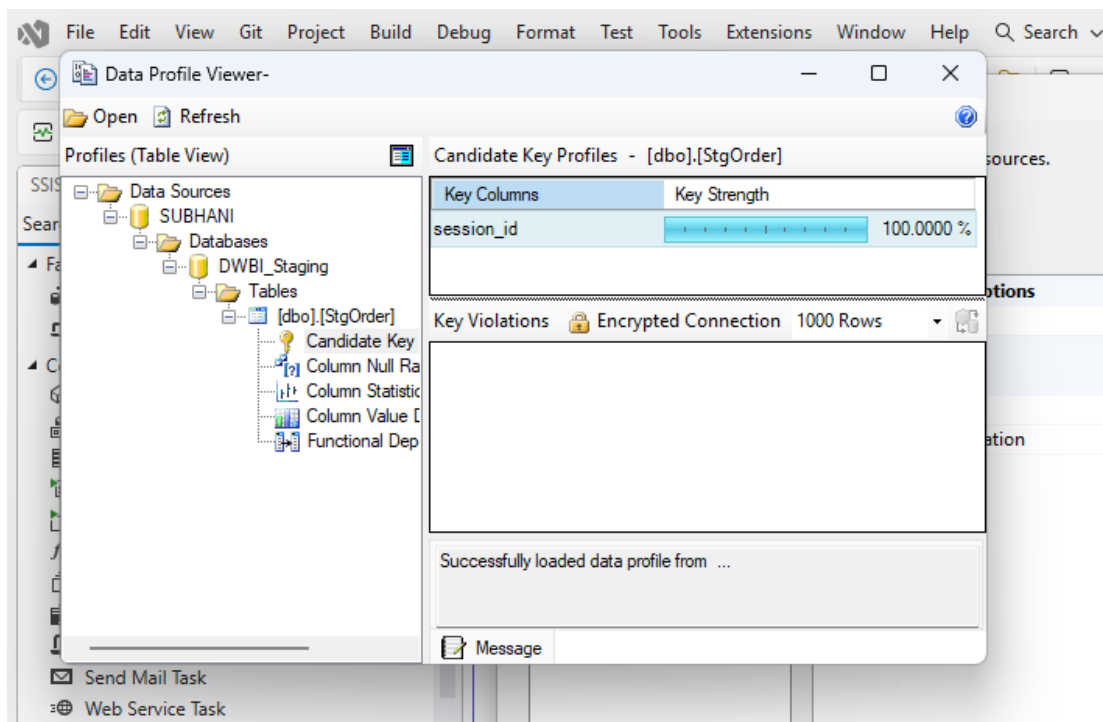
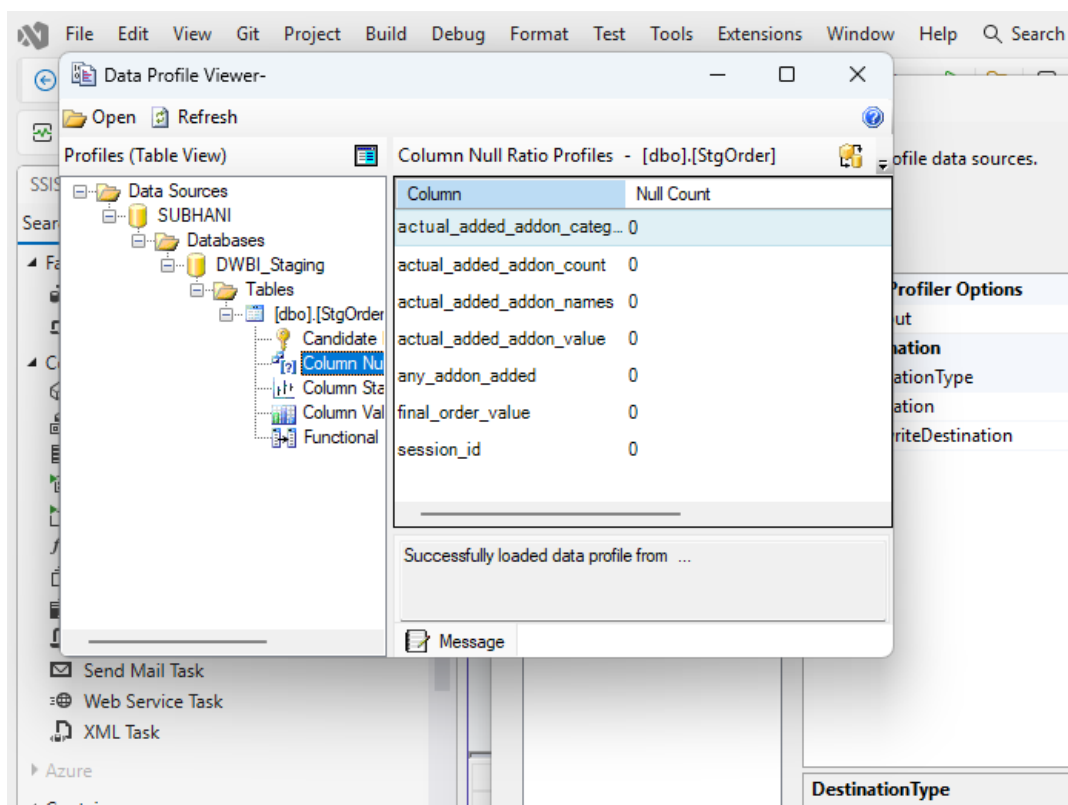


Staging table data was used to analyzed and determined what types of transformation was needed to perform on the data.

- Right click on SSIS Packages and New SSIS Package selected.
- In the Control Flow of Data_Profiling.dtsx, drag and drop the Data Profiling Task and double click to open the configuration.
- Clicked on Quick Profile button to open Single Table Quick Profile Form.
- Clicked on New button and create the connection to DWBI_Staging

- Once the green tick appeared, double click the Data Profiling Task and Click on the Open Profile Viewer to view the analyzed data





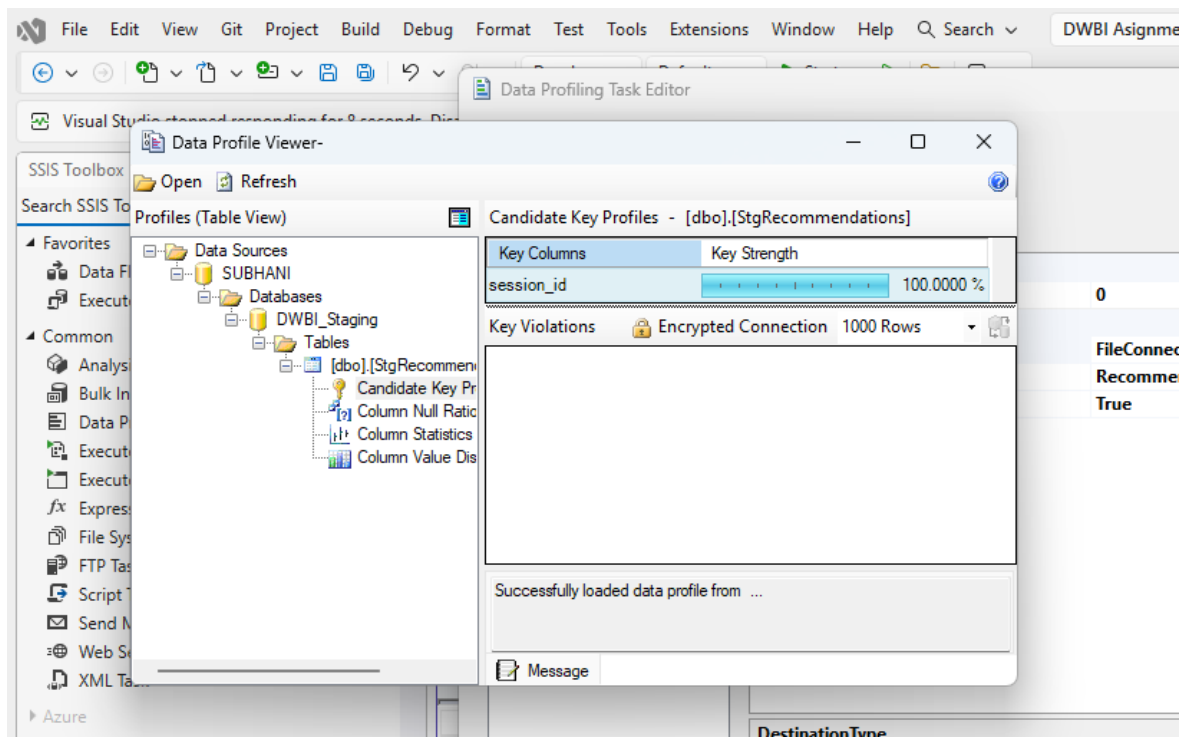
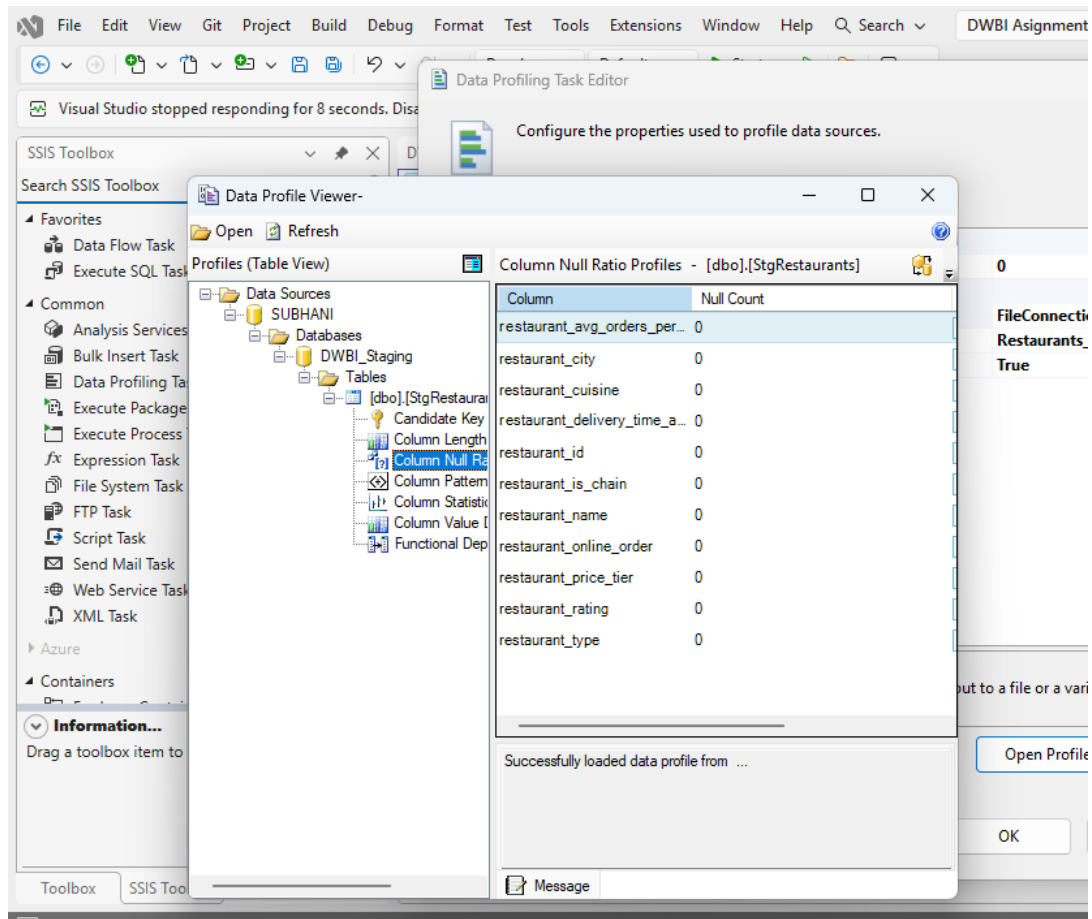
Visual Studio Code interface showing the Data Profile Viewer window.

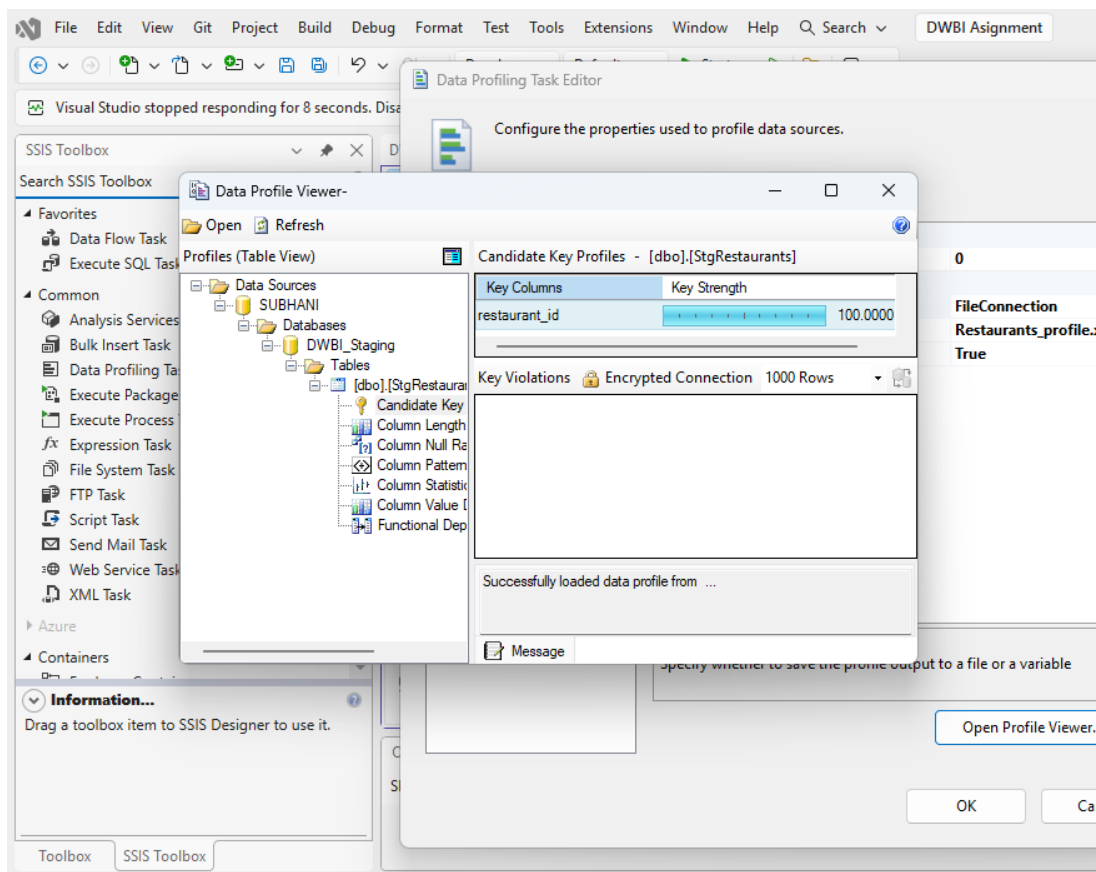
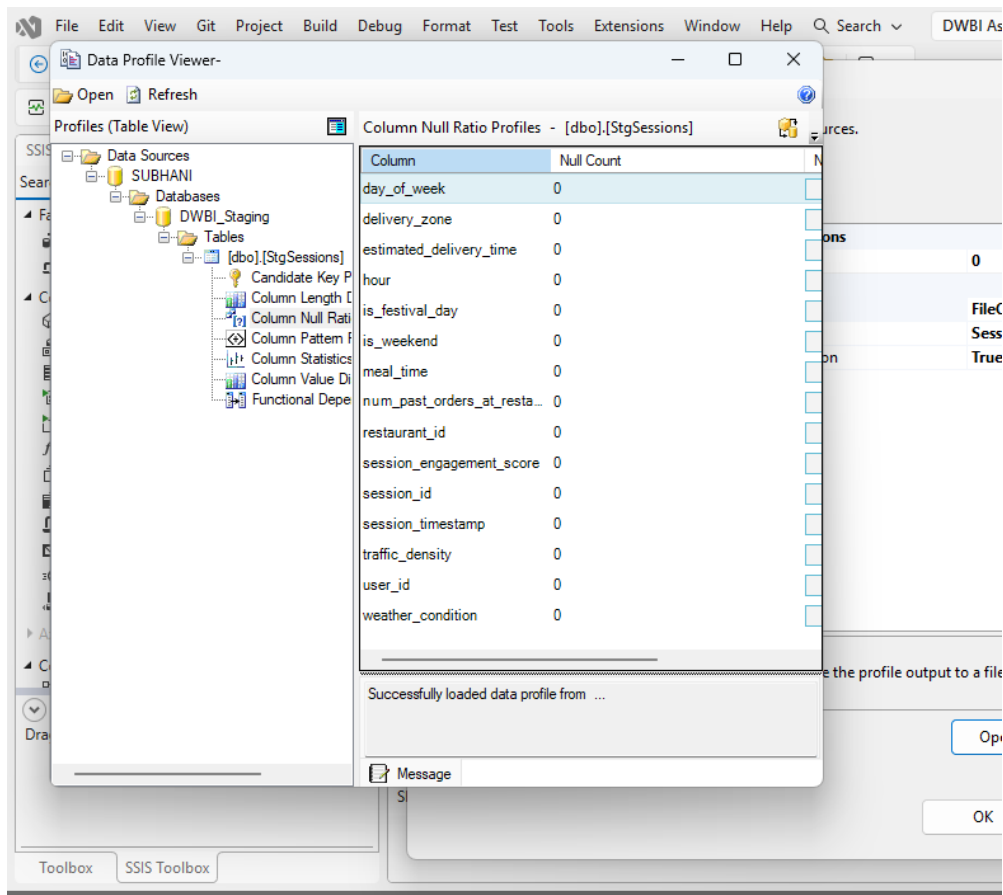
The Data Profile Viewer window displays the following table:

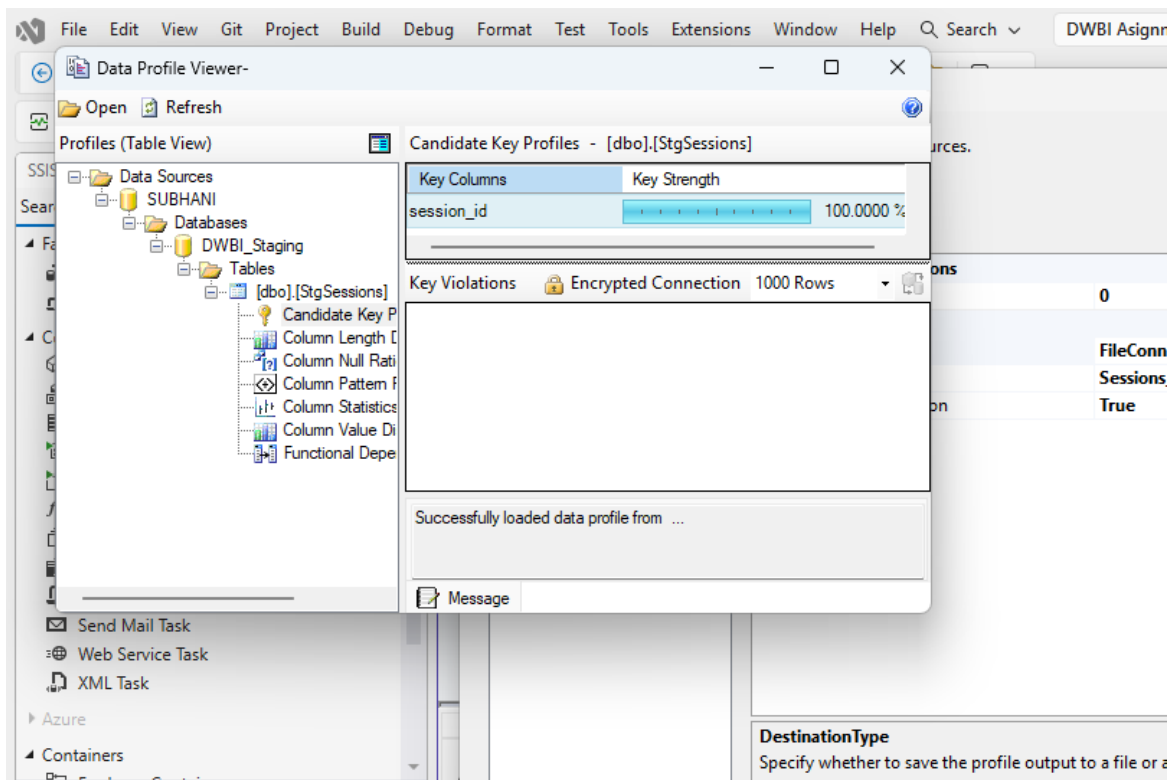
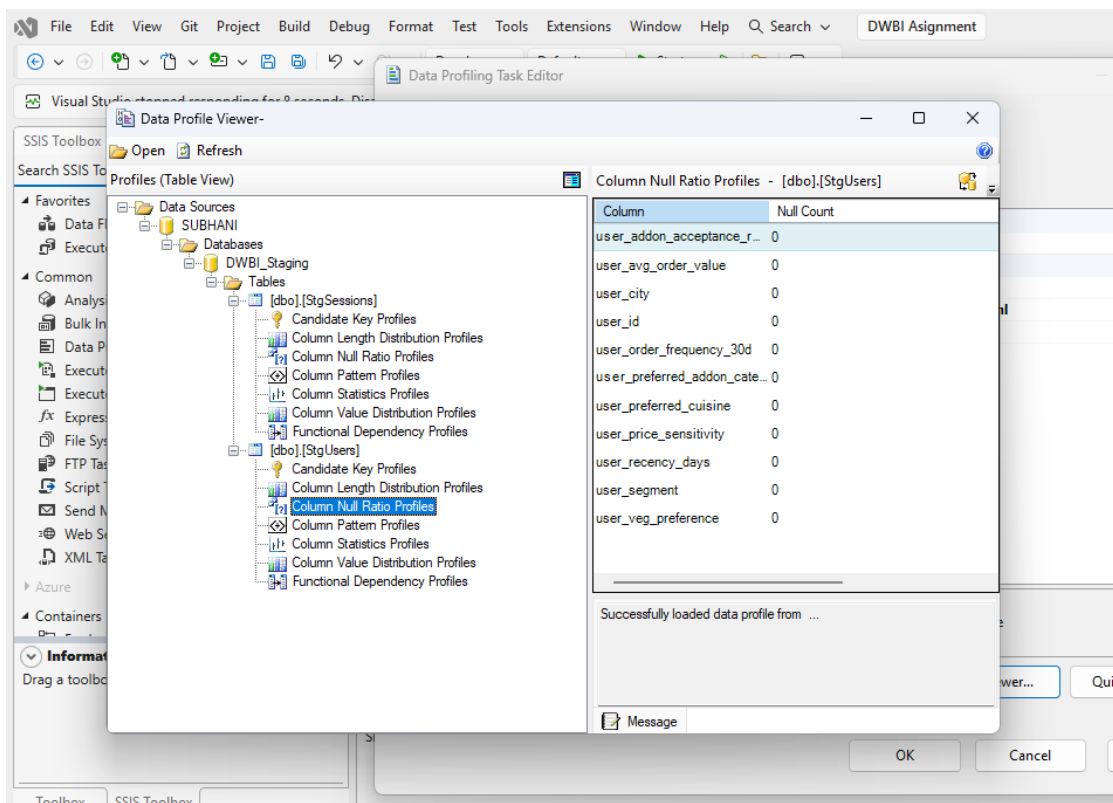
Column	Null Count
avg_reco_is_complementary	0
avg_reco_is_complementary	0
avg_reco_popularity	0
avg_reco_popularity	0
avg_reco_price_ratio	0
avg_reco_price_ratio	0
avg_reco_score	0
avg_reco_score	0
recommended_addon_cate...	0
recommended_addon_cate...	0
recommended_addon_nam...	0
recommended_addon_nam...	0
recommended_addon_pric...	0
recommended_addon_pric...	0
session_id	0
session_id	0

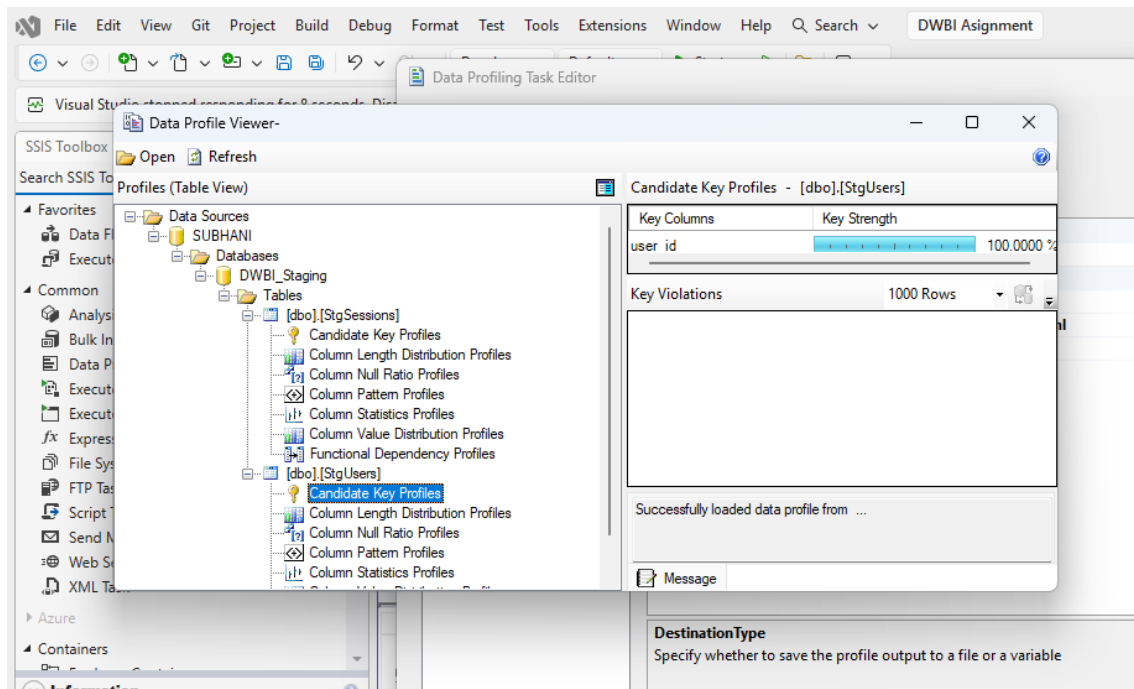
Message: Successfully loaded data profile from ...

Buttons: OK

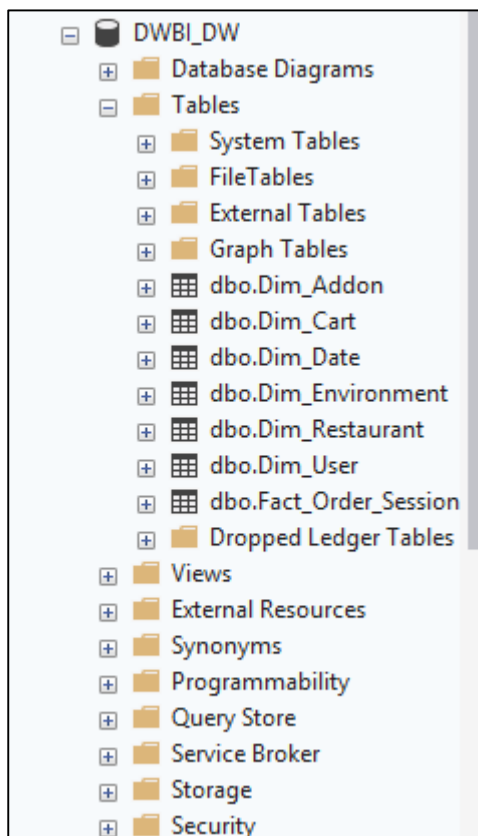


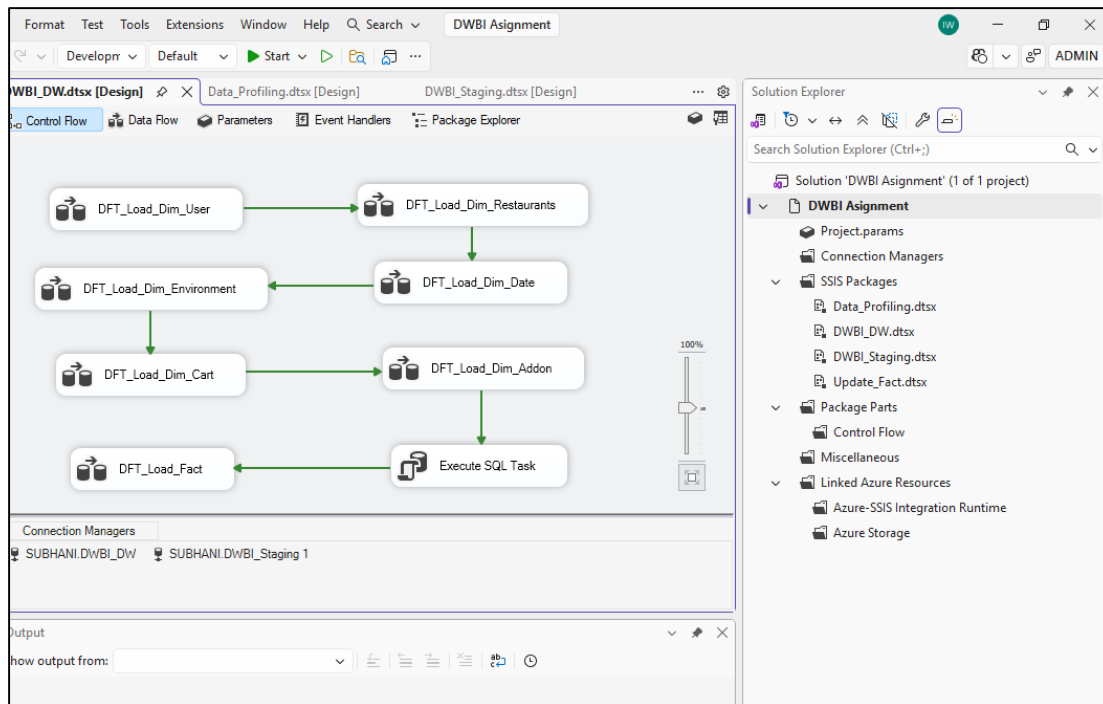






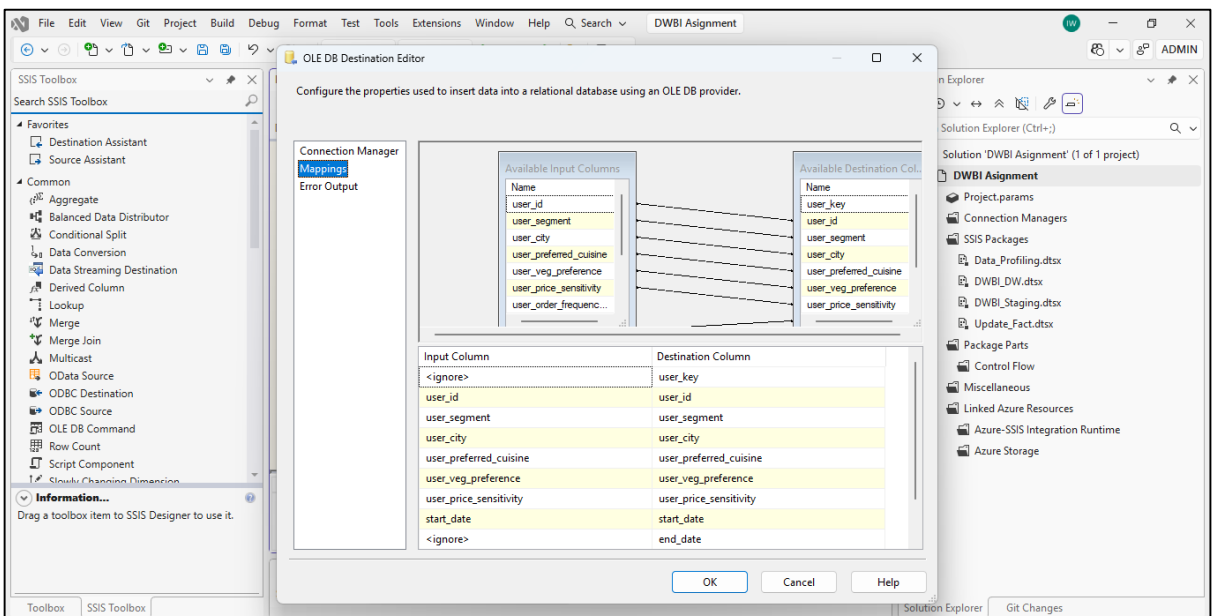
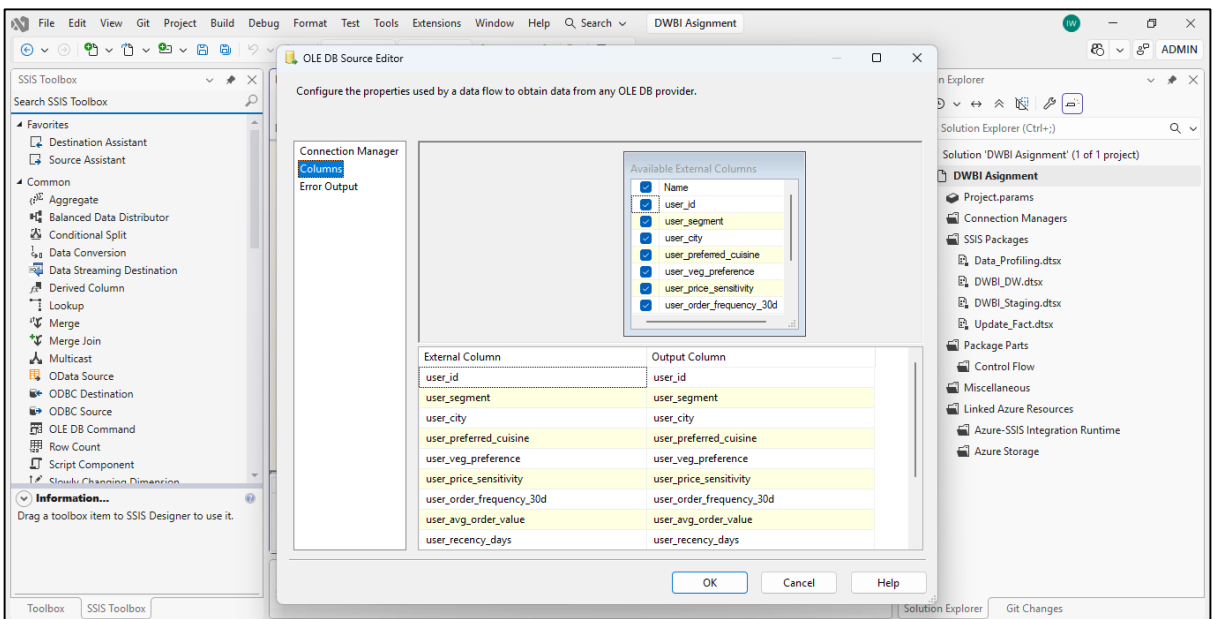
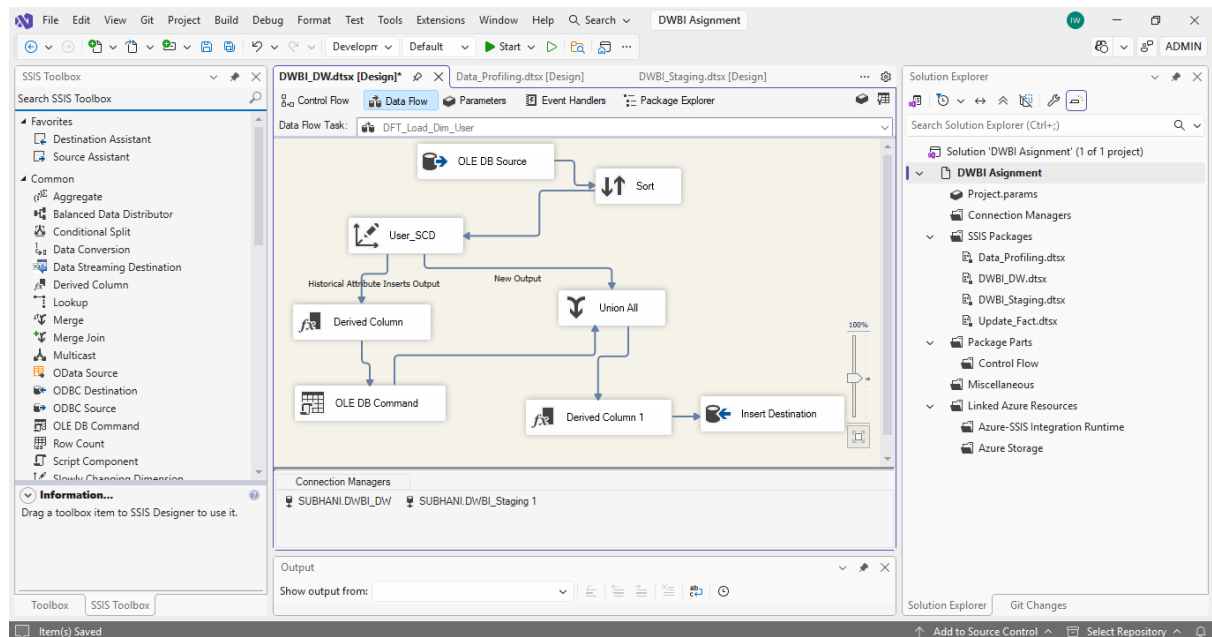
Data Transformation

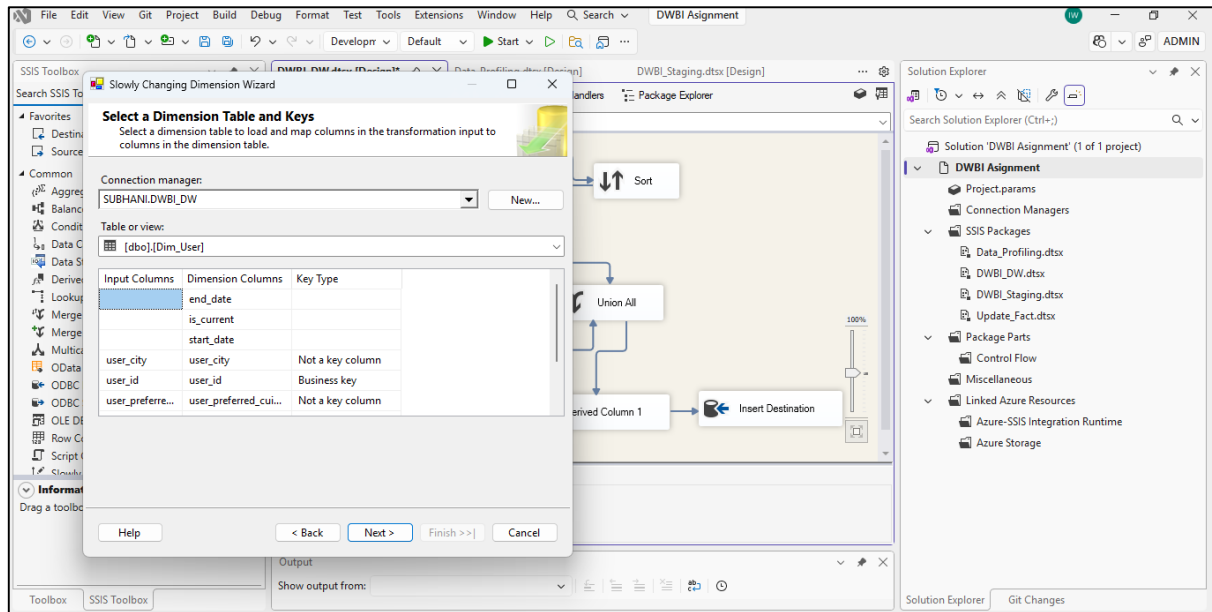




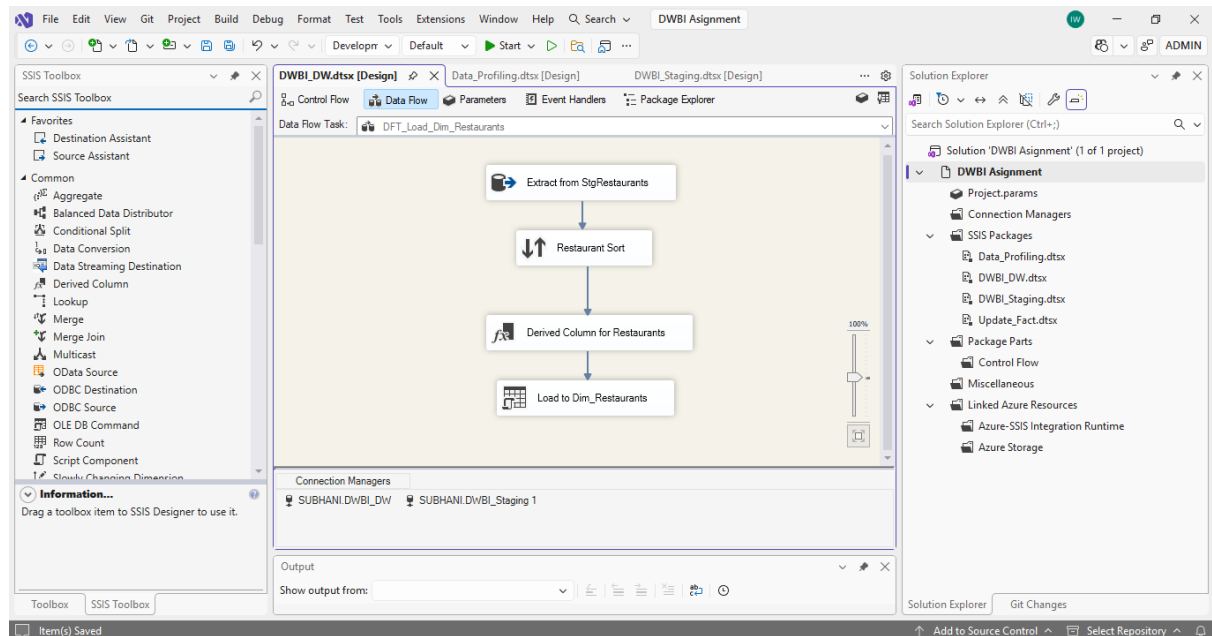
- Finally extracted all the data from tables which are in the DWBI_Staging into DB named DWBI_DW database as shown in the below using SQL Server Integration Service Software.
- OLE DB Command SSIS tool used to execute procedures, it used to insert data from Population details which is in DWBI_Staging to tables in DWBI_DW. Stored Procedure ensure no duplicates are entered into the data warehouse table. If there is an existing Population record, it will be updated with the latest record coming from staging table. Else, if it is a new record, it will be inserted into Dimensional Tables.
- Used sort to sort details of data which are coming from staging tables to dimensional table using primary keys.
- Fact_Order_Sessions - The fact table stores measurable data related to business processes and is linked to multiple dimension tables through foreign keys. In this project, the fact table captures session/order-level data such as transaction values and timestamps. It is designed as an **accumulating fact table**, where records are initially inserted with a creation time and later updated with completion time and processing duration as the transaction progresses. This structure enables tracking of the entire lifecycle of each transaction for analytical purposes.

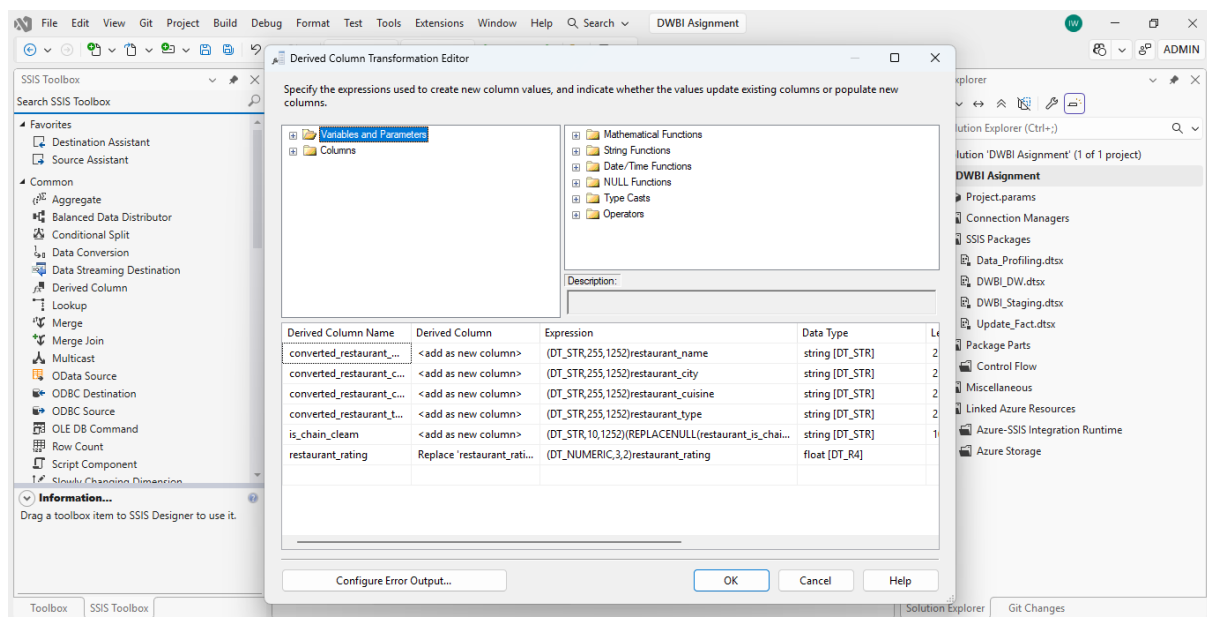
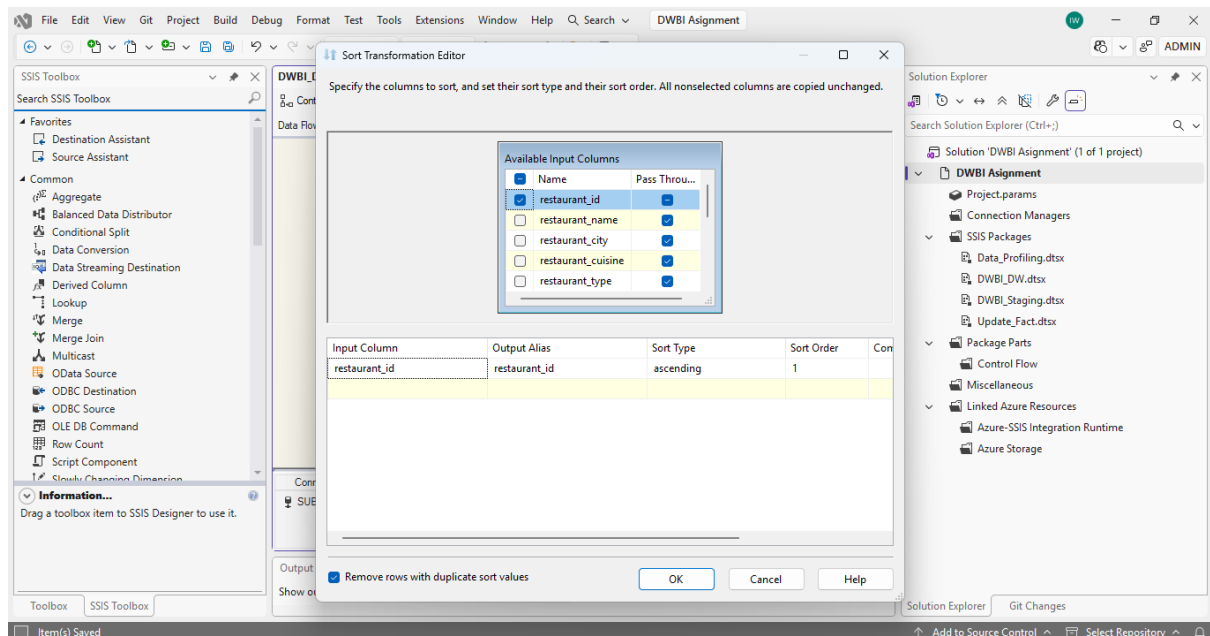
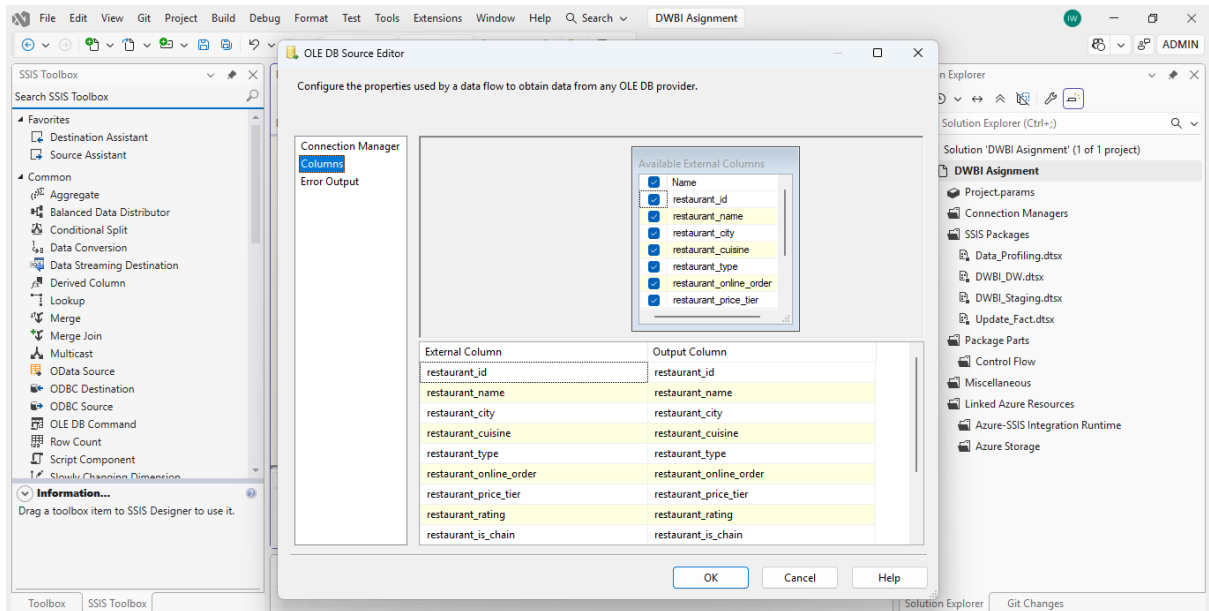
Dim_User

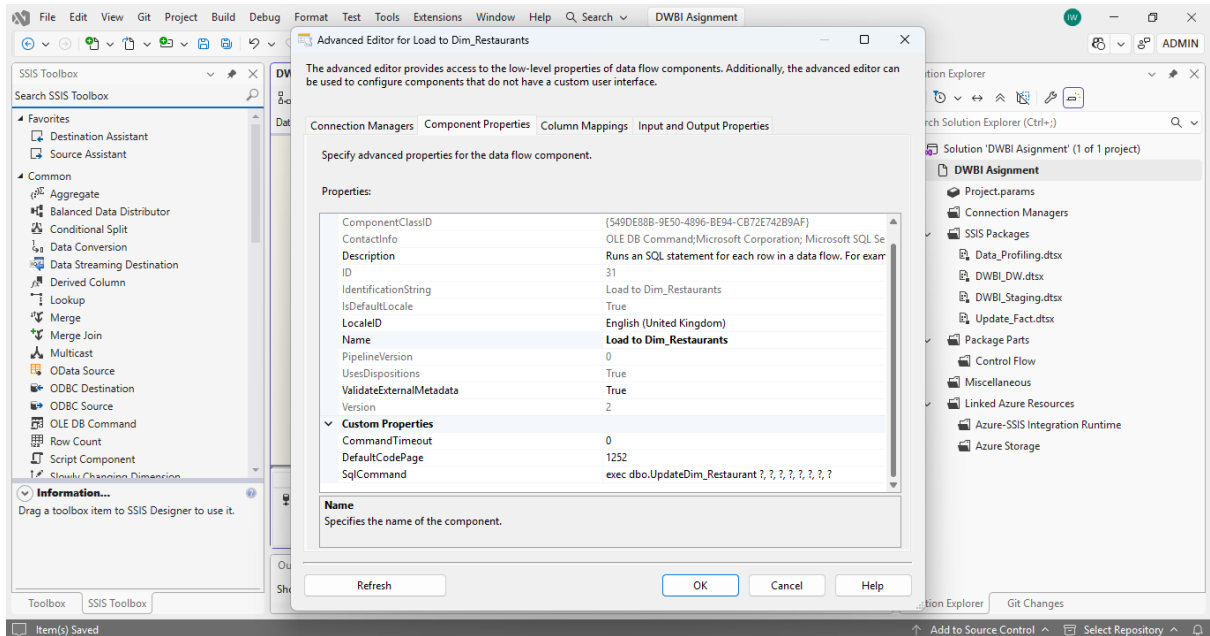




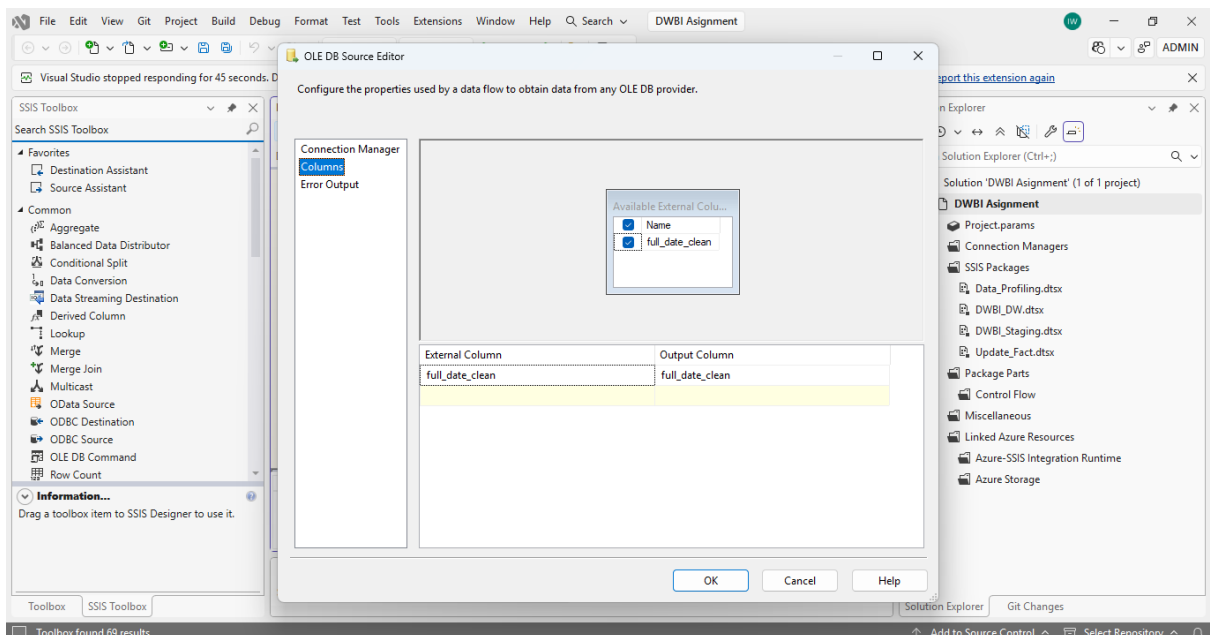
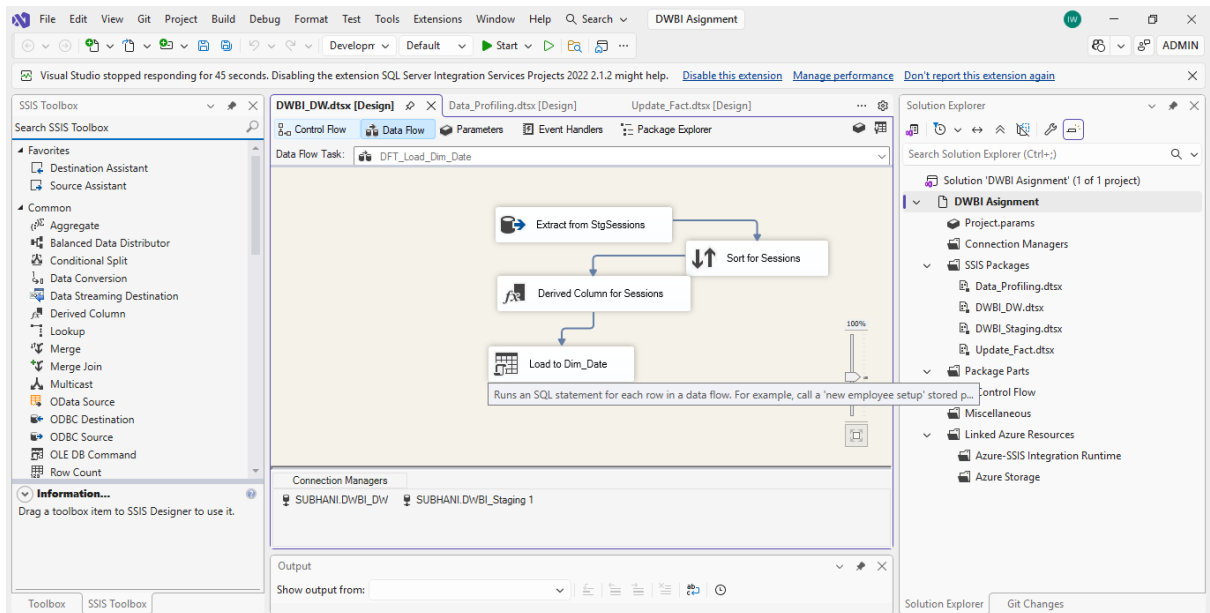
Dim_Restaurant

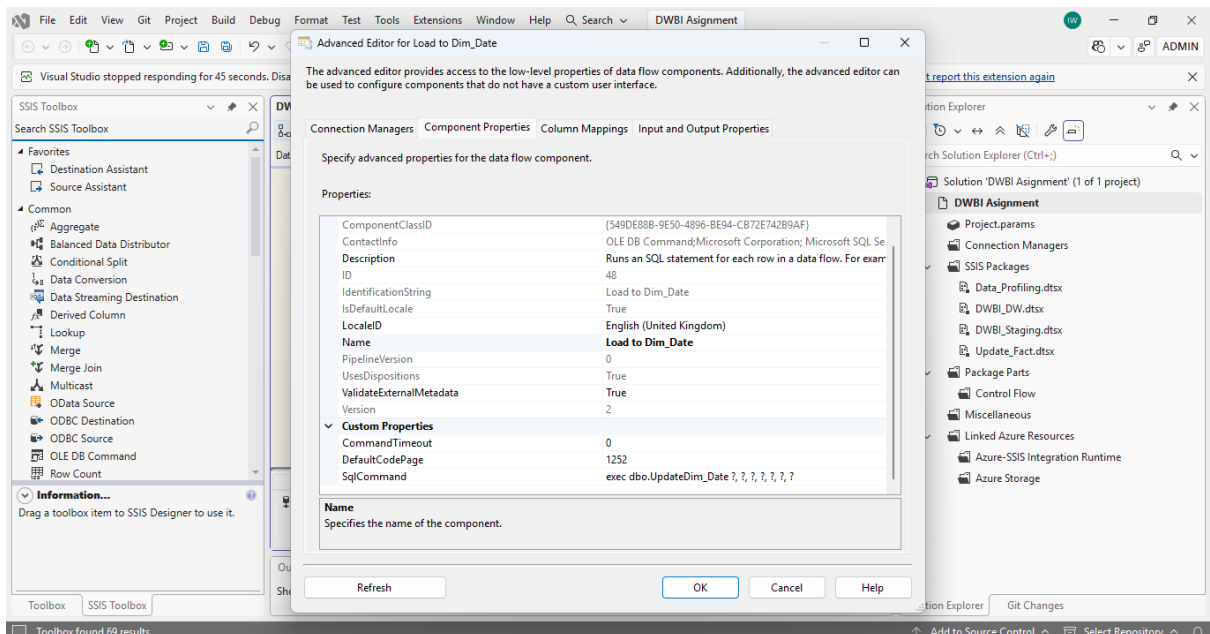
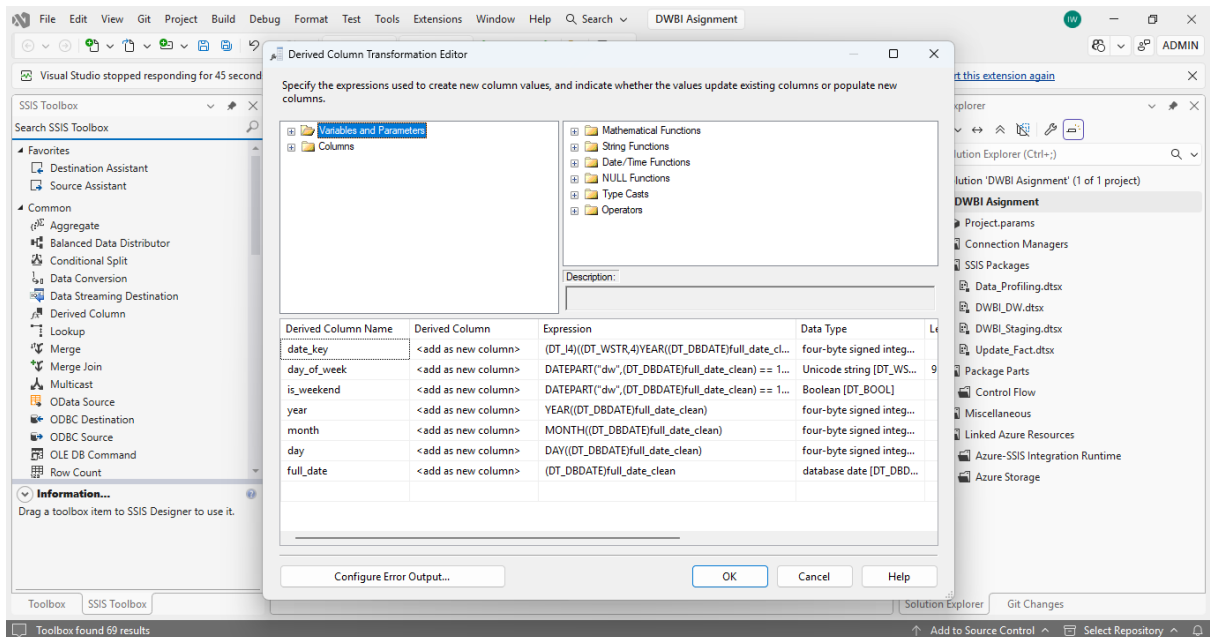
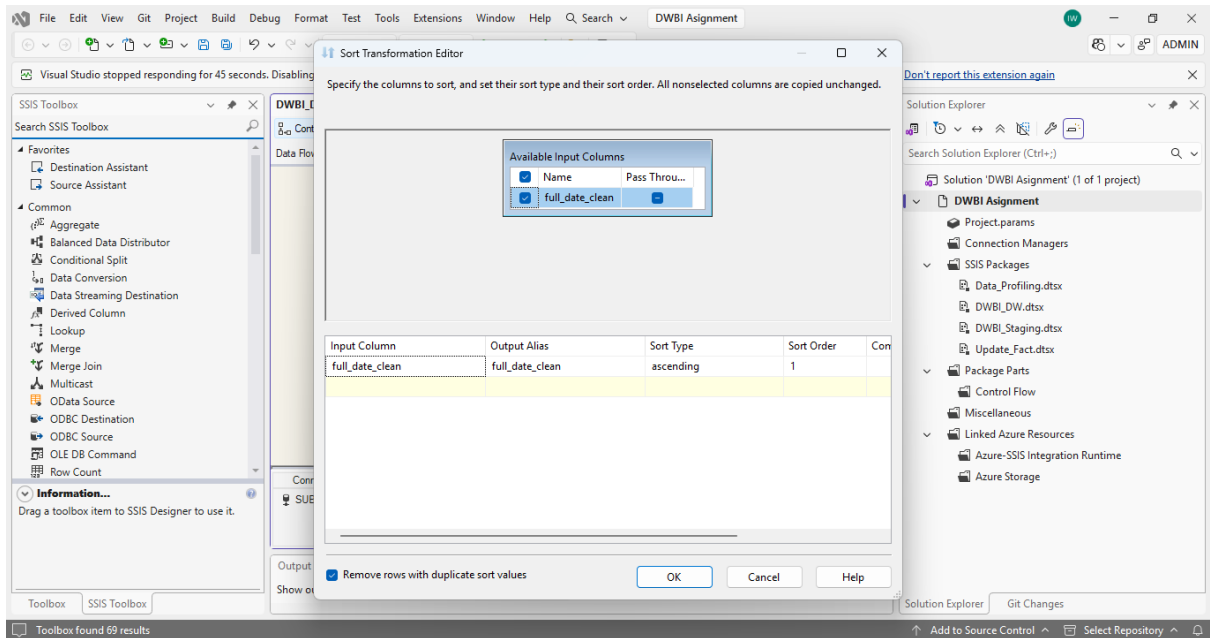




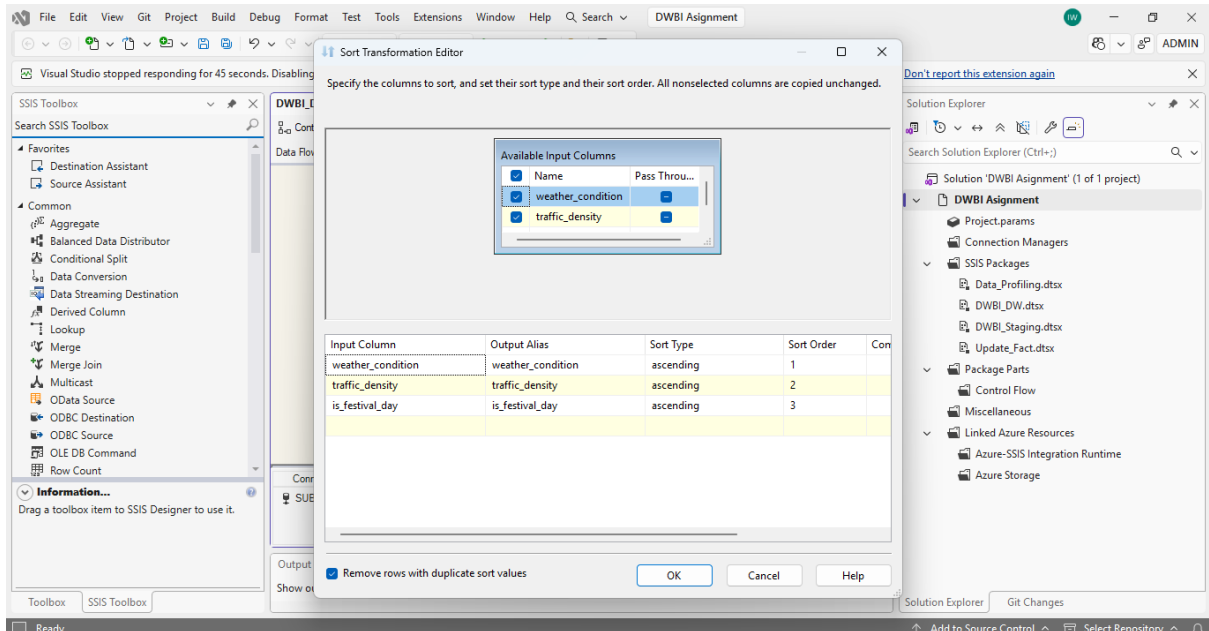
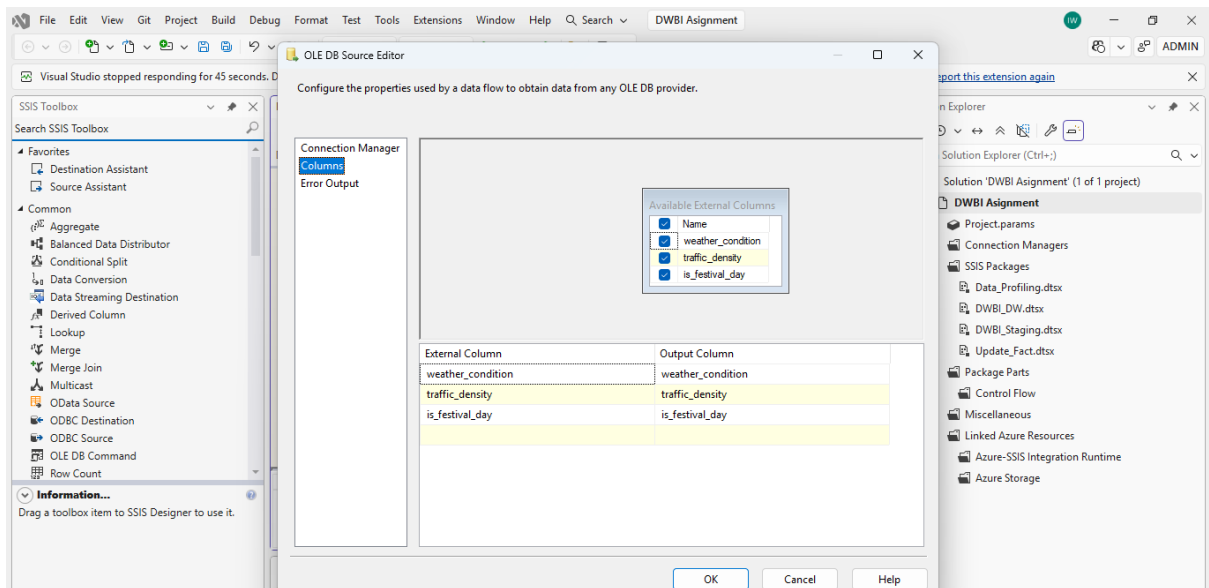
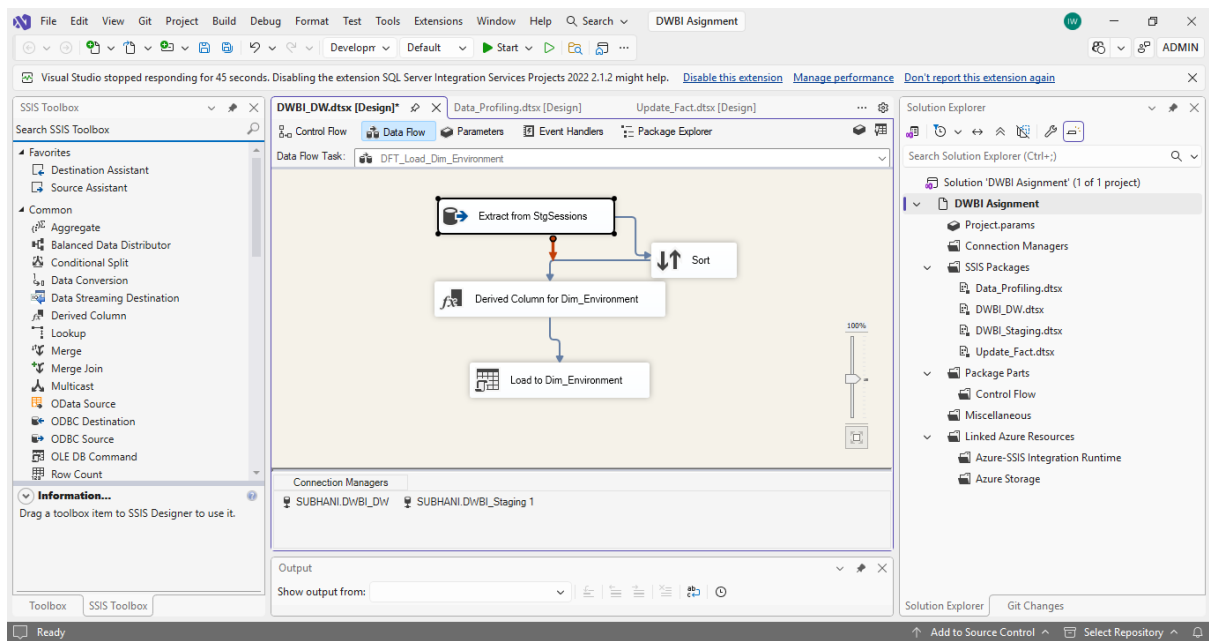


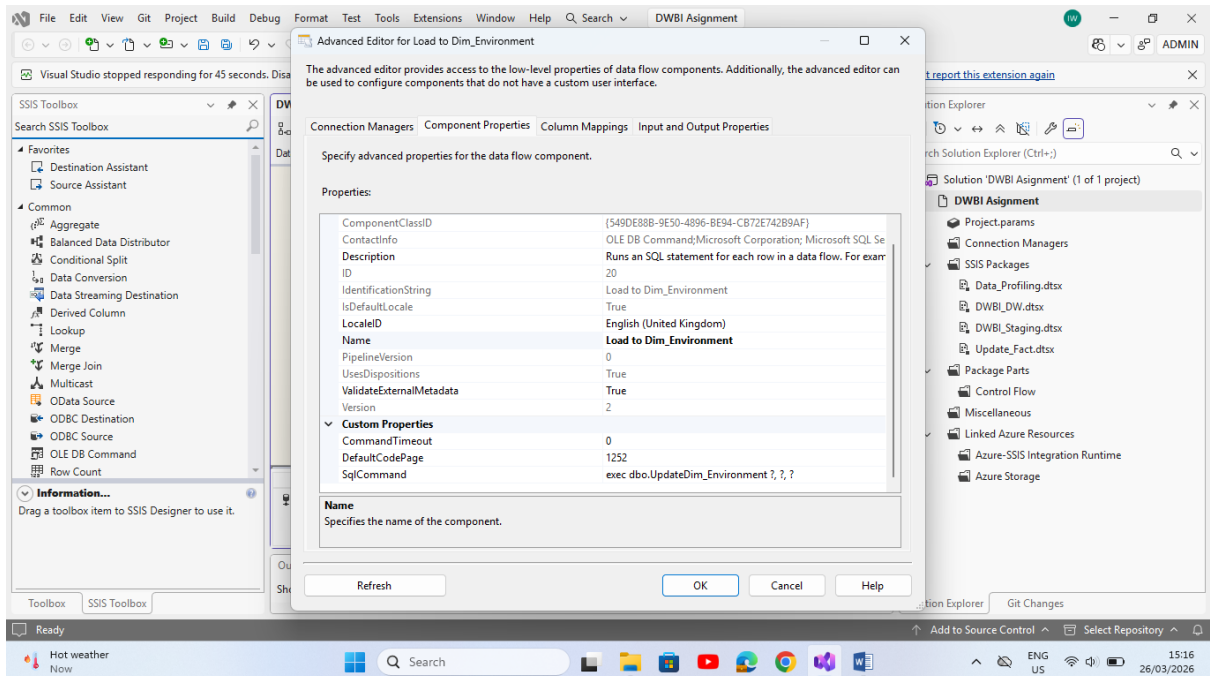
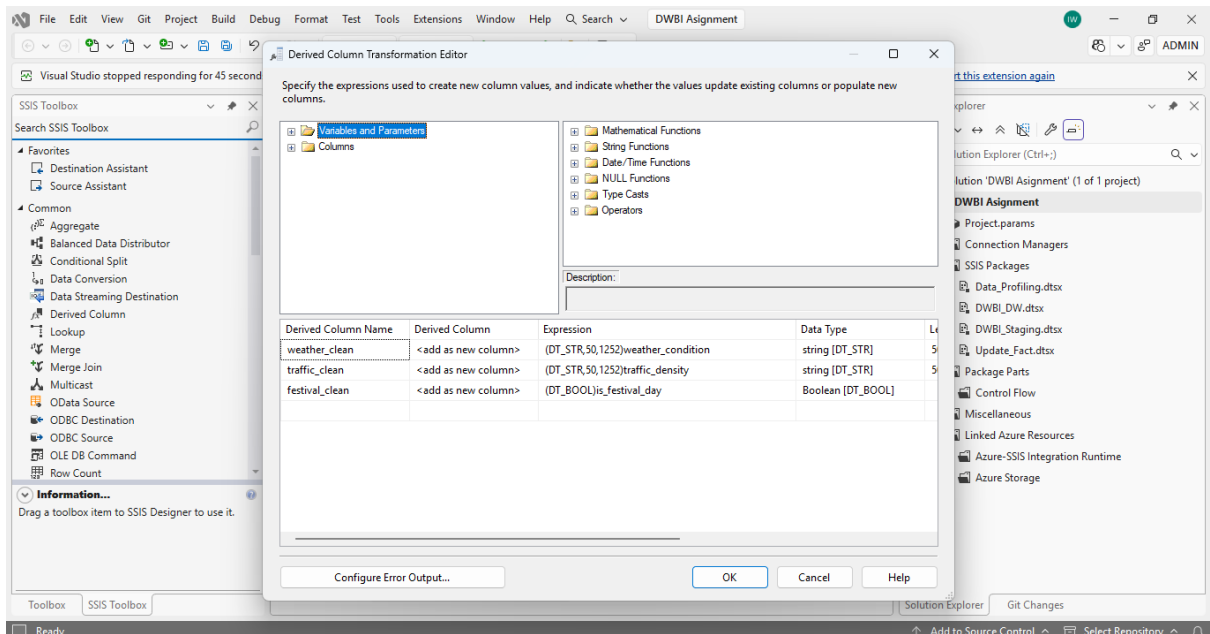
Dim_Date



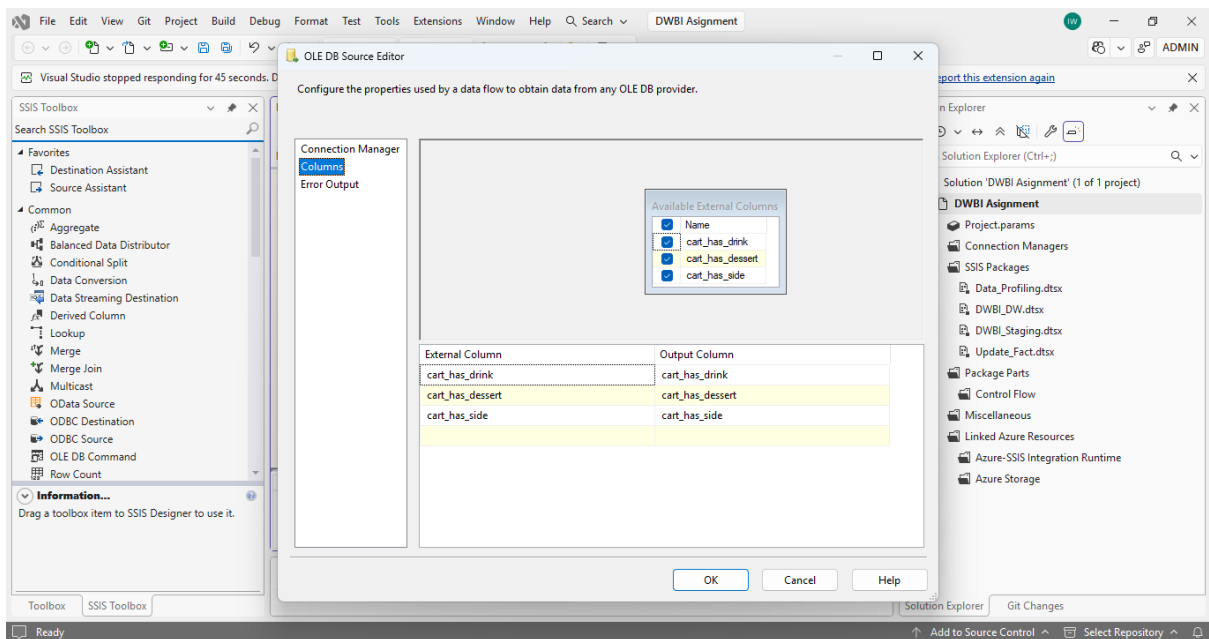
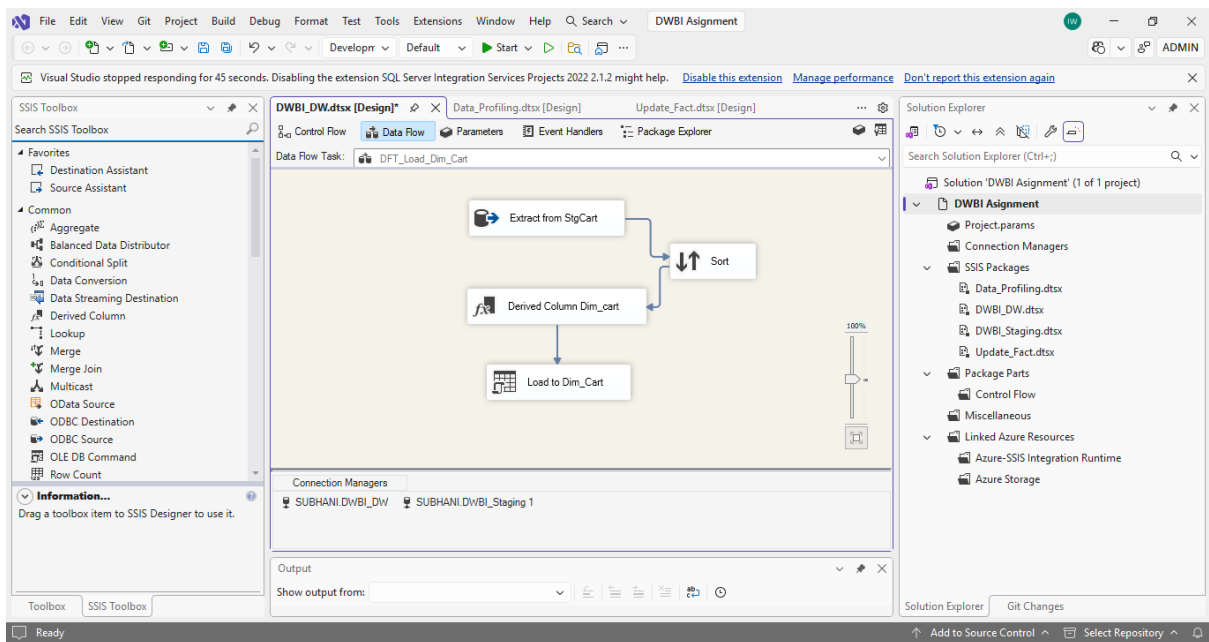


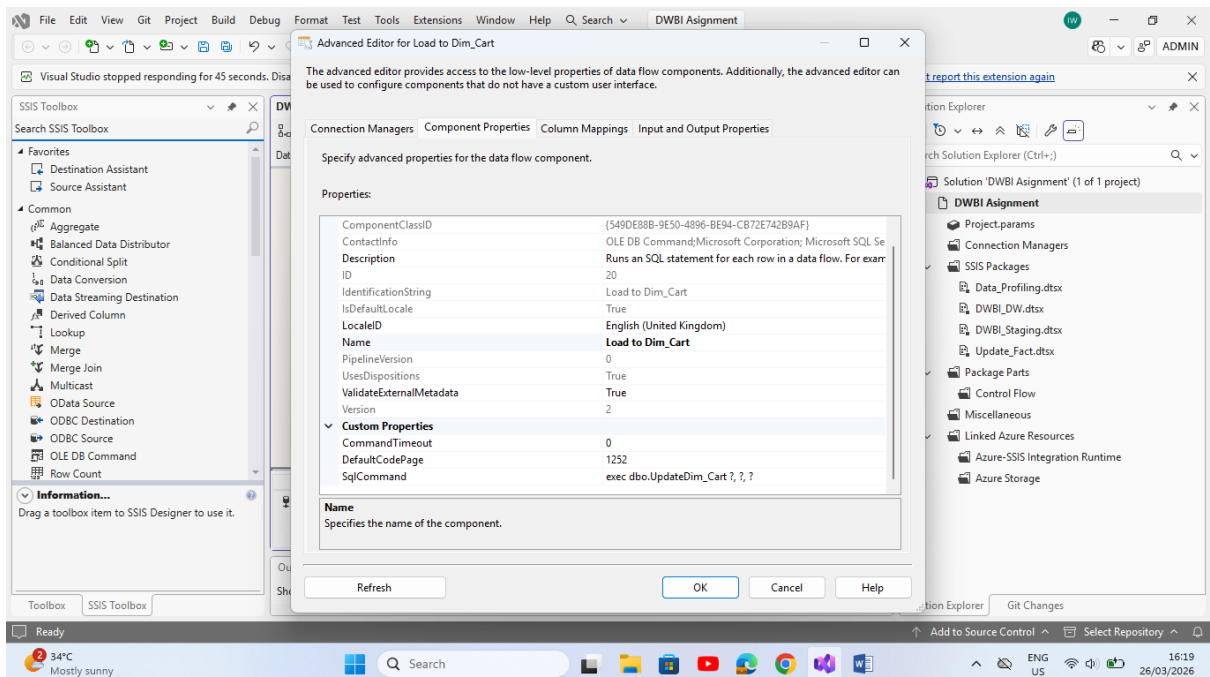
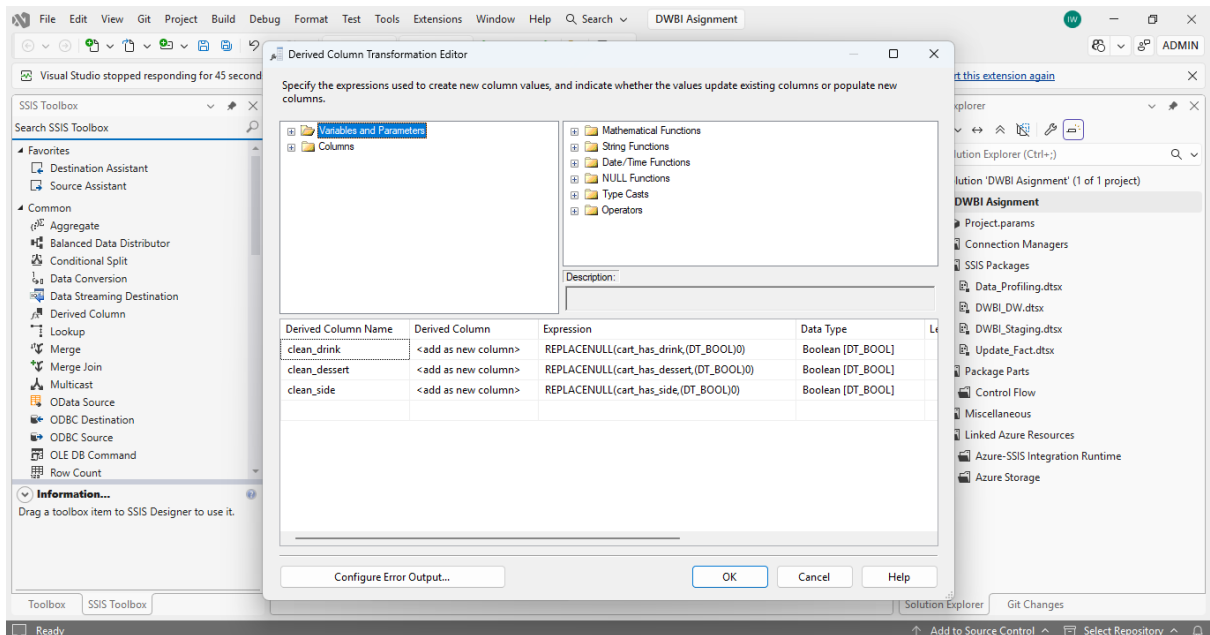
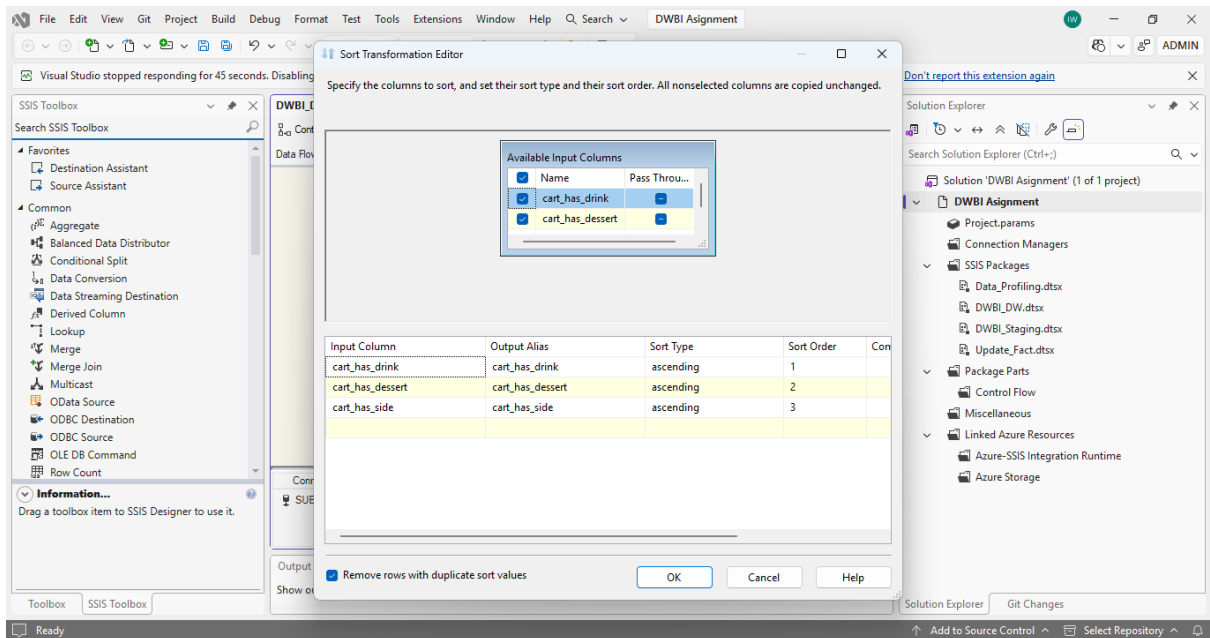
Dim_Environment



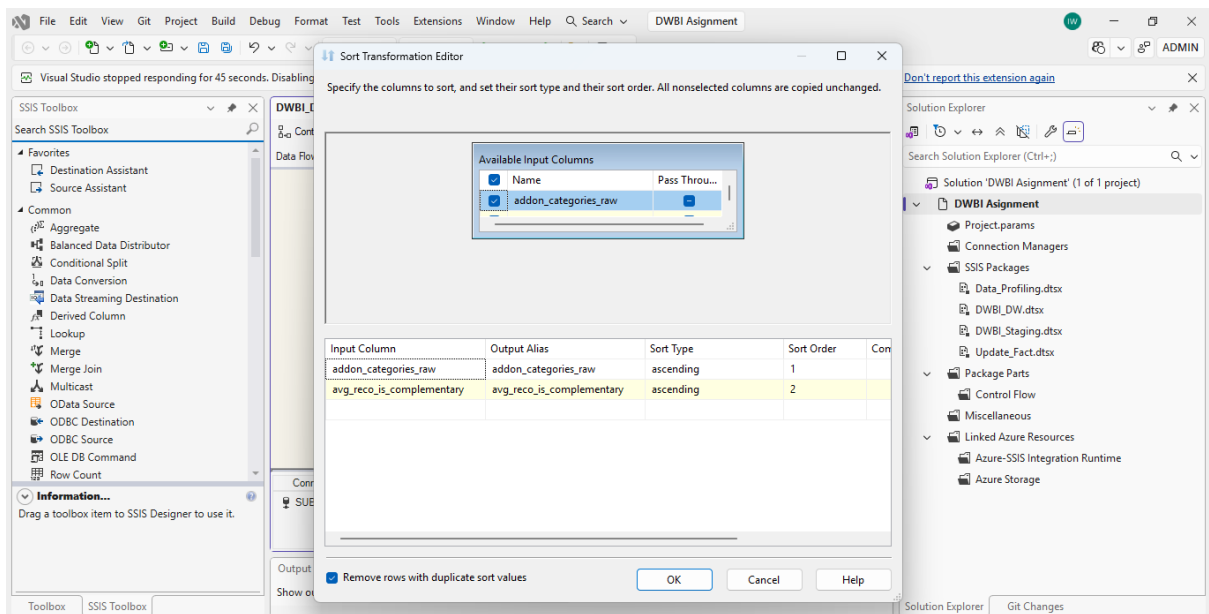
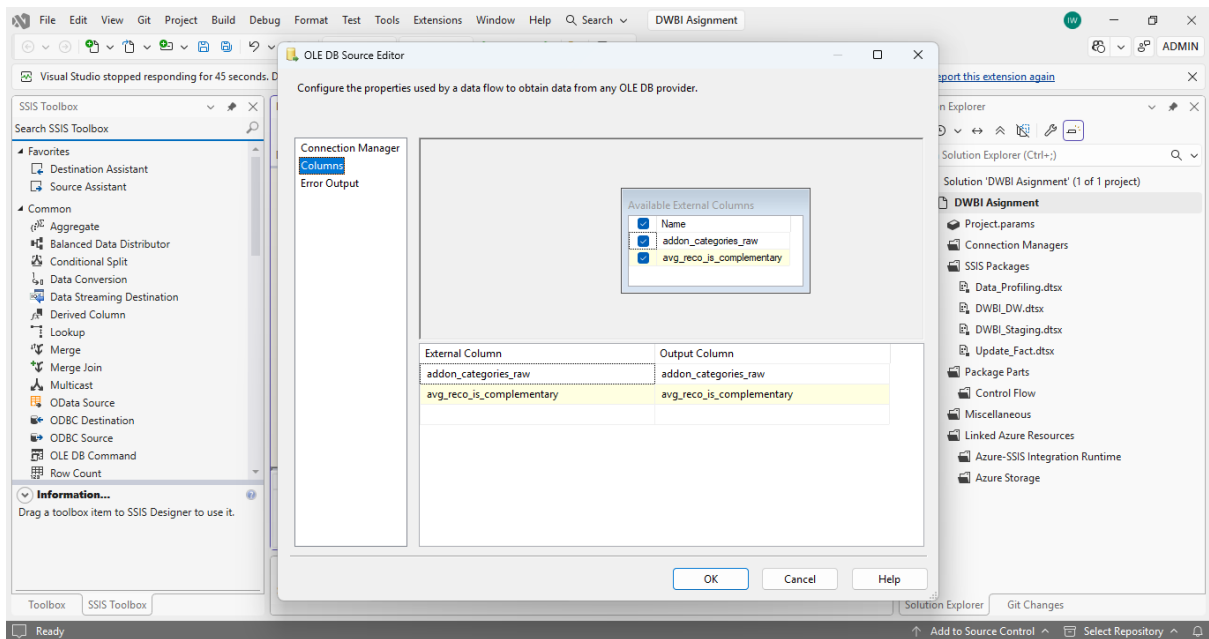
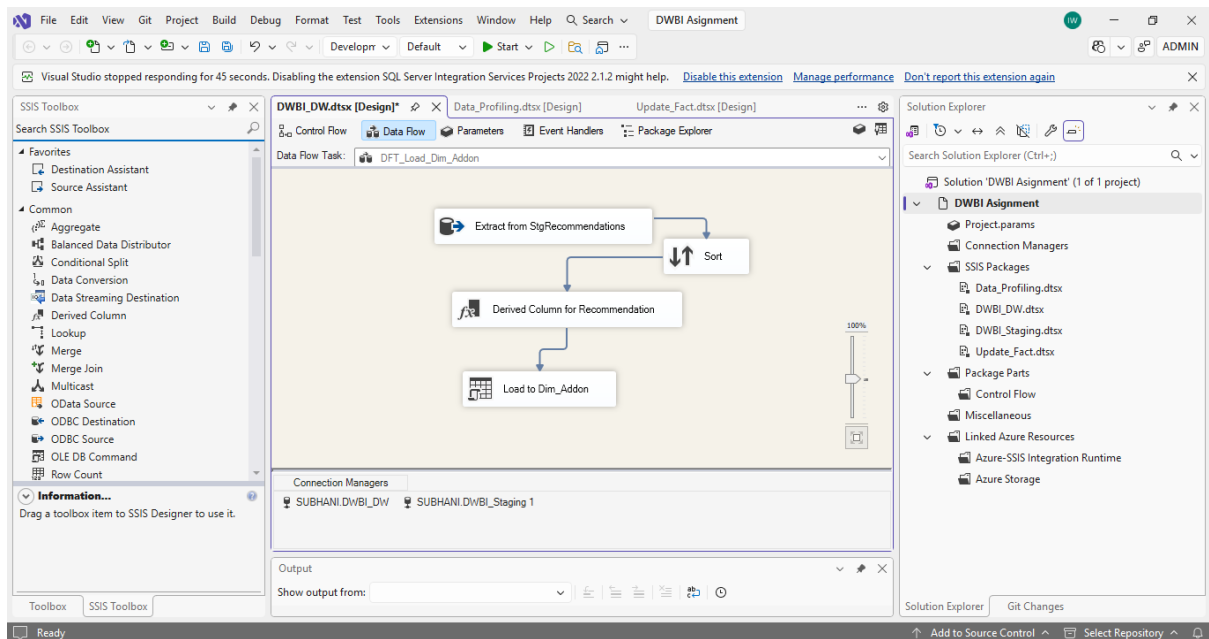


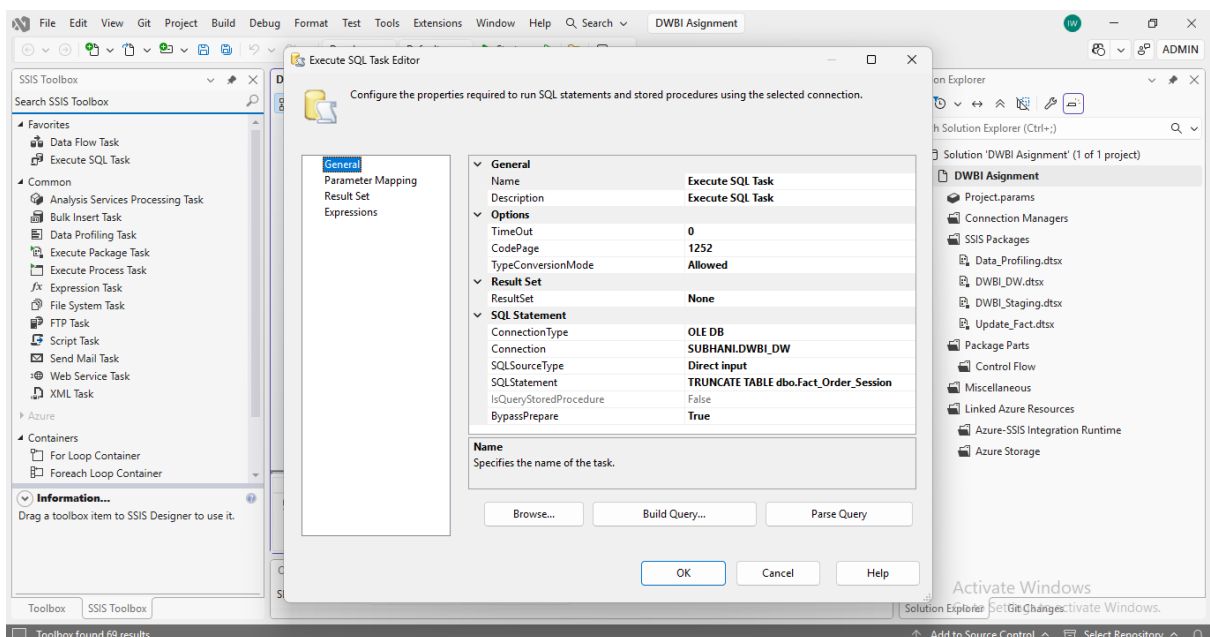
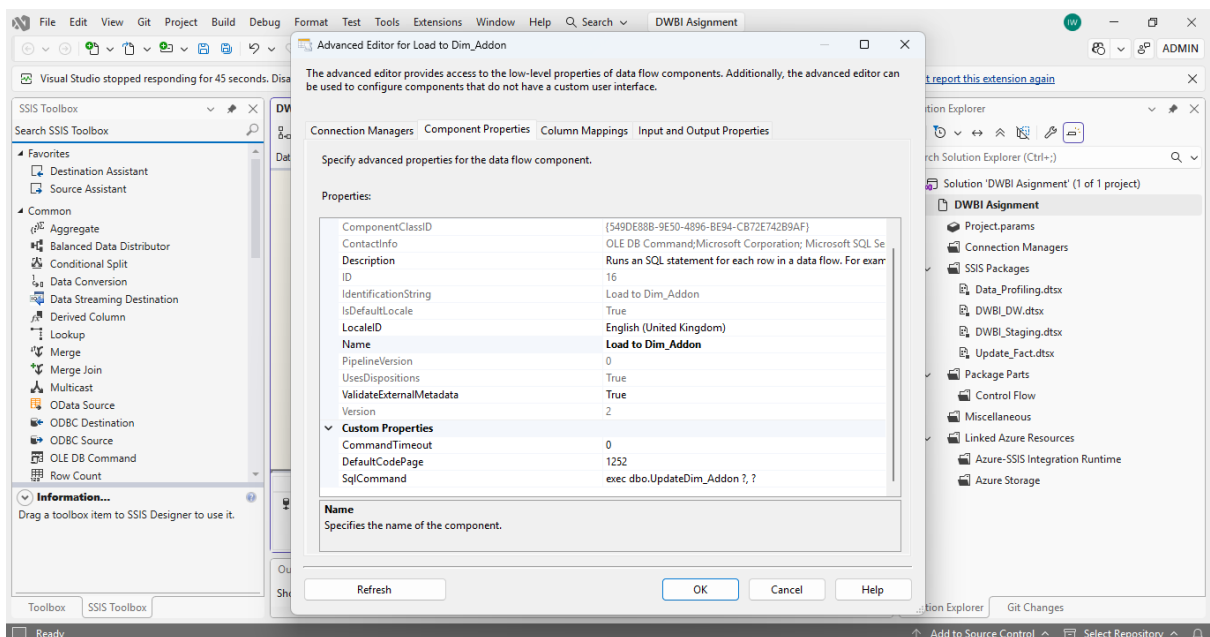
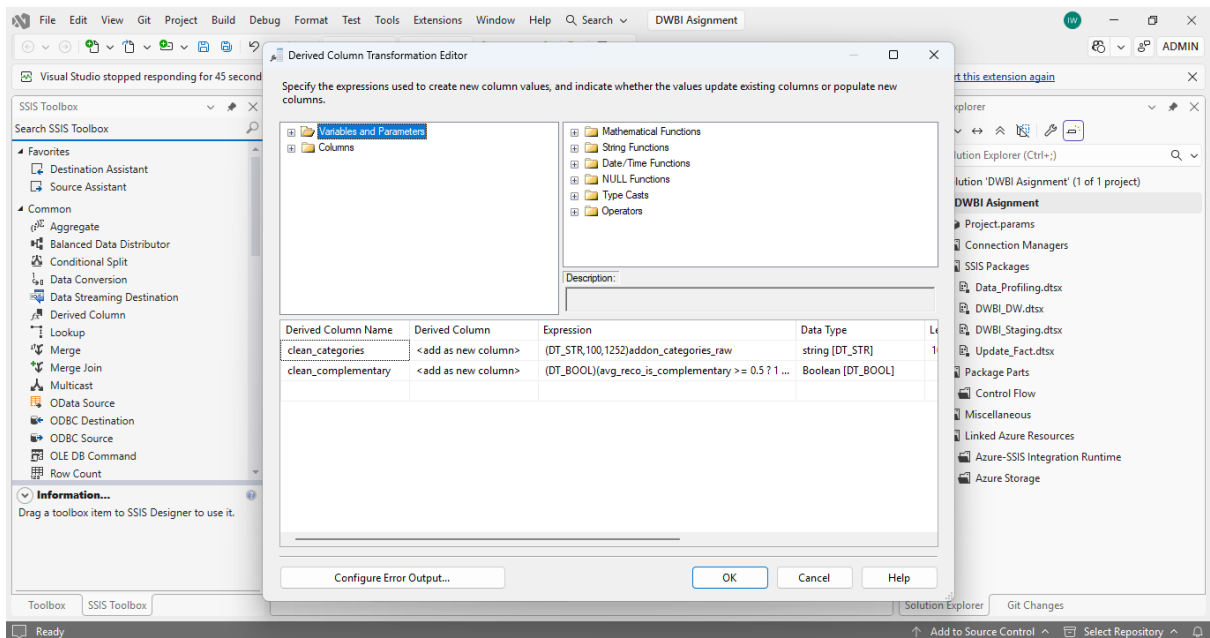
Dim_Cart





Dim_Addon

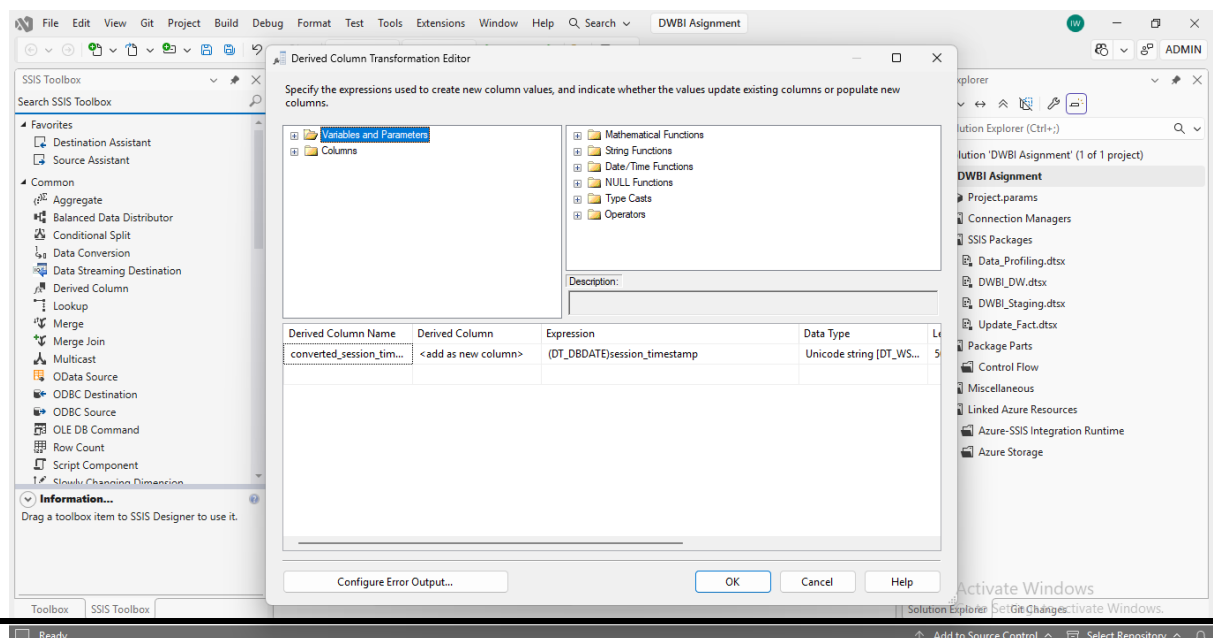
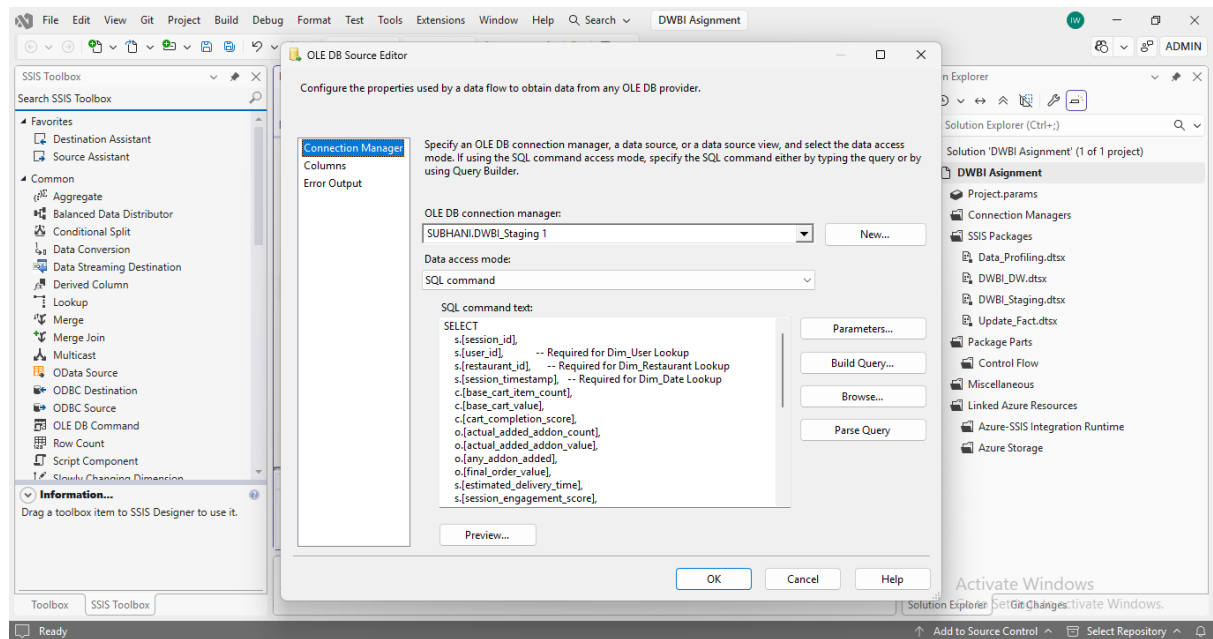


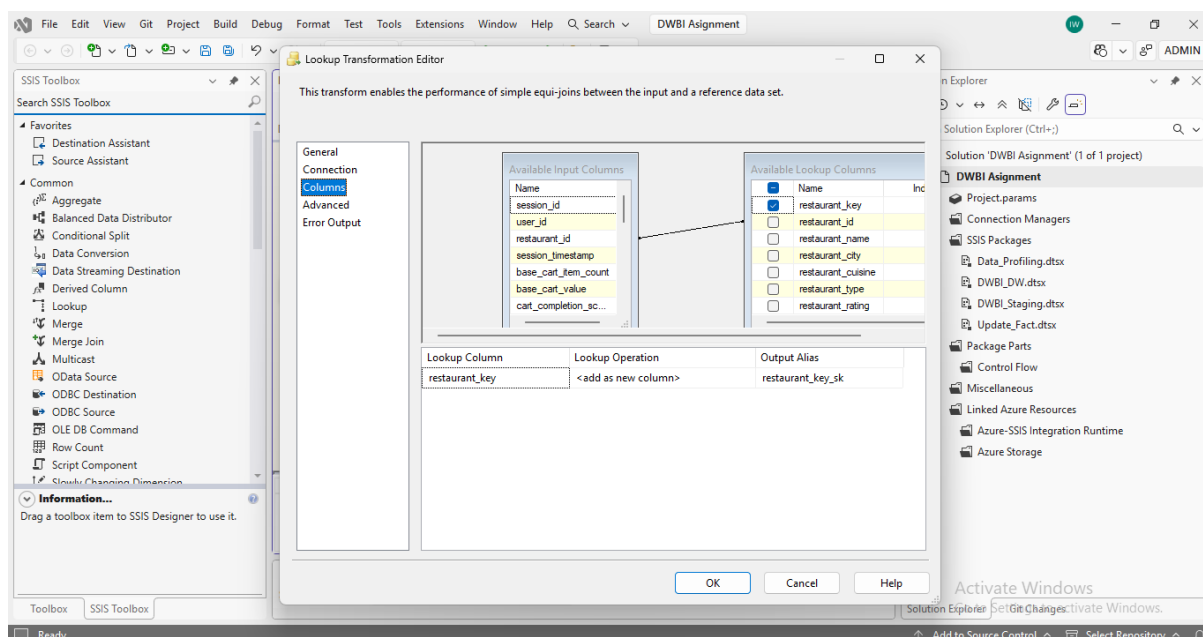
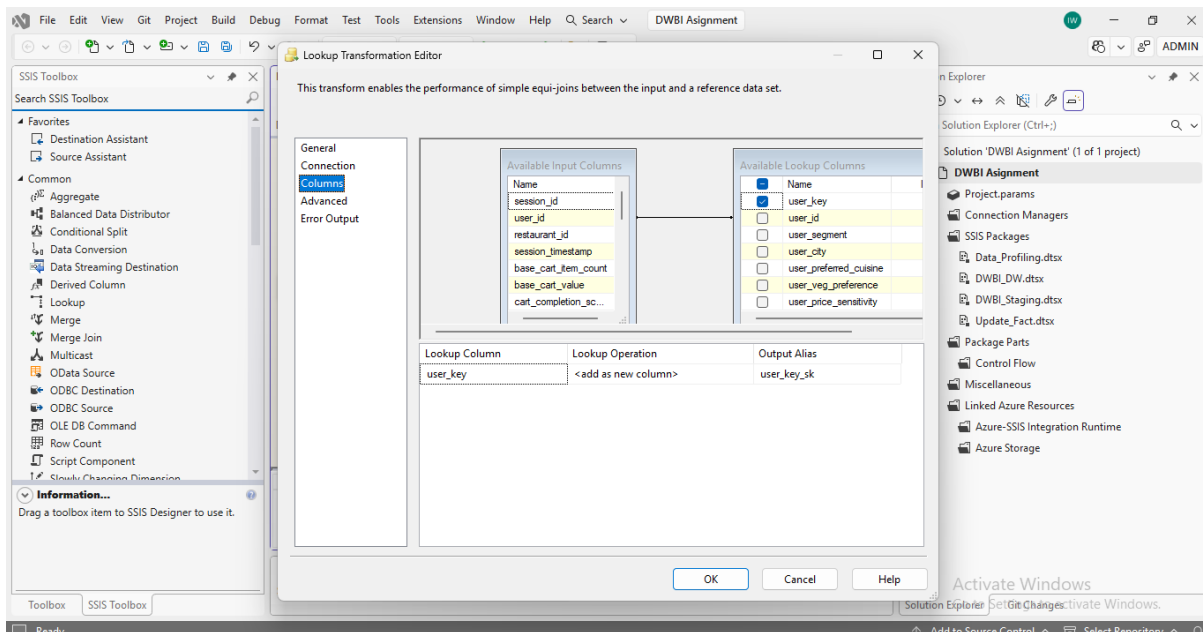
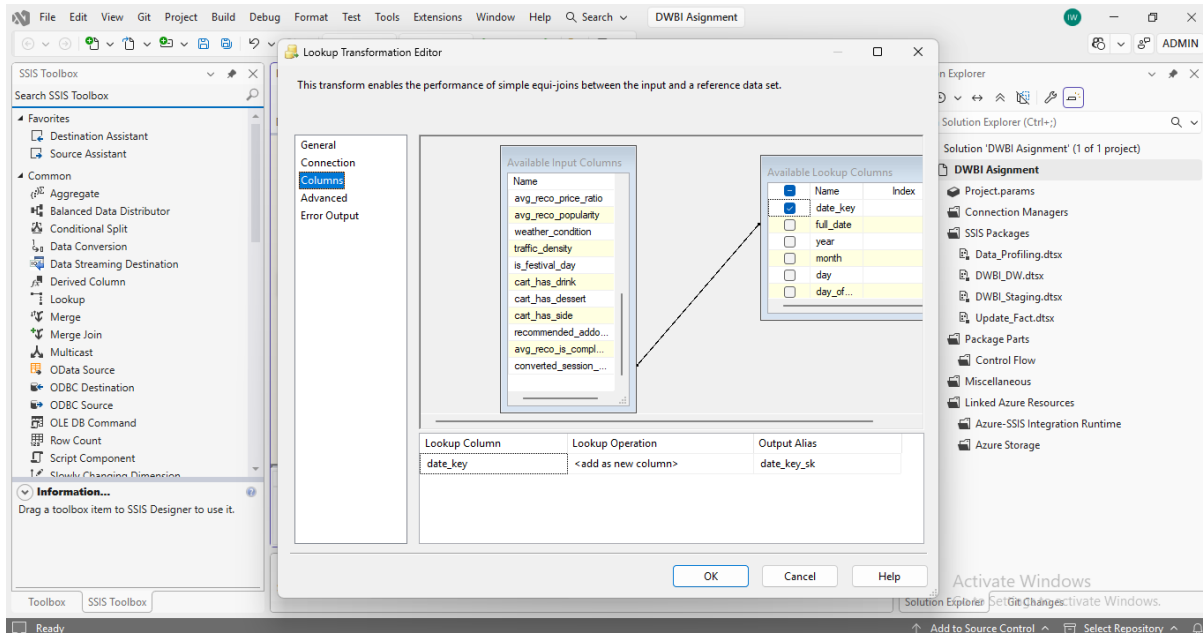


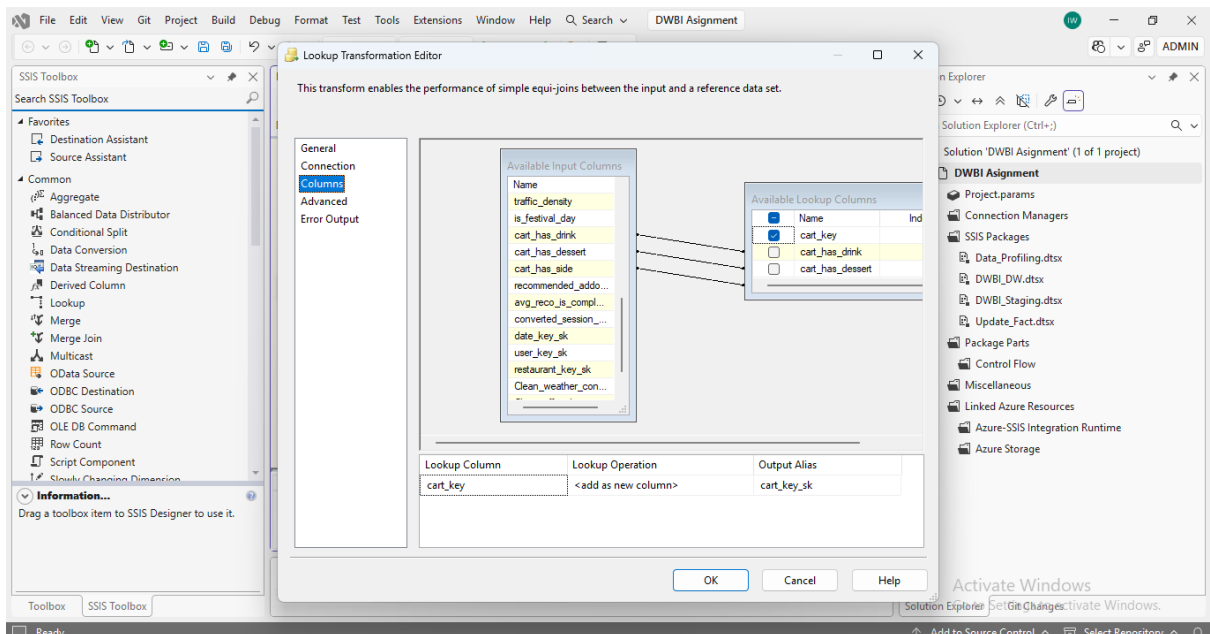
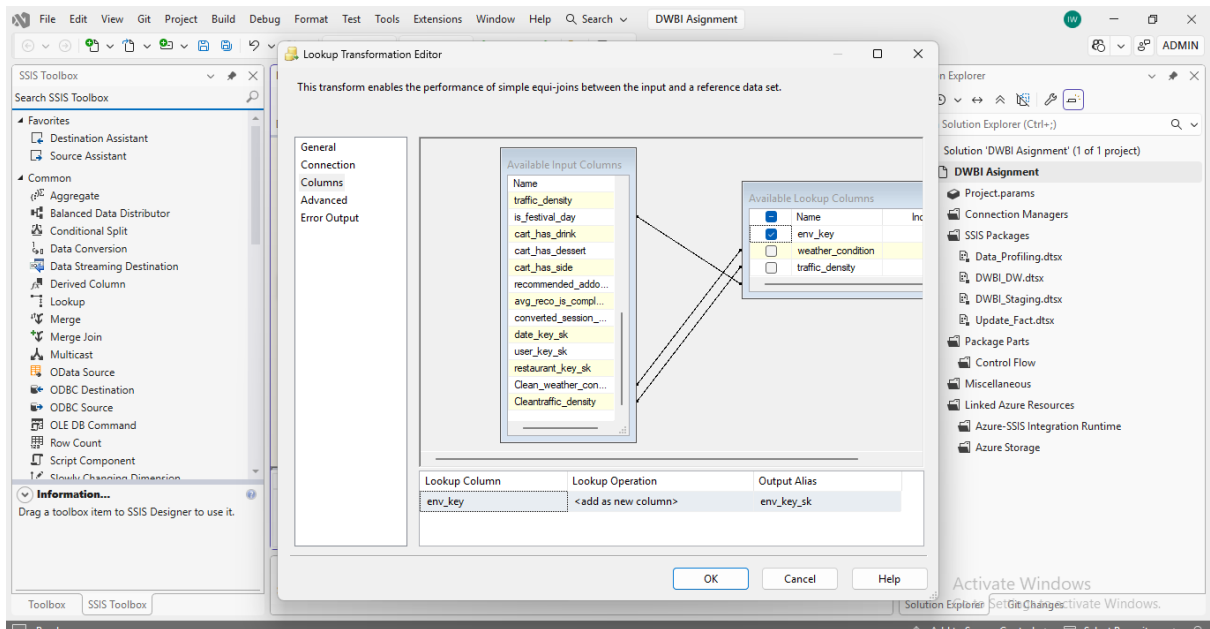
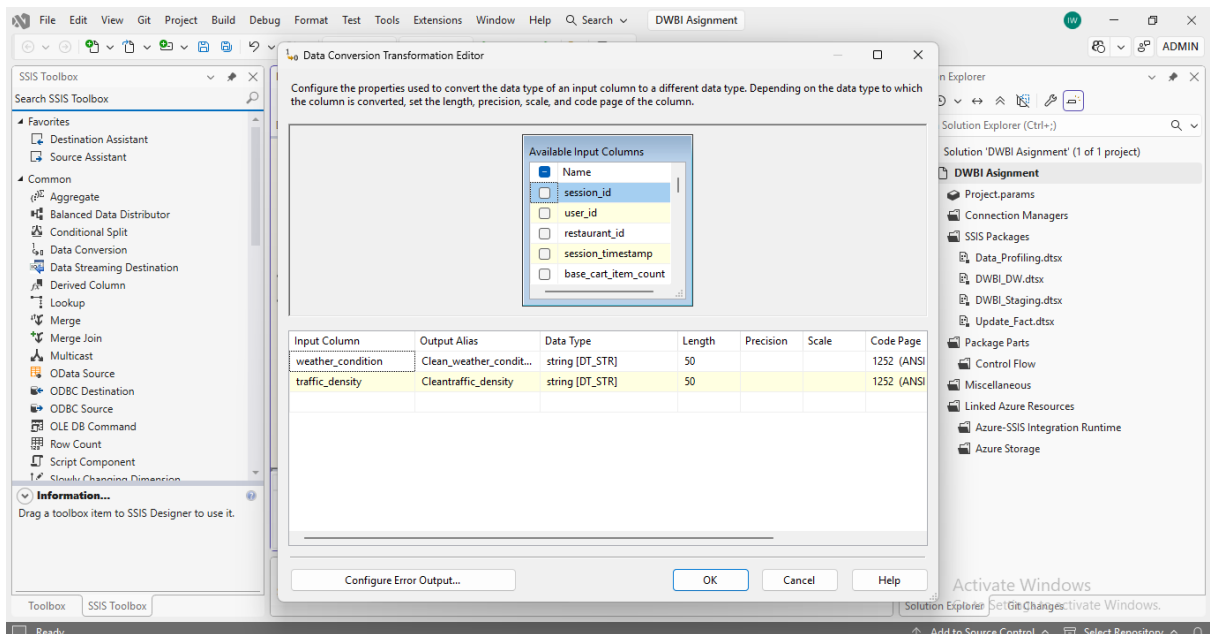
The screenshot displays the SSIS Designer interface for a project named 'DWBI_Assignment'. The main canvas shows a Data Flow Task named 'DFT_Load_Fact'. The task is composed of several data sources and a destination:

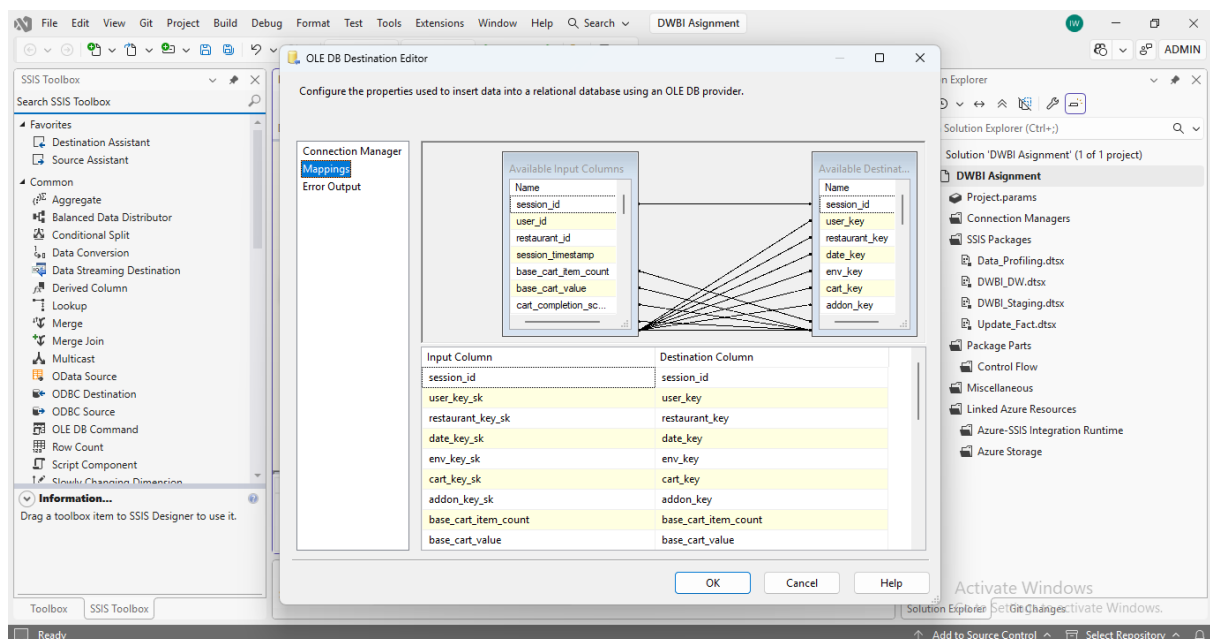
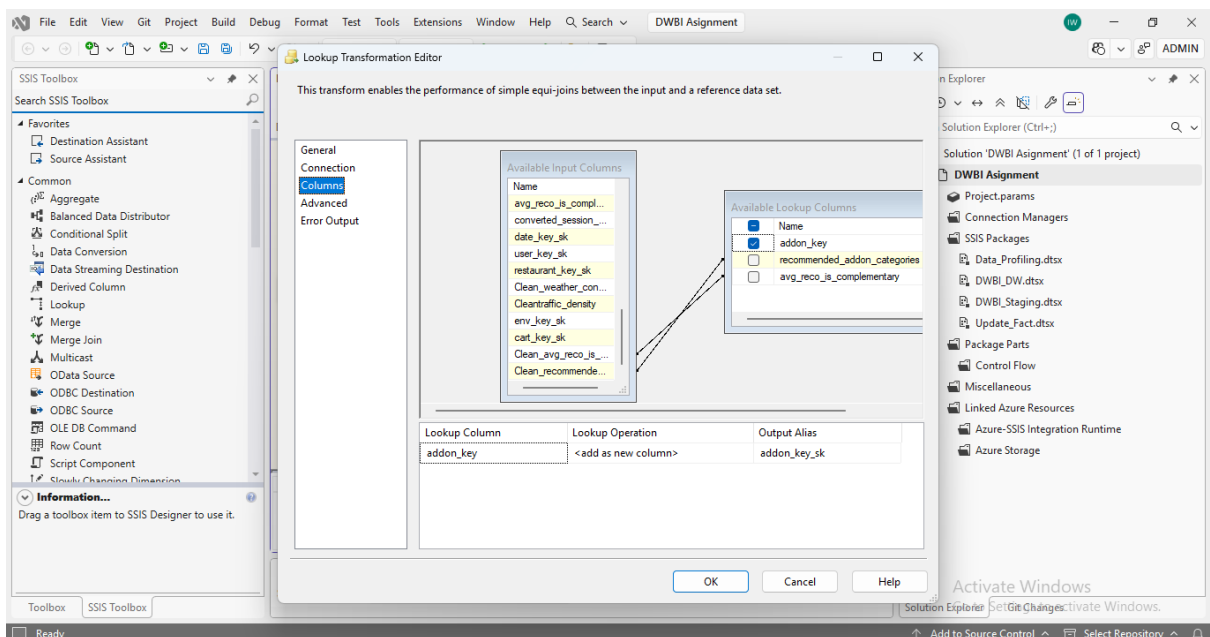
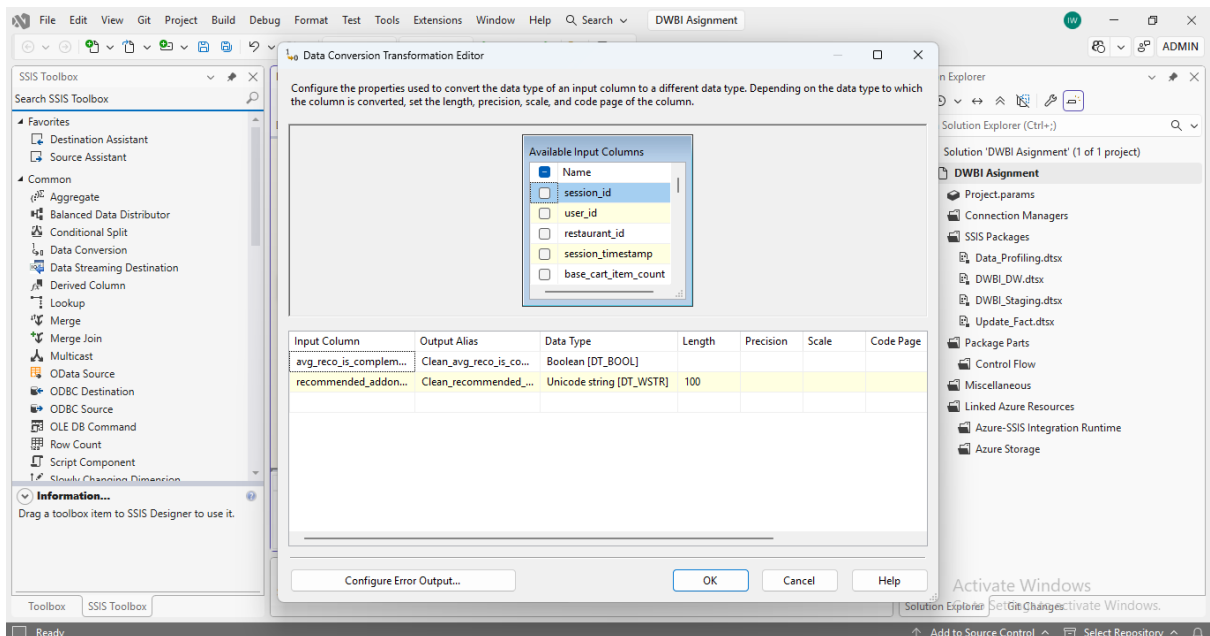
- Data Sources:**
 - Restaurant_Lookup**: Connected to **LookUp Match Output**.
 - User_Lookup**: Connected to **LookUp Match Output**.
 - Data_Lookup**: Connected to **LookUp Match Output**.
 - Data_Conversion_for_environment**: Connected to **LookUp Match Output**.
 - Environment_Lookup**: Connected to **LookUp Match Output**.
 - Data_Conversion_for_Addition**: Connected to **LookUp Match Output**.
 - Car_Lookup**: Connected to **LookUp Match Output**.
 - Addition_Lookup**: Connected to **LookUp Match Output**.
 - Derived Column**: Connected to **LookUp Match Output**.
- Destination:** **OLE DB Destination**.

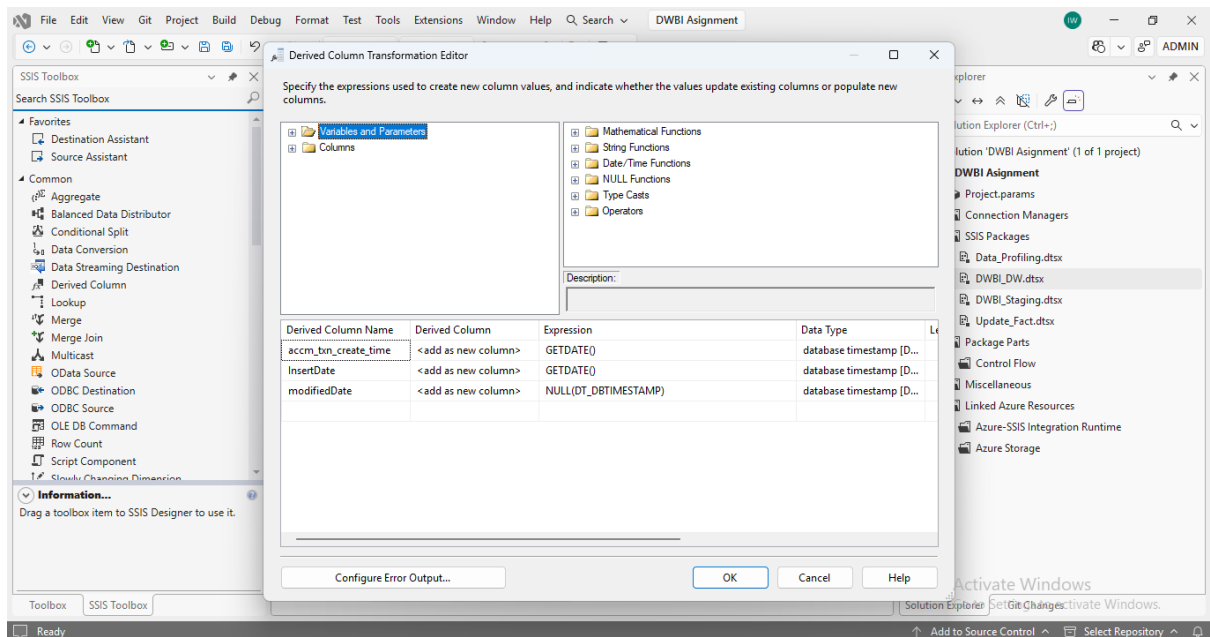
The interface includes a Solution Explorer on the right showing the project structure, and a Task Designer on the left with various data flow components. The bottom status bar indicates 'Zoom Change: 70%'.



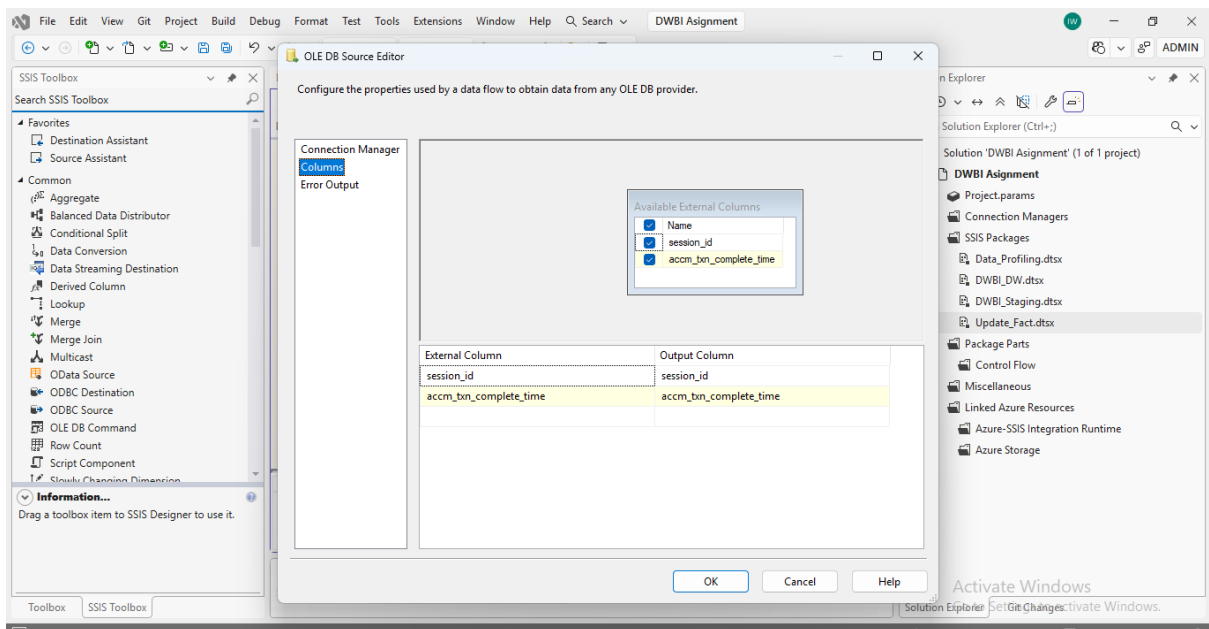
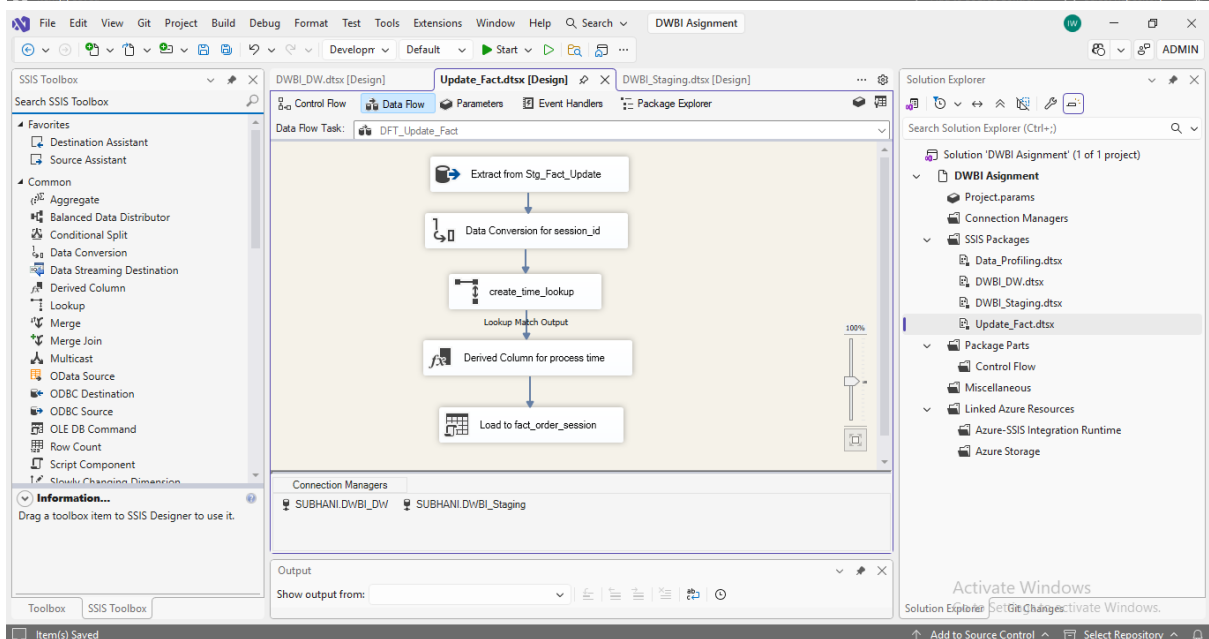
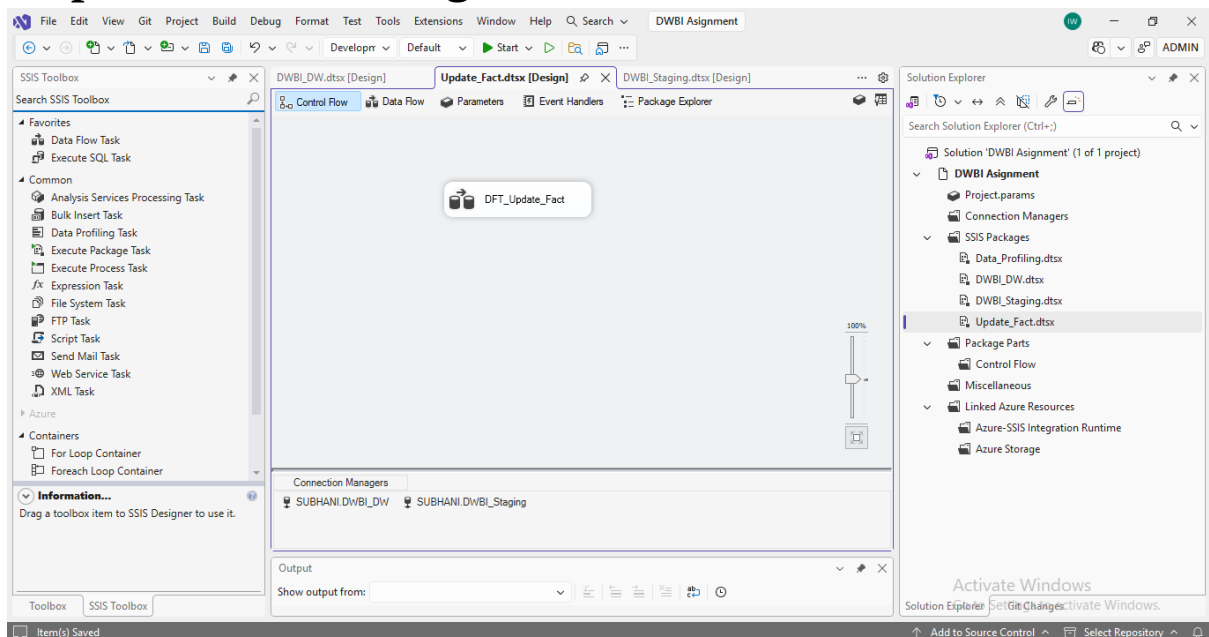


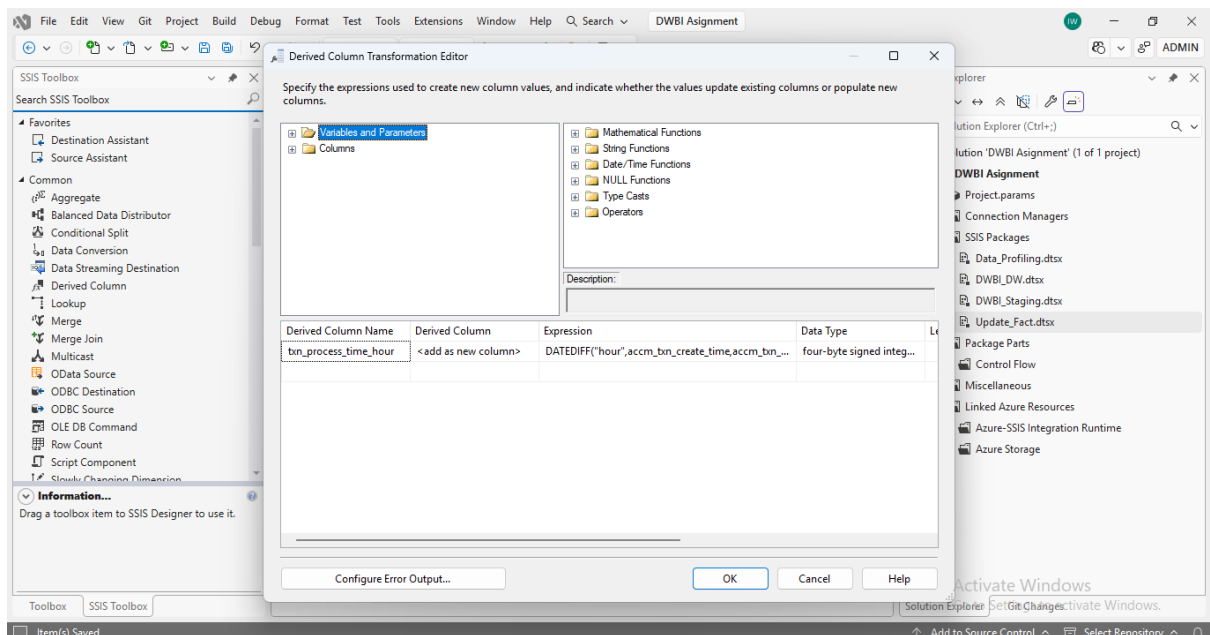
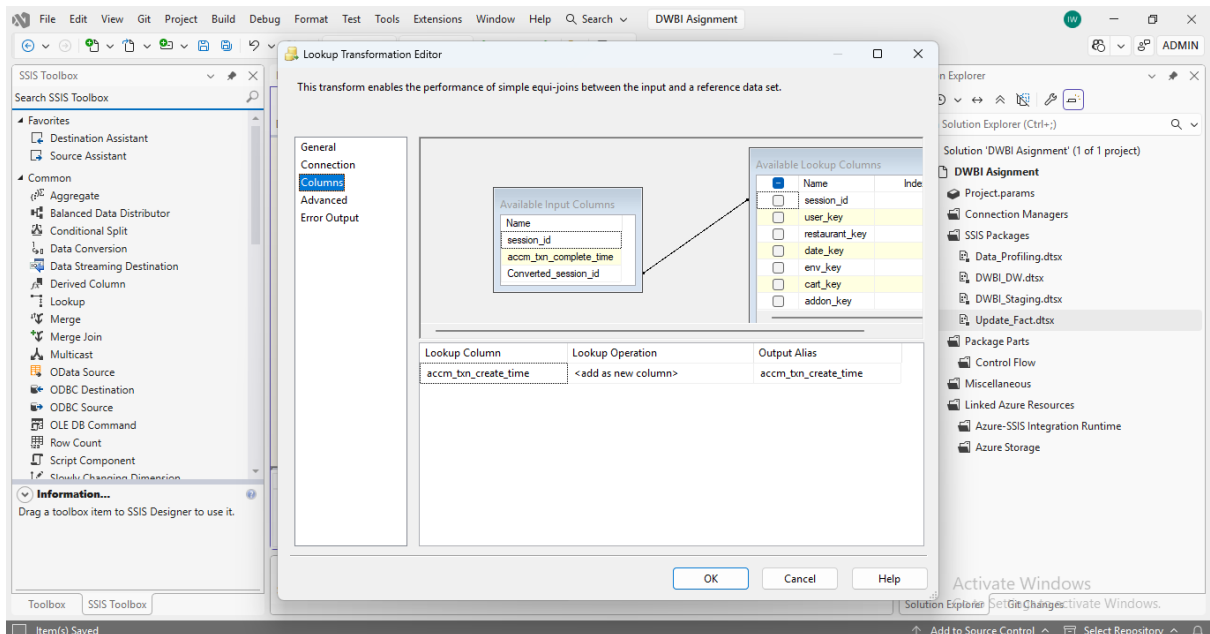
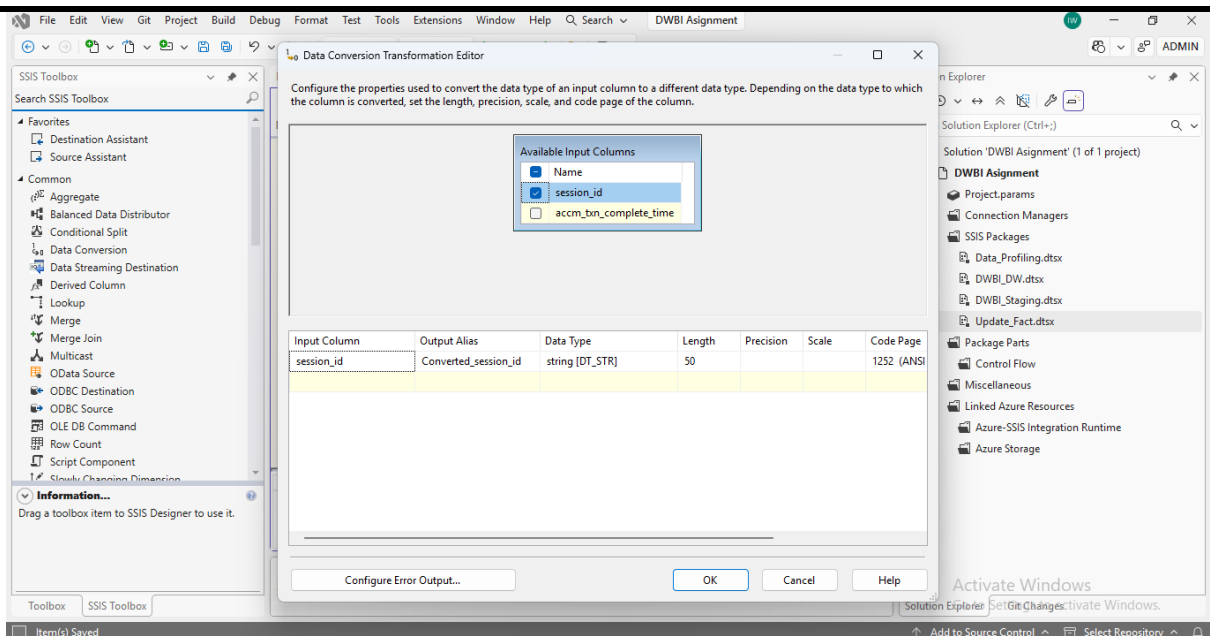


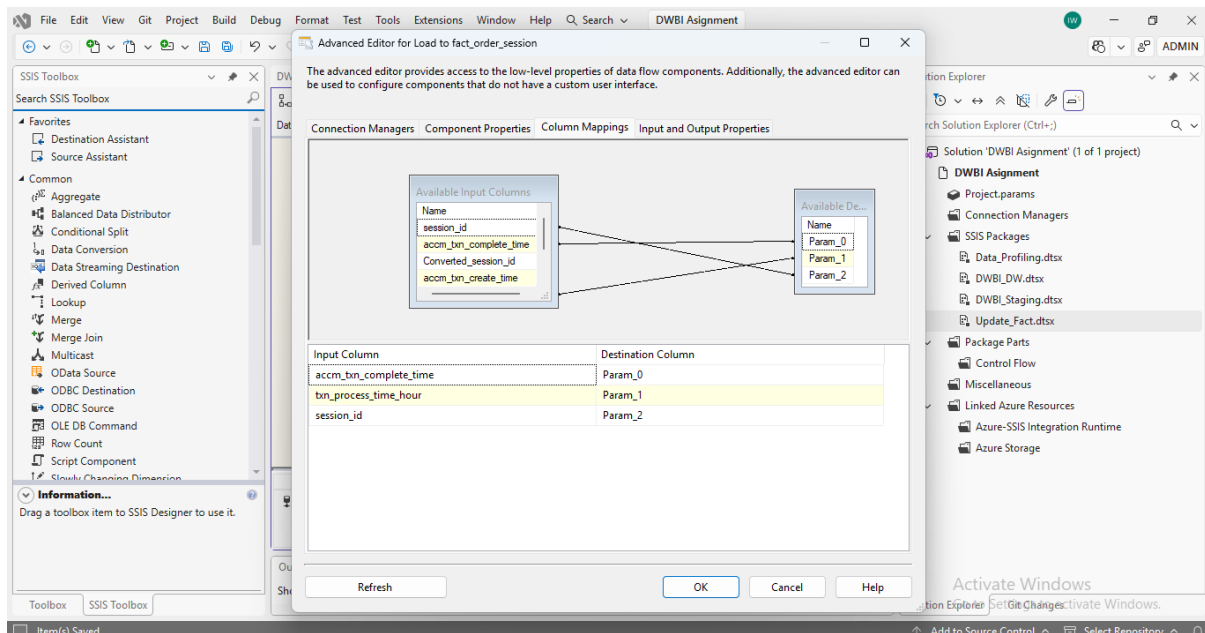




Step 6 – Accumulating Fact Table







Step 1: Create Staging Table for Updates

A staging table was created to store intermediate update data such as completion time.

```
CREATE TABLE Stg_Fact_Update (
    session_id INT,
    accm_txn_complete_time DATETIME
);
```

Step 2: Add Derived Columns in SSIS

A **Derived Column Transformation** was used to generate additional fields required for the accumulating fact table.

Columns Added:

- accm_txn_create_time → GETDATE()
- InsertDate → GETDATE()
- ModifiedDate → GETDATE()

Step 3: Load Initial Data into Fact Table

Initial data was loaded into the fact table using data from staging tables.

Description:

- Each record represents a session.
- At this stage:
 - accm_txn_create_time is populated
 - accm_txn_complete_time is **NULL**

Step 4: Perform Lookup Transformation

A **Lookup Transformation** was used to match records between:

- Fact table
- Staging update table

Description:

- Matching is done using:

session_id

- This ensures that updates are applied to the correct records.

Step 5: Update Fact Table

An **OLE DB Command Transformation** was used to update existing records in the fact table.

Example:

```
UPDATE Fact_Table
SET
    accm_txn_complete_time = ?,
    ModifiedDate = GETDATE()
WHERE session_id = ?
```

Step 6: Calculate Processing Time

A **Derived Column Transformation** was used to calculate duration.

Example:

```
txn_process_time_hours =
(DATEDIFF("hour", accm_txn_create_time, accm_txn_complete_time))
```

Step 7: Execute and Verify Results

After executing the SSIS package:

- Records with matching `session_id` were updated
- Other records remained unchanged

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the Object Explorer with the database structure of 'SUBHANI (SQL Server 17.0.1000.7 - sa)'. The central pane shows a query window with the following SQL statement:

```
1 | Select * from Fact_Order_Session
```

The bottom pane shows the query results in a grid format. The status bar indicates 'Query executed successfully.' and 'SUBHANI (17.0 RTM) sa (67) DWBL_DW | 00:00:02 | Rows: 1, Col: 1 | 100,000 rows'.

	core	avg_reco_score	avg_reco_price_ratio	avg_reco_popularity	acom_bn_create_time	acom_bn_complete_time	bn_process_time_hours	InsertDate	ModifiedDate
1	0.3661	0.3751	0.5733	2026-03-25 18:44:44.137	NULL	NULL		2026-03-25 18:44:44.137	2026-03-25 18:44:44.137
2	0.4492	0.3872	0.4496	2026-03-25 18:44:44.137	2026-03-26 01:31:36.353	7		2026-03-25 18:44:44.137	2026-03-25 21:52:15.767
3	0.4361	0.4549	0.6395	2026-03-25 18:44:44.137	NULL	NULL		2026-03-25 18:44:44.137	2026-03-25 18:44:44.137
4	0.4759	0.1344	0.5575	2026-03-25 18:44:44.137	NULL	NULL		2026-03-25 18:44:44.137	2026-03-25 18:44:44.137
5	0.5229	0.264	0.6561	2026-03-25 18:44:44.140	NULL	NULL		2026-03-25 18:44:44.140	2026-03-25 18:44:44.140
6	0.4463	0.5407	0.5742	2026-03-25 18:44:45.557	NULL	NULL		2026-03-25 18:44:45.557	2026-03-25 18:44:45.557
7	0.3882	0.4135	0.4823	2026-03-25 18:44:45.557	NULL	NULL		2026-03-25 18:44:45.557	2026-03-25 18:44:45.557
8	0.4283	0.1731	0.5424	2026-03-25 18:44:45.557	NULL	NULL		2026-03-25 18:44:45.557	2026-03-25 18:44:45.557
9	0.4163	0.1029	0.6418	2026-03-25 18:44:45.557	NULL	NULL		2026-03-25 18:44:45.557	2026-03-25 18:44:45.557
10	0.4229	0.3403	0.5799	2026-03-25 18:44:45.557	NULL	NULL		2026-03-25 18:44:45.557	2026-03-25 18:44:45.557

